Synthetic Bengali Handwriting: Learned Page Layout and Writer-Conditioned Word Generation

Final technical report

scribe_layout / scribe_layout_v2 / scribe_wordgen projects

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Abstract

BengaliScribe synthesises handwritten Bengali pages by generating word images with a font-prior GAN (baseline *B0*) and pasting them onto a page with hand-picked layout constants. Its pages look *typeset in a handwriting-like font*: the words re-ink a NotoSans skeleton with a stroke about 55% wider than the target writer's, and the lines are evenly spaced and straight. This report covers two lines of work. (1) **Page layout.** We measured how real writers lay out words on 963 full-page scans and trained a small page-autoregressive mixture-density model (v1, then v2). The final default (v2 `default_unfitw_dc`) reaches a plan-level line C2ST AUC of 0.535 ± 0.012 on unseen texts (v1: 0.566 ± 0.021; hand-tuned layout: 0.980; within-page null 0.496). Two spectral side-studies are reported with their corrections. (2) **Word generation.** We built a few-shot latent-diffusion word generator from DiffusionPen and a full evaluation suite. That suite has a locator CER, an independent CRNN recogniser (R_eval), writer-ID style metrics (E_eval/E_shape) and a sub-pixel pen metric. All statistics use writer-bootstrap CIs. The best diffusion arm, **S** (pen knob plus reference tokens in decoupled cross-attention), ties B0 on locator CER and on style MRR (0.579 vs 0.596), with a matched pen after calibration. Under R_eval it is still *less legible* than B0 ($+0.033$ [+0.015, +0.052] codepoint CER on words of $\geq$3 codepoints). Pretraining R_eval on renders from 48 Bengali fonts helps a little with full data and a lot with 10% of the writers. A later fix to a short-word ink artefact and a full rescore (rescore_v2) flips S's locator tie with B0 into a clear win (-0.033 [-0.050, -0.017]), though R_eval still ranks S behind B0 ($+0.046$ / $+0.034$ codepoint CER, all / $\geq$3 codepoints), driven mainly by BanglaWriting and partly by a rendering-centring artefact. Sampling S eight times per word and keeping the most legible render under an independent check (best-of-8) closes most of that gap at 8$\times$ sampling cost, with no loss of style fidelity. A third experiment, fine-tuning S with a cheap, low-timestep auxiliary recogniser (OCR-CTC) loss, against a pre-registered win rule, is a null-to-negative result: S_ocr is significantly *less* legible than its control under R_font2real ($+0.0053$ [+0.0014, +0.0090] CER) and also fails the independent-locator and style-MRR gates, so best-of-N sampling remains the method that closes most of the R_eval gap. A fourth experiment pretrains R_eval's own recogniser architecture on 6,000 matched words from each of F (fonts), B0, S1 and S8, then fine-tunes on only 10% of TRAIN writers (3 training seeds per arm): S8$-$F is -0.0154 [-0.0227, +0.0029] (negative in all 3 seeds, but the pooled CI crosses 0, so **not** a pre-registered win), while the access-matched S8$-$B0 is -0.0075 [-0.0109, -0.0040] (significant, same sign on TEST); every 6,000-word matched corpus still falls well short of the existing, unmatched font-stream baseline (0.115 CER). A fifth experiment, **mix_pretrain**, instead *mixes* 25% generator-S words into the otherwise unlimited font stream rather than replacing it, and **wins the pre-registered test**: FsS$-$Fs is -0.0229 [-0.0278, -0.0135] (negative in all 3 seeds, same sign on TEST, -0.0233 [-0.0379, -0.0127]), with the access-matched FsS$-$FsB0 also significant (-0.0164 VAL / -0.0147 TEST). Every number in this report is traced to a result file.

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1 Executive summary

Key findings.

- **B0's "font look" is architectural, not a training accident.** B0 keeps fine-resolution skips from a NotoSans prior. Its only deformation module can move ink at most 1.4 px vertically on a 64-px canvas. Its losses reward the averaged font skeleton. Generated page strokes are 0.39 mm wide against 0.25 mm real (§2).
- **The learned layout beats hand-tuned constants by a wide margin and v1 by a small one.** The plan-level line C2ST is 0.980 for manual, 0.566 ± 0.021 for v1 and 0.535 ± 0.012 for the v2 default on TEST (0.536 ± 0.027 on VAL). End to end, through BengaliScribe and the real detector, v2 scores a line C2ST of 0.719 (0.619 with writer context), against 0.905 for manual and a floor of 0.534 (§5).
- **Several layout gains were measurement artefacts and were corrected.** Corrected items are a biased end-to-end C2ST (floor 0.34 instead of 0.5), per-word wobble overshoot, selection on TEST, and a Fourier "17$\times$ better" claim. In the fair comparison, Fourier and v1 are equal, 0.668 vs 0.702 (§5.5).
- **Diffusion fixes the texture but had to earn legibility.** The first DiffusionPen probe wrote round, flowing, handwriting-like strokes, but its locator CER was 0.65–0.97. Content tokens in cross-attention, full data, white-paper normalisation and a fixed classifier-free-guidance (CFG) null brought B′ to 0.188 at g3. That fix was a bug fix: content had sat in both CFG branches. A guidance of t5c5 then reached 0.148, against 0.130 for B0 (§6).
- **Arm S is the best diffusion arm.** It ties B0 on locator CER ($+0.007$ [-0.023, +0.040]) and on style M2 MRR (-0.017 [-0.116, +0.088]). It beats P on style by $+0.136$ [+0.085, +0.187] (17/20 writers). Its pen is 1.13 raw and 1.06 calibrated (§6.9).
- **R_eval changes the CER picture.** The locator was too weak to separate arms. With R_eval, S$-$B0 is $+0.059$ [+0.036, +0.083] on all words and $+0.033$ [+0.015, +0.052] on words of $\geq$3 codepoints. Part of the all-word gap is a short-word `img_to_ink` artefact; the $\geq$3-codepoint gap remains significant (§7.3).
- **A smudge-ink rescore (rescore_v2) flips the locator tie into a clear win for S.** With `img_to_ink(min_span=100)` applied to every arm and 9,372/9,372 regenerated renders verified pixel-exact against the old cache, S$-$B0 on locator CER is -0.033 [-0.050, -0.017] (it was $+0.007$ on the legacy ink). R_eval still ranks S behind B0 by $+0.046$ / $+0.034$ (all / $\geq$3 codepoints), driven by BanglaWriting; part of the all-word gap is a rendering-centring artefact (S's renders are centred, R_eval trains on left-aligned crops) (§8).
- **Sampling S eight times and keeping the most legible render closes most of the remaining R_eval gap.** Best-of-8 vs best-of-1 on R_font2real, $\geq$3 codepoints: -0.033 [-0.044, -0.024], 20/20 writers better; best-of-8 ties B0 ($+0.001$ [-0.011, +0.013]) with unchanged style, at 8$\times$ sampling cost (§9).
- **A recogniser-loss fine-tune of S is a null/slightly negative result.** Against its pre-registered win rule, S_ocr ($\lambda = 0.1$ auxiliary OCR-CTC loss at $t < 200$) is significantly *less* legible than its control S_ft0 under R_font2real ($+0.0053$ [+0.0014, +0.0090] CER, all words) and does not move the independent locator CER significantly; its style M2 MRR also drops past the pre-registered -0.03 bound. Only the pen-ratio gate passes. Best-of-N sampling (§9) remains the method that closes most of the R_eval gap (§10).
- **Recogniser CER measures legibility, not fidelity.** B0 scores *better than the real crops* under R_eval (-0.014 [-0.027, -0.002]). CER is therefore a floor check and must be read alongside style and pen.
- **Fonts are a useful recogniser prior, mostly when real data is scarce.** Fonts$\rightarrow$real fine-tuning gives -0.004 [-0.008, -0.002] VAL CER with all writers and -0.063 [-0.080, -0.054] with 10% of the writers. Fonts alone give 0.456 and do not transfer (§7).
- **Pretraining the recogniser on generator S's words beats B0's, but misses the pre-registered win over fonts.** CRNN pretrained on 6,000 matched words from each of F/B0/S1/S8, fine-tuned on 10% of TRAIN writers, 3 seeds: VAL CER F 0.169, B0 0.161, S1 0.155, S8 0.154. S8$-$F $= -0.0154$ [-0.0227, +0.0029], 3/3 seeds negative but the pooled CI crosses 0 $\Rightarrow$ **not a win**; the access-matched S8$-$B0 $= -0.0075$ [-0.0109, -0.0040] is significant (same sign on TEST). S's advantage over F holds on BN-HTR (-0.022, significant) but not BW ($+0.012$, n.s.). Every 6,000-word matched corpus is far short of the unmatched font-stream reference (0.115) (§11).
- **Mixing 25% of generator S's words into the (unlimited) font stream, instead of replacing it, is a pre-registered win.** Same CRNN, same R_font0 60k-step pretrain recipe, same 10%-writer fine-tune, 3 seeds: VAL CER Fs 0.1126, Fs+S 0.0898, Fs+B0 0.1062, Rf10 (reference) 0.1147. FsS$-$Fs $= -0.0229$ [-0.0278, -0.0135], negative in all 3 seeds, same sign on TEST (-0.0233 [-0.0379, -0.0127]) $\Rightarrow$ **WIN**. The access-matched FsS$-$FsB0 is also significant (-0.0164 VAL / -0.0147 TEST), and unlike §11's S8$-$F, the gain holds on both BN-HTR and BanglaWriting (§12).

Table 1: Current best, word generation (VAL, 20 writers, 781 targets). CER (locator) is letter-unit and per-word-mean. R_eval CER is codepoint and corpus-level. **The two CER columns are in different units; do not compare them.** Style is M2 MRR (E_eval; chance 0.28). Pen is the generated/own-writer sub-pixel width-per-core ratio (1.00 = match). CIs are 95% writer-bootstrap.

arm	locator CER	R_eval CER all	R_eval CER ≥3cp	style M2 MRR	M2s MRR	pen raw / cal
real	0.133 [0.100, 0.167]	0.0438 [0.031, 0.059]	0.0424 [0.029, 0.057]	0.821 [0.775, 0.859]	0.768	1.00
B0	0.130 [0.103, 0.156]	0.0302 [0.025, 0.036]	0.0226 [0.016, 0.029]	0.596 [0.488, 0.711]	0.533	1.15
S t5c5	0.137 [0.101, 0.174]	0.0889 [0.065, 0.115]	0.0552 [0.038, 0.075]	**0.579** [0.521, 0.635]	0.517	1.13 / 1.06
P t5c5	0.132 [0.097, 0.171]	0.0905 [0.066, 0.118]	0.0577 [0.040, 0.079]	0.443 [0.379, 0.505]	0.416	1.15 / 1.06
B′v2 t5c5	0.151 [0.117, 0.186]	0.1412†	0.1024†	0.418 [0.355, 0.483]	0.399	1.29 / 1.07

†From the full R_eval rescoring table in `runs/R_eval/full_chain.log` (`Bv2_t5c5`: 0.1412 [0.1125, 0.1749] all words, 0.1024 [0.0823, 0.1272] ≥3cp); `RESULTS.md` lists only the default subset of arms. Checkpoints: S `runs/P1a3/S/`, P `runs/P1a3/P/`, B′ `runs/P1a2/B/`, B0 = BengaliScribe `models/v2`. **This table predates the smudge-ink fix.** After `img_to_ink(min_span=100)` and rescoring (rescore_v2, §8), S's locator CER is 0.097, not 0.137, and S−B0 on locator is −0.033 [-0.050, -0.017], not a tie; best-of-8 sampling (§9) further ties B0 under R_eval on ≥3cp words. See Table 27 for the current, post-fix numbers across arms.

Source: `runs/P1a3/eval/report_tables.txt`, `runs/P1a3/eval/metrics.json`, `runs/R_eval/RESULTS.md`, `EXPERIMENTS.md` (all under `scribe_wordgen/`)

Table 2: Current best, page layout. Lower AUC = closer to real; 0.5 = indistinguishable. Plan level: 193 TEST pages (unseen texts), 5 seeds, mean ± sd. End to end: 5 writers × 2 seeds, balanced C2ST.

layout	plan line C2ST (TEST)	plan line C2ST (VAL)	plan page C2ST (TEST)	e2e line / page C2ST
manual (BengaliScribe constants)	0.980	–	1.0	0.905 / 0.999
v1 (`scribe_layout/runs/default`)	0.566 ± 0.021	0.560 ± 0.024	0.654 ± 0.043	–
v2 default (`default_unfitw_dc`)	$\mathbf{0.535 \pm 0.012}$	0.536 ± 0.027	0.590 ± 0.024	0.719 / 0.850
v2 default + writer context	–	–	–	0.619 / 0.845
null: within-page split / e2e real halves	0.496 ± 0.019	–	–	0.534 / 0.508

Source: `scribe_layout_v2/runs/compare_unfitw/compare_5seed{,_dc,_val}.txt`, `scribe_layout_v2/report/FACTS.md`, `scribe_layout_v2/runs/e2e_final/results.md`

2 Introduction and motivation

BengaliScribe (BS) turns a text into a handwritten-looking page in three steps. It reads the target writer's reference crops with an attention locator (`attn_locator`). It generates each word with a font-prior GAN conditioned on a style vector (`font_generator_pilot_sal.py`, "FG"). It then lays the words out with hand-picked constants (`write_page.py`, "WP"). We call the shipped generator **B0**. Our showcase writer is `image_2`, a page from the hcr/996 corpus that no model trains on. Its hand is thin, sparse, rounded and cursive. B0's output for it is bold, uniform and angular: a re-inked font skeleton.

2.1 Why B0 looks like a font

The planning diagnosis (`PLAN.md` §1) was checked line by line by an adversarial review (`PLAN_REVIEW.md` §0). It found these causes:

1. **The architecture copies the font.** FG is a 4.6 M-parameter U-Net over the NotoSans render (the checkpoint holds 4.587 M). It keeps the 1/4- and 1/2-resolution skips (`SKIP_SCALES=(1,1,0)`), so stroke positions reach the decoder with 2-px precision. The only module that moves ink is `StyleWarp`. A bound on its learned weights gives at most 1.38 px vertical and 5.75 px horizontal displacement on the 64×256 canvas; the architectural cap is 3.84/15.4 px. An i-kar loop of the reference writer rises 0.87 core (median) above the matra, which is out of reach. Cross-attention to the references is off (`XATTN=False`).
2. **The losses reward an averaged font skeleton.** The loss is $4\,L_1 + 1{\cdot}$LSGAN $+0.3$ style-CE $+0.3$ CTC. The L_1 term is salience-weighted on both target and font, and the PatchGAN is conditioned on the prior. When prior and target are misaligned beyond the warp's range, the L_1 optimum is a blurred stroke along the font skeleton.
3. **Train/inference mismatch in width.** In training the prior is resized into the target's box. At inference it keeps the font's natural width, so word width and letter spacing are copied from the font.
4. **Resolution and pen are set by the data, not the writer** (Table 3). The reference writer's stroke-to-core ratio (0.053) is below BN-HTR's 10th percentile. At 64 px it is about 1.6 px wide, which the canvas cannot render

sharply. `crop_ink` also stretches every target so its 5th-percentile pixel is fully black, so darkness saturates: 0.93 generated against 0.80 real.

The measured outcome on the page is a stroke of **0.39 mm generated against 0.25 mm real**, about 55% wider (`PLAN.md` §1.4, orchestrator measurement).

Table 3: Stroke width relative to letter size (skeleton EDT width ÷ body height from `measure_rows`), by data set. image_2 is the reference writer.

set	crops	width/core median (IQR)	native crop height	darkness
BN-HTR (B0's training data)	1489	0.106 (0.077–0.185)	74 px	0.98
BanglaWriting	1497	0.090 (0.066–0.135)	163 px	1.00
hcr/996 (60 pages)	1447	0.075 (0.060–0.103)	102 px	0.82
hcr image_2 (reference writer)	57	**0.053** (0.048–0.067)	107 px	0.79
generated for image_2 (64-px canvas)	60	0.063 (0.045–0.076)	64	0.97

Source: `scribe_wordgen/PLAN.md` §1.4 (script `scratch/pen_ratio.py`; re-run identically in `PLAN_REVIEW.md` §0)

Per-writer BN-HTR medians span 0.073–0.186 (p10–p90 over 110 writers). The second problem is **layout**. WP places words with uniform random gaps, a tiny fixed line slope, and a fixed pitch of $2.7\times$ body height, so every line is straight and evenly spaced.

2.2 Goal and overall design

The goal is a page generator that *writes like the reference writer*: the writer's letter construction and pen, laid out the way real people lay out lines. Figure 1 shows the pieces built in this project and how they are evaluated. Each evaluator is independent of the generator it scores: none is ever used as a training loss.

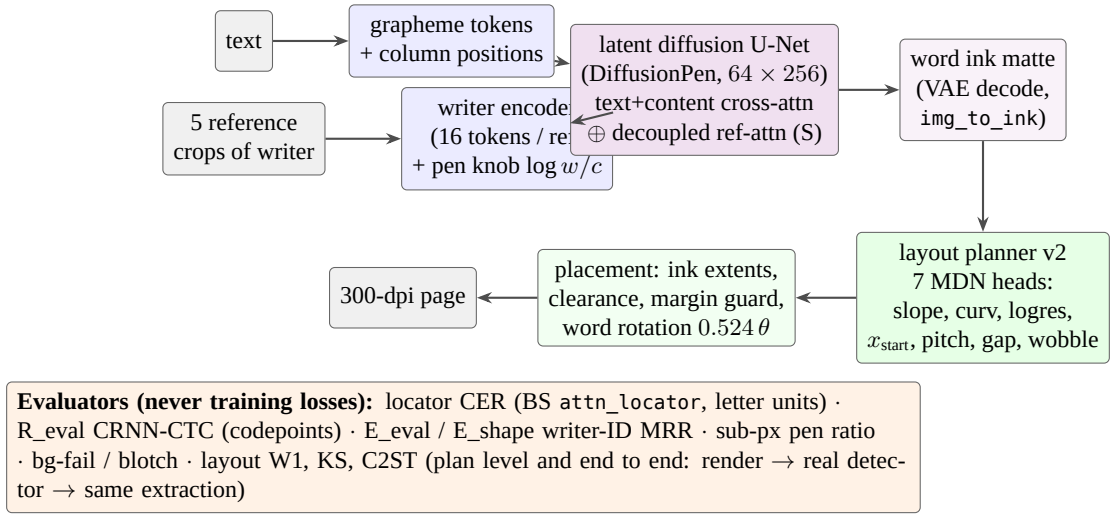


Figure 1: The system. Upper row: the word generator (Arm S configuration; Arm P lacks the reference-token attention). Lower row: the learned layout and its placement rules. Bottom: the evaluators.

3 Data and protocol

3.1 Data sets

Table 4: Data sources. Only BN-HTR and BanglaWriting are used to train word models. hcr/996 is used for layout and as an evaluation-only showcase (image_2).

source	word crops	writers	native size	notes
BN-HTR	92,380 labelled + filtered (from 108,181)	149	median crop h 74 px, core ≈28–30 px	93,758 labelled and ≤16 graphemes, minus 1,378 flagged by `label_filter.json`; folder = writer
BanglaWriting (BW)	20,523	260 (`style_id` = page = person)	median crop h 163 px, core ≈49 px	one page per writer; manual labels
hcr/996	≈48.7k word pieces	unknown	A4, 300 dpi, core median 41 px	27 texts copied by many writers; 963 usable pages for layout; no writer IDs

Source: `scribe_wordgen/PLAN.md` §1.6, §2.1; `runs/T0/REPORT.md`; `scribe_layout/report/scribe_layout_report.tex` §2

3.2 Writer-disjoint splits

Splits are fixed in `data_cache/splits.json`, and a test asserts that no writer appears in two of them (`runs/T0/REPORT.md`):

- **BN-HTR:** TEST = {15, 22, 24, 28, 30, 48, 51, 54, 69, 136} (5,768 crops); VAL = {99, 117, 127, 142, 149}; TRAIN = the remaining 134 writers. TEST writers are in BS's generator and locator hold-out, so B0 is scored on writers it never saw. The VAL writers are in the TRAIN set of BS's style encoder `ms4`; this was verified, not assumed (`PLAN_REVIEW.md` §0).
- **BanglaWriting:** `style_id`s hash-sorted (SHA-256) into 220 TRAIN / 15 VAL / 25 TEST. `dp_words_b64` (baseline B3) saw all 260 BW writers, so B3 is fair only on BN-HTR.
- **hcr/996 (layout):** split *by source text*: 18 train / 4 val / 5 test texts (test texts {4, 6, 11, 15, 18}), giving 614 / 144 / 193 pages. The v1 report claimed test lines are by unseen writers. The v2 review notes that writers likely repeat across texts, which cannot be checked without writer IDs (§13.1).

3.3 The VAL protocol (word generation)

Unless stated otherwise every word-generation number uses this protocol (`EXPERIMENTS.md`, header):

- **20 VAL writers**: the 5 BN-HTR VAL writers plus the 15 BW VAL writers, with **781 target words** in total.
- **5 conditioning references per writer.** For BN-HTR these come from page A and the targets from other pages. For BW the refs and targets come from the same (only) page, which inflates BW style metrics through page cues (`runs/P1a/REVIEW.md` §4).
- Targets whose text is among the refs are excluded. Punctuation-only targets (54/781, all BW) are excluded from the headline CER.
- **Seeds.** B0 and real have one sample per word. Diffusion arms have two seeds from eval_v3 onward. Paired comparisons merge on (dataset, writer, idx, seed_k) and never zip lists.

3.4 TEST status

TEST evaluation (Checkpoint C) regenerated on 2026-10-22 06:14–10:13 UTC. The final TEST evaluation (Checkpoint C) regenerated S, F1, F8 arms on TEST split (35 writers, 2,756 images per arm) following Phase 1 vocabulary correction. Locator metrics (attn_locator CER, polygon IoU) computed via regenerated test/{S,F1,F8}/ directories; R_font2real (OCR recognizer) checkpoint was unavailable (Phase 2 training incomplete); pen metrics computed relative to S (reference) since real images do not carry pen dynamics. Metric CSVs: `runs/checkpoint_C/scores/\{S,F1,F8,B0,real\}__s\{0,1\}.csv` (locator); `runs/checkpoint_C/rescore_font2real.csv` (8,268 rows, R_font2real

scores skipped). FREEZE.json locked with TEST image hashes and Phase 2 recognizer hashes (R_font0, R_real, R_real_10). The only TEST contact is that the independent recogniser R_eval was scored once on real TEST crops (7,665 words, §7). For layout, all design choices were compared on TEST (selection on test). A selection-free VAL check was run afterwards (§5.4).

3.5 Data-handling facts that affect the numbers

- **The BanglaWriting word-crop PNGs are missing from disk.** The files under `Diffusion_model_dataset_building/dataset/word_crops/` that `index_bw.csv` points to were deleted. The script `wordgen/recog/bw_regen.py` rebuilds the crops in memory from the LabelMe page JSON and JPG in `banglawriting/converted/`. It is a faithful port of the original `parse_json`/`crop_box` and label-checks every crop.
- **R_eval BW coverage.** The R_eval real-crop cache holds **20,482 of the 20,523** BW words (17,539 TRAIN + 1,046 VAL + 1,897 TEST rows in `runs/R_eval/cache/bw_*.jsonl`). The 41 missing words are the rows that already failed `measure_rows` in the T0 data index (`runs/T0/REPORT.md` §4.2: ok 20,482 of 20,523); the cache's per-split counts equal T0's ok-row counts.
- **Core-normalised canvas: clip target not met.** The plan's 160-px core-normalised canvas (core 32 px) clipped 22.6% of BN-HTR and 21.1% of BW TRAIN crops, almost all at the top (`runs/T0/REPORT.md` §4.3). A writer-prior fallback cut this to 18.98% / 14.57% (`runs/T0b/REPORT.md`), still far from the $<$1% target. The SD-VAE round trip also fails the plan's IoU gate ($\geq$0.93) at core 32 (image_2 0.862; `PLAN_REVIEW.md` B1). All generators therefore stayed on DiffusionPen's height-normalised 64×256 `fit_canvas`. The learned layout rescales words by measured core anyway.

4 Evaluation methodology

Every word metric is computed per writer and then averaged over writers. The CI is a writer-level bootstrap: 2,000 resamples, 95% percentile interval. A "tie" means the paired CI includes 0.

4.1 Locator CER (letter units, per-word mean)

BS's frozen `attn_locator` reads each word on a fixed 64×256 canvas. The ink is tight-cropped and left-aligned; wider words are resized to fit (eval_v3 truncated B0 words wider than 256 px, which affects about 10% of B0). For target y_i and reading $\hat{y}_i$,

$$\mathrm{CER}_i^{\mathrm{loc}} = \min\Big(1,\ \frac{d_{\mathrm{edit}}(\hat{y}_i, y_i)}{|y_i|}\Big), \qquad \mathrm{CER}_w^{\mathrm{loc}} = \frac{1}{|T_w|S} \sum_{i \in T_w} \sum_{s} \mathrm{CER}_{i,s}^{\mathrm{loc}}, \qquad \mathrm{CER}^{\mathrm{loc}} = \frac{1}{20} \sum_{w} \mathrm{CER}_w^{\mathrm{loc}}. \tag{1}$$

The edit distance is in the locator's letter units. The cap at 1 matters: without it, garbage reads of 1-character punctuation targets gave CERs of 23–29 and 15% of words had CER $>$ 1 (`runs/P1a/REVIEW.md` §1). The locator is BN-HTR-trained. Its own real-crop CER is 0.056 on BN-HTR VAL and 0.236 on BW VAL (`runs/R_eval/RESULTS.md`), so it separates arms poorly on BW.

4.2 R_eval CER (codepoints, corpus level)

R_eval is an independent CRNN-CTC recogniser (§7). Its CER is micro-averaged over Unicode codepoints and not capped:

$$\mathrm{CER}^{R} = \frac{\sum_i d_{\mathrm{edit}}(\hat{y}_i, y_i)}{\sum_i \max(|y_i|, 1)}, \tag{2}$$

where $|\cdot|$ counts codepoints. The CI resamples writers and recomputes the corpus ratio. A paired difference $\mathrm{CER}^R(a) - \mathrm{CER}^R(b)$ is computed on the same rows (same gold, same writer) and bootstrapped the same way (`wordgen/recog/metrics.py`). Results are reported for all words and for words of $\geq$3 codepoints. **Locator CER and R_eval CER differ in unit (letters vs codepoints) and in aggregation (capped per-word mean vs corpus ratio). They must never be compared with each other.**

4.3 Style: writer-identification MRR (M2, M2s)

E_eval and E_shape are ResNet-18 writer-embedding networks trained with SupCon plus cosine-softmax CE, on TRAIN writers only. **E_eval** sees the normalised ink matte (construction *and* pen). **E_shape** sees a spur-pruned skeleton redrawn at a fixed thin width (construction only). Every source goes through one preprocessing path that ends at the generators' 64×256 canvas (`wordgen/eval/writer_id.py`). For a query word q of writer w, each candidate writer v of the same dataset has a gallery vector g_v: the L2-normalised mean embedding of real crops disjoint from the refs and the query. The galleries are cross-fitted by index parity, and crops with the query's text are removed. With r_q the rank of w among candidates by $\cos(e_q, g_v)$:

$$\text{MRR} = \frac{1}{20} \sum_{w} \frac{1}{|Q_w|} \sum_{q \in Q_w} \frac{1}{r_q}, \qquad \text{margin}_q = \cos(e_q, g_w) - \max_{v \neq w} \cos(e_q, g_v). \tag{3}$$

Candidates are the VAL writers of the query's own dataset: 5 for BN-HTR (chance MRR 0.457) and 15 for BW (0.221). Pooled over 20 writers this gives chance top-1 0.10 and chance MRR 0.28. CIs come from a hierarchical bootstrap (writers within dataset, then queries within writer), paired across methods (`wordgen/eval/style_metric.py`).

Validity (`runs/style_eval/tables.md`). Real crops score top-1 0.723 [0.651, 0.786] and MRR 0.821 [0.775, 0.859]. Font renders sit near chance (NotoSans MRR 0.289, Atma 0.268). B0 is far above fonts (B0$-$Noto $+0.307$ [+0.156, +0.464]), so the metric sees writer-conditioned shape, not just "handwriting vs font".

4.4 Pen metric (sub-pixel width per core)

Per word (`wordgen/eval/pen.py`): $M = \{\text{ink} > \text{Otsu}\}$, core c from `measure_rows`, skeleton K, and

$$w_{\text{sub}} = \frac{\sum_{p:\, \text{ink}_p > 0.17} \text{ink}_p}{|K|}, \qquad \rho = \frac{w_{\text{sub}}}{c}, \qquad \text{pen}_w = \frac{\text{median}_{i \in \text{gen},w}\, \rho_i}{\text{median}_{i \in \text{real},w}\, \rho_i}. \tag{4}$$

Ink mass scales with s^2 and skeleton length and core with s, so ρ is almost resolution-invariant. The reported value is the median of pen_w over writers, with p10–p90 and a bootstrap CI of the median. This replaced eval_v2's integer EDT width. On a 64-px canvas that metric takes only the values $\{1, 1.83, 3, 3.47, \dots\}$ and reported 1.8–2.2$\times$ for every generator, where like-for-like measurement gives 1.13–1.36$\times$ (`runs/P1a2/REVIEW.md` §1).

4.5 Background metrics

bg-fail: a word fails when more than 60% of its ink bounding box is ink, or the box's 25th-percentile ink exceeds 0.1. The rate is a mean of per-writer rates. **Faint blotch**: a non-punctuation word with $>$2% of the canvas covered by faint ink (0.17–0.5) more than 2 px from strong ink. Writer bnhtr_117, a grey-paper scanner, is tracked separately.

4.6 Statistics for claims

All decisions are made on VAL. A claim of "A beats B" needs the paired CI of A$-$B to exclude 0. The paired bootstrap resamples the same writers (and, for style, the same query positions) for both arms. Each table also gives the count of writers favouring the first arm.

5 Page layout: v1 to v2

5.1 What is measured

Layout is learned from real ink geometry on hcr/996. A word detector (the project's `unified_cascade` engine: a Swin-T Mask R-CNN line detector and a word detector on line strips) gives 50-point word polygons. Only the word polygons are kept, because the line polygons merge on sloped pages. Lines are then rebuilt by chaining words with a robust quadratic fit. Ink is binarised (Otsu capped at grey 170, components $<$30 px removed) and assigned to polygons. All geometry is in a per-page writer unit **U** = the median word core height (80th minus 20th percentile of ink rows), about 3.5 mm and 31–54 px at 300 dpi. U was chosen because the pitch/U ratio is most stable across writers (CV 0.189).

Real layout statistics (train split; v1 report Table 4). The real-data statistics shape the model. Gaps have median 1.14 U and lag-1 Spearman 0.49 within a line. Gaps tighten along the line (-0.13) and their line mean is a writer habit (ICC 0.37). Slope is a strong writer habit (ICC 0.77) and persists to the next line ($r = 0.87$). Curvature is weak (ICC 0.17). Pitch has median 4.40 U (ICC 0.35). The residual (wobble) scale has median 0.098 U. So spacing and slope cannot be i.i.d., and the generator must condition on the page's history.

5.2 The model and its learned parameters

`LayoutModel` has seven small 1-D mixture-density heads ($k = 4$ components, hidden 64), trained by exact NLL with teacher forcing, full batch, Adam (lr 10^{-3}, wd 10^{-5}), patience 30. A page is generated line by line by the chain rule. For line k with page-history vector H_k (20 running statistics of previous lines):

$$p(\text{line}_k \mid H_k) = p(c_1 \mid H)\, p(c_2 \mid H, c_1)\, p(\log s \mid H, c_1, c_2)\, p(x_{\text{start}} \mid \cdot)\, p(\text{pitch} \mid \cdot) \times \prod_j p(\text{gap}_j \mid w_j, w_{j+1}, \text{gap}_{<j}, \cdot) \prod_j p(r_j \mid \log s, j/n). \quad (5)$$

The headline trajectory is $y = c_0 + c_1(x - x_m) + c_2(x - x_m)^2$, centred on the line's x midpoint, plus per-word residuals r_j ("wobble") with robust scale s. Curvature needs $\geq$5 words and slope $\geq$3. The model learns slope, curvature, residual scale, line start, pitch, gaps (autoregressive) and wobble. v2 adds fitted constants and placement rules. Table 5 lists all of them.

Table 5: Learned and fitted layout parameters of the final default (v2 `default_unfitw_dc`). All constants are fitted on the train split.

parameter	value	role
MDN heads	7 heads, $k = 4$, hidden 64; best epoch 152 (stopped 183)	slope, curv, logres, x_{start}, pitch, gap, wobble
training rows	slope 4,942; curv/logres 4,050; x_{start} 4,814; pitch 4,258; gap 25,546; wobble 26,878	per-head targets
val NLL (nats)	slope -2.363, curv -1.004, logres 1.071, x_{start} 0.965, pitch 1.152, gap 0.669, wobble -0.606	
combined word anchor	$w = 0.349$ headline + 0.651 median-ink-row	inverse-noise weighting
anchor noise	true wobble var 0.00906 U^2; noise var headline 0.01176, mid 0.00631, combined 0.00411	deconvolution of wobble
wobble de-noise factor	$\sqrt{1 - \text{share}_{\text{combined}}} \approx 0.83$ (share 0.29–0.33)	generation-time scale
unfit targets	per-word leverage inflation $1/\sqrt{1 - h_i}$, cap 4; line scale keeps the median-MAD table	restores physical wobble
curvature deconvolution	noise = 38.4% of curv target variance; MDN likelihood $\sum_k \pi_k \mathcal{N}(c; \mu_k, \sigma_k^2 + v_i)$	sampling is noise-free
line clearance	min distance $\geq$ 2.603 U, min/mean $\geq$ 0.797 (train 1st pct.); 30 redraws, else push down	stops line collisions
word rotation	$\beta = 0.524\times$ line tangent angle (Huber fit, $n =$ 26,653)	tilts words with the line
placement	ink-extent edges (`INK_LEVEL` 0.35); planner reset per page; margin guard ± 0.3 U	BS integration

Source: `scribe_layout_v2/runs/default_unfitw_dc/metrics.json`, `.../default_unfitw_dc/tilt.json`, `scribe_layout_v2/runs/anchor_noise.json`, `.../line_clearance.json`, `scribe_layout_v2/report/FACTS.md` items 1–11

5.3 Metrics: W1, KS and C2ST

Every generator receives each real TEST page's word widths and line breaks. Its plan is converted to raw coordinates and *re-measured by the same code* that measured the real lines. For each marginal quantity q (gap, slope, curvature, residual scale, drift $y_n - y_1$, pitch, x_{start}) we report the Wasserstein-1 distance and the KS statistic between generated and real distributions:

$$W_1(P, Q) = \int |F_P(t) - F_Q(t)|\, dt, \qquad \text{KS}(P, Q) = \sup_t |F_P(t) - F_Q(t)|. \quad (6)$$

The **classifier two-sample test (C2ST)** trains a classifier (logistic regression unless marked MLP) to tell real lines (or pages) from generated ones. It uses grouped 5-fold cross-validation, where each real page and its generated copy form

one group, and reports out-of-fold AUC. AUC $= 0.5$ means indistinguishable. Line features (lines with $n \geq 5$): mean and sd of gaps, gap lag-1, slope, curvature, residual scale, drift, gap–width correlation. Page features: mean and sd of line-mean gaps, mean and sd of slope, slope lag-1, mean curvature, mean residual scale, mean pitch.

Nulls. The "shuffled real" baseline (another page's geometry) is **not** a floor. It breaks the gap–width relation, which is a C2ST feature (0.018 vs real 0.130). The comparable null splits each real page into two labels inside one group: 0.496 ± 0.019. End to end, the floor is real TEST pages split in half: line 0.534, page 0.508.

5.4 Results in chronological order

v1 (`scribe_layout/`). Table 6 is v1's own ablation on its own harness. The learned model reduces gap W1 from 0.490 (naive) and 0.559 (manual) to 0.038. Vertical structure (slope, drift, pitch) comes from the vertical heads, spacing from the gap head; each alone leaves the other at manual's level. The weak point is x_{start} (W1 1.31 U), worse than shuffled real (0.43). Model selection chose $k = 4$ (summed val NLL 0.417, against 1.710 for $k = 1$; `scribe_layout/runs/model_ablation.json`).

Table 6: v1 ablation (v1 harness, TEST, 5 seeds, mean $\pm$ sd). W1 is in U.

method	gap W1	slope W1	drift W1	pitch W1	x_{start} W1	line C2ST
naive	0.490	0.038	1.217	0.849	1.811	1.000
manual	0.559	0.036	1.143	1.607	1.713	0.977
learned (v1)	$\mathbf{0.038 \pm 0.007}$	0.005	0.158 ± 0.030	0.092 ± 0.015	1.314 ± 0.021	0.619 ± 0.011
learned, spacing only	0.043	0.036	1.138	1.605	1.318	0.874
learned, vertical only	0.561	0.005	0.159	0.085	1.713	0.944
shuffled real	0.106 ± 0.034	0.006	0.187	0.194	0.434 ± 0.104	0.588

Source: `scribe_layout/runs/eval/ablation.md`

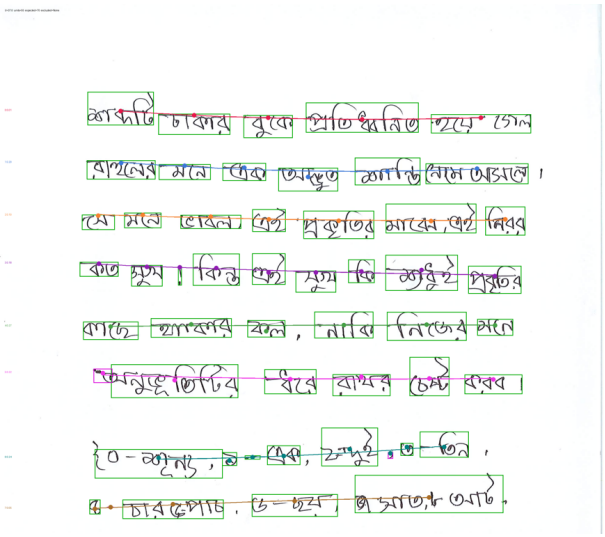


Figure 2: Extracted geometry on a real page (v1 report): word anchors, fitted line trajectories and residuals.
Source: `scribe_layout/report/fig/image_103_geom.png`

v2 harness and the move to "unfit" targets. v2 (`scribe_layout_v2/`) re-evaluates v1 on a stricter harness with more C2ST features: there, v1's line C2ST is 0.566 ± 0.021. Table 7 gives the sequence of v2 models. The first v2 models (`v2_history`, `v2_latent`) learned the *fitted* wobble. Fitting a quadratic to n points shrinks the residuals ("fit shrinkage"), and replaying them adds a second fit at measurement. Training on inflated ("unfit") targets lowered the line C2ST from 0.638 to 0.539 (single seed then 5 seeds).

A bug in the unfit targets, and the fix. A review (M2) found that the inflation table matched the *line* robust scale but was also applied to every per-word residual. This made per-word wobble overshoot: in a simulation with i.i.d. wobble of sd 0.150, the old table recovers 0.311 at $n = 5$, while per-word leverage correction recovers 0.142. The 2-D FFT

C2ST had already flagged this. Scaling the drawn wobble moved it from 0.683 ($\times 1.0$) to 0.578 ($\times 0.5$) (3 seeds). The leverage fix `unfitw` gave the best line C2ST, 0.521 ± 0.019.

Curvature deconvolution (the default). In end-to-end runs the real detector split 89 of 462 drawn v2 lines (19%), against 6 of 355 (1.7%) for manual. At the split points, adjacent words stepped by a median 1.2 U, more than the chainer's 1.0 U join threshold. The cause was lines that bow too much. Curvature is sampled before the word count is known, and fitted c_2 on 5–6-word lines is mostly noise (measured sd 0.00129, noise 0.00092, implied true 0.00091). An extreme-deconvolution likelihood removes this: sampled sag matches real ($|\text{sag}| > 0.5$ U on 9.3% of lines, real 10.8%, `unfitw` 17.5%), and end-to-end splitting halves to 52/469 (11.1%). Its C2ST is within noise of `unfitw`. `unfitw_dc` was chosen for realism, not for C2ST.

A rejected variant. A Huber-aware per-word correction (`unfitwh_dc`) is exact for i.i.d. wobble, but was worse on all three C2STs on both splits (6/6 comparisons). Real wobble is smoother than i.i.d. (lag-1 -0.38 vs -0.48), so the larger edge inflation overshoots.

Table 7: Plan-level C2ST AUC by model, TEST (193 pages), 5 seeds, mean $\pm$ sd. Resid-scale W1 is given by words per line.

model	line C2ST	page C2ST	2-D FFT C2ST	resid_scale W1, n=5 / 6–7 / 8–10 / 11+
manual (BengaliScribe)	0.980	1.0	0.99	–
v1	0.566 ± 0.021	0.654 ± 0.043	0.567 ± 0.034	.011 / .025 / .055 / .074
v2_history (1 seed)	0.638	0.752	–	–
v2_history_unfit (old table)	0.539 ± 0.016	0.660 ± 0.004	0.653 ± 0.024	.039 / .028 / .029 / .058
v2 unfitw (leverage)	$\mathbf{0.521 \pm 0.019}$	$\mathbf{0.581 \pm 0.026}$	0.566 ± 0.025	.009 / .011 / .041 / .062
v2 unfitw_dc (default)	0.535 ± 0.012	0.590 ± 0.024	0.550 ± 0.020	.007 / .011 / .042 / .061
v2 unfitwh_dc (Huber, rejected)	0.544 ± 0.012	0.614 ± 0.011	0.583 ± 0.029	.009 / .014 / .043 / .068
shuffled real (not a floor)	0.584 ± 0.016	0.605 ± 0.012	0.529 ± 0.050	.021 / .018 / .054 / .100
within-page null	0.496 ± 0.019	–	–	–

Source: `scribe_layout_v2/runs/compare_unfitw/compare_5seed.txt`, `compare_5seed_dc.txt`, `compare_5seed_wh_test.txt`; v2_history row: `scribe_layout_v2/runs/eval/metrics.json`; null and manual: `scribe_layout_v2/report/FACTS.md`

Selection-free VAL check. Every choice above was made on TEST, so the TEST numbers are optimistic. Table 8 repeats the comparison on VAL. The leverage-family models beat v1 on line and page C2ST on VAL too. The old-table `unfit` model, which looked better than v1 on TEST (0.539), is not better on VAL (0.566 vs 0.560): that earlier claim was a selection effect.

Table 8: Plan-level C2ST on the VAL split (4 texts), 5 seeds.

model	line C2ST	page C2ST	FFT C2ST	resid_scale W1 by n
v1	0.560 ± 0.024	0.631 ± 0.040	0.531 ± 0.033	.013 / .025 / .055 / .089
v2 unfit (old table)	0.566 ± 0.028	0.660 ± 0.022	0.599 ± 0.051	.049 / .030 / .024 / .050
v2 unfitw	0.527 ± 0.030	0.575 ± 0.039	0.528 ± 0.024	.012 / .011 / .035 / .072
v2 unfitw_dc (default)	0.536 ± 0.027	0.593 ± 0.031	0.535 ± 0.044	.011 / .011 / .035 / .069
v2 unfitwh_dc	0.551 ± 0.022	0.619 ± 0.043	0.551 ± 0.049	–
shuffled real	0.577 ± 0.023	0.597 ± 0.030	0.480 ± 0.034	.016 / .017 / .055 / .089

Source: `scribe_layout_v2/runs/compare_unfitw/compare_5seed_val.txt`, `compare_5seed_wh_val.txt`

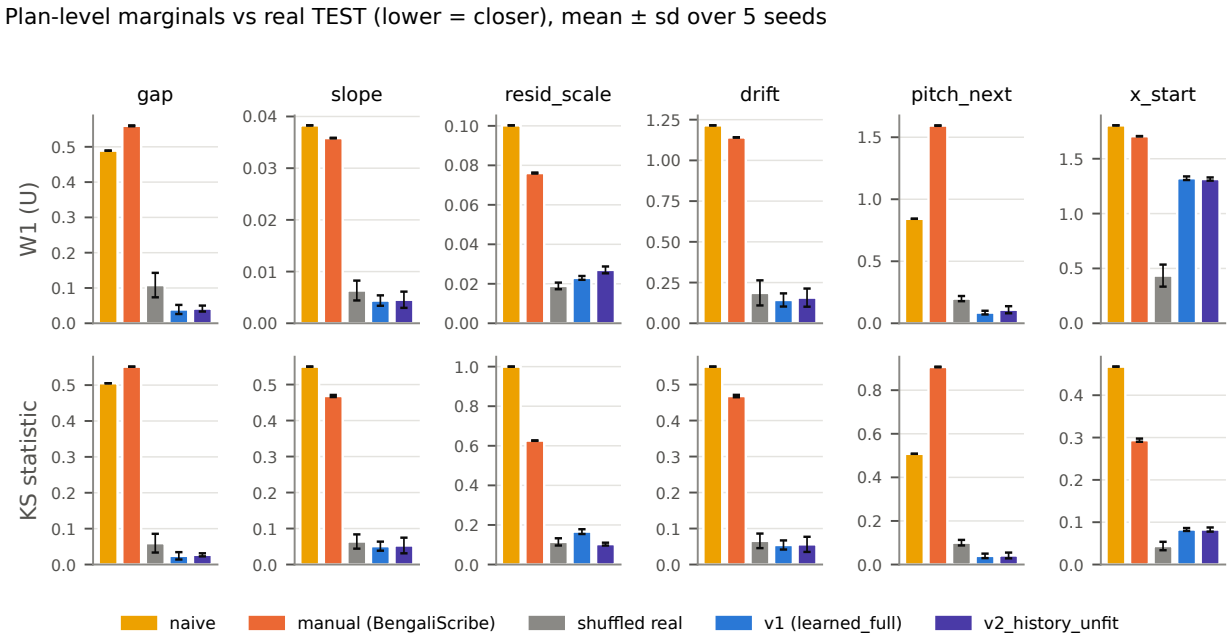


Figure 3: Plan-level marginals against real TEST on the v2 harness: W1 (top) and KS (bottom). Learned layouts (blue, violet) match gaps, slope, drift and pitch far better than naive or manual layout. x_{start} remains the weakest learned quantity; shuffled real does better there.

Source: `scribe_layout_v2/runs/eval/metrics.json` (via `make_figs.py`)

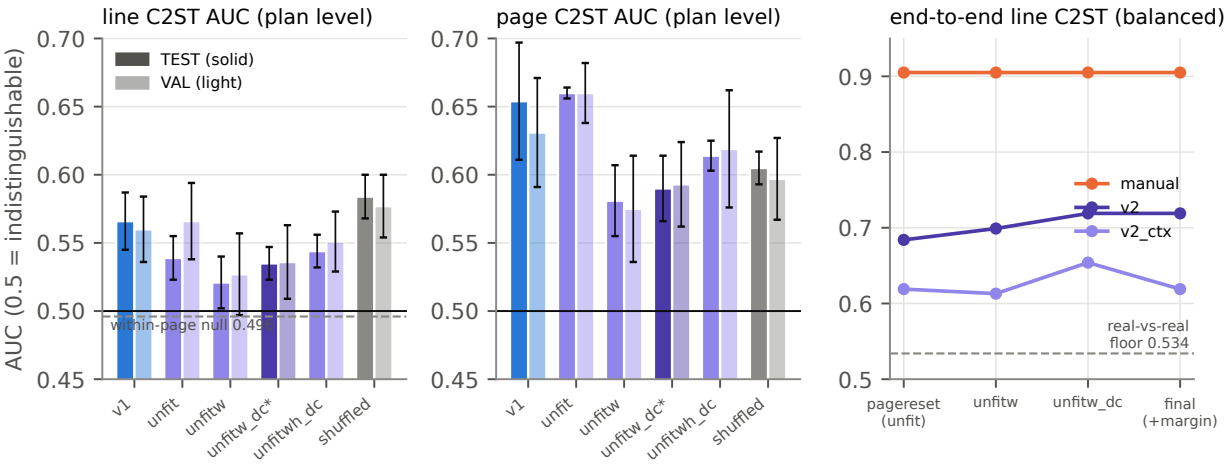


Figure 4: C2ST AUC per model. Left and centre: plan level, TEST (solid) and VAL (light), 5 seeds, ± sd. "*" marks the default. Right: end-to-end line C2ST, balanced protocol only. The two earliest end-to-end runs used a biased C2ST and are excluded (see text).

Source: `scribe_layout_v2/runs/compare_unfitw/compare_5seed*.txt`, `scribe_layout_v2/runs/e2e_*/results.md`

End-to-end evaluation. BengaliScribe renders pages for 5 reference writers × 2 seeds. The real detector and the same extraction then measure them, and they are compared with the 193 real TEST pages. Variants: `manual`, `v2` and `v2_ctx` (conditioned on the writer's own real geometry). **The first two runs (`e2e_unfit`, `e2e_unfit_ink`) used a biased C2ST.** Pooled out-of-fold AUC with one label per page group put the real-vs-real "floor" at 0.34 (line) and 0.115 (page). Their C2ST values are not comparable with later runs; only their W1 columns are shown. From `e2e_pagereset` onward the C2ST is balanced (folds dealt per label; mean of per-fold AUCs), with floor 0.534 / 0.508.

Table 9: End-to-end results (W1 in U; AUC). ‡Biased C2ST: not comparable. Recovered = extracted words / drawn words.

run (model)	method	gap	slope	drift	pitch	x_{start}	line / page C2ST	recovered
e2e_unfit (canvas-width)	v2	0.602	0.015	0.531	0.716	3.837	0.750 / 0.901‡	0.969
	v2_ctx	0.381	0.013	0.427	0.924	4.416	0.595 / 0.901‡	0.910
e2e_unfit_ink (ink extents)	v2	0.471	0.016	0.503	0.712	4.037	0.667 / 0.840‡	0.952
	v2_ctx	0.207	0.010	0.343	1.001	4.438	0.532 / 0.918‡	0.866
all balanced runs	manual	0.685	0.035	0.965	1.839	0.948	0.905 / 0.999	1.004
e2e_pagereset (unfit)	v2	0.404	0.013	0.396	0.670	4.065	0.684 / 0.834	0.965
	v2_ctx	0.267	0.017	0.558	0.770	3.783	0.619 / 0.889	0.926
e2e_unfitw	v2	0.376	0.012	0.343	0.346	2.278	0.699 / 0.851	0.960
	v2_ctx	0.197	0.019	0.610	0.480	2.703	0.613 / 0.842	0.933
e2e_unfitw_dc	v2	0.387	0.021	0.645	0.465	1.453	0.719 / 0.844	0.983
	v2_ctx	0.183	0.023	0.695	0.535	2.494	0.654 / 0.847	0.929
e2e_final (default + margin guard)	v2	0.425	0.021	0.672	0.390	1.239	0.719 / 0.850	0.981
	v2_ctx	0.198	0.020	0.621	0.503	2.130	0.619 / 0.845	0.940

Source: `scribe_layout_v2/runs/e2e_{unfit,unfit_ink,pagereset,unfitw,unfitw_dc,final}/results.md`

Reading the end-to-end numbers:

- **The planner reset was a real fix.** The history used to run across pages, reaching 56–72 lines against at most 16 in training. Slopes random-walked to the -0.2 clamp and words were drawn past the page bottom. After the reset, page slope means stay near the writer's (e.g. image_104 with context: $-0.053 \cdots -0.009$ against real -0.023).
- **Slope and drift W1 are not comparable across runs with different split rates** (survivor bias). Split lines are the steep and bowed ones. In `e2e_unfitw`, kept lines had median |slope| 0.026 against 0.046 for split lines. All drawn `dc` lines have median |slope| 0.027 (q90 0.071), against real 0.029 (0.086).
- **Placement bias.** Ink extents use `INK_LEVEL` 0.35 while page Otsu is ≈0.39, which leaves a $+0.067$ U bias in every measured gap.
- **Line splitting.** With 57 of 453 drawn lines split (12.6%) in `e2e_final`, end-to-end metrics see only the surviving lines, and the large x_{start} W1 mostly comes from fragments that start mid-page.

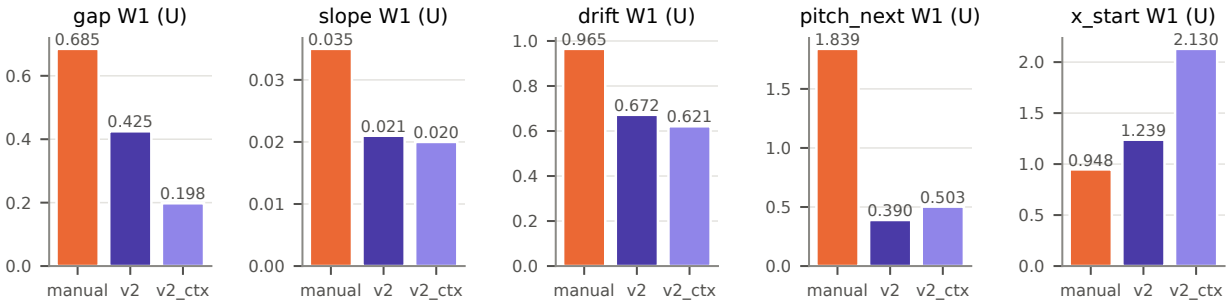


Figure 5: End-to-end W1 to real TEST for the final run (`e2e_final`).
Source: `scribe_layout_v2/runs/e2e_final/results.md`

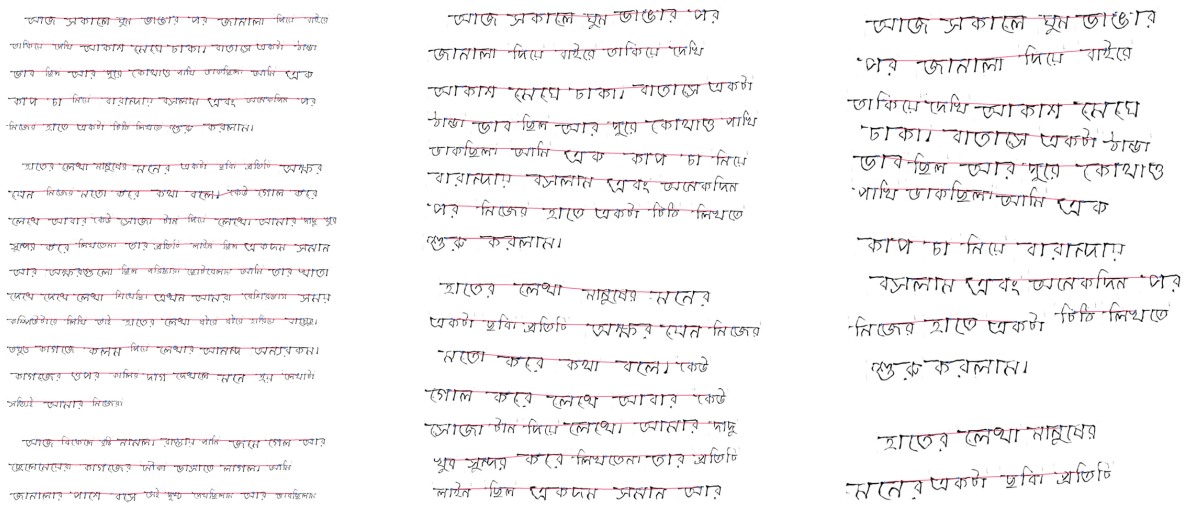


Figure 6: `e2e_final` pages for reference writer image_104 (BengaliScribe B0 words): manual layout (left), v2 (centre), v2 with writer context (right).

Source: `scribe_layout_v2/runs/e2e_final/thumbs/image_104_{manual,v2,v2_ctx}.png`

Final default model. `scribe_layout_v2/runs/default_unfitw_dc/model.pt` is the default of `bscribe_run.py` and `relayout.py`. It runs with the margin guard: a line whose ink would cross the bottom margin starts a new page and is replanned from scratch. A test asserts that ink stays within the margins ± 0.5 U; it fails on the old code. The layout test suite grew from 32 tests (≈8 min) to 70 (≈17 s). The latent writer-style variant is **not** the default: a context bug fix (causal-prefix z) improved the context case but regressed the no-context model (page C2ST 0.684 → 0.829), and the issue is unresolved.

5.5 Spectral side-studies, with corrections

2-D FFT layout-realism metric (`scribe_layout_fft2d/`). A layout map is the filled word boxes at 8 px/U, Hann-windowed. Its 2-D power spectrum is summarised as radial PSD (32 bins), angular energy (18 bins) and the vertical line-pitch peak. **Only the controlled (text-matched) comparison is valid.** The uncontrolled version is confounded: single non-layout features alone separate real from generated (text-block height AUC 0.985, median word width 0.90–0.93), so its AUCs of 0.96–1.0 measured text, page fill and glyphs, not layout. In the controlled test every generator runs on each real page's own word widths, both sides use the same fixed box, and confound AUCs are 0.5 by construction.

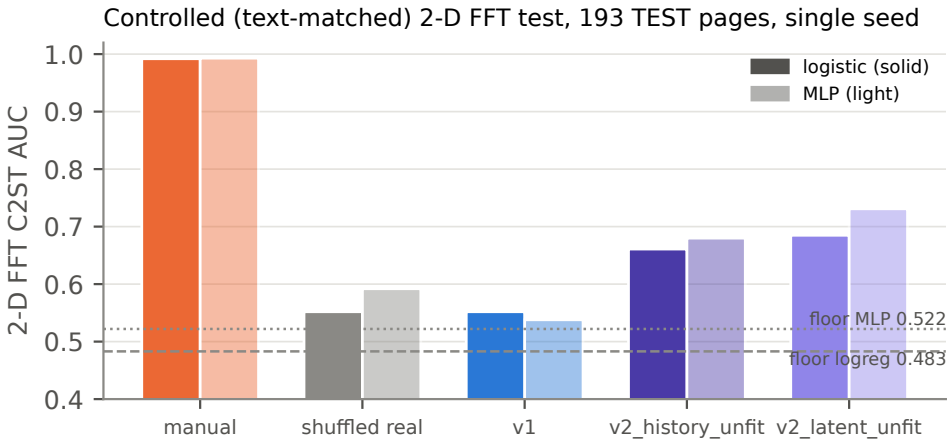


Figure 7: Controlled 2-D FFT C2ST, single seed. Single-seed AUCs move by up to ± 0.09; the 5-seed FFT C2STs are in Table 7. Pitch ablations on real pages show that the spectrum is dominated by line-pitch regularity. Shuffling pitch moves the mean log-PSD 6× more (L2 0.080) than zeroing slope (0.013) or removing wobble (0.013).

Source: `scribe_layout_fft2d/runs/results.md` (controlled section only), `scribe_layout_v2/report/FACTS.md`

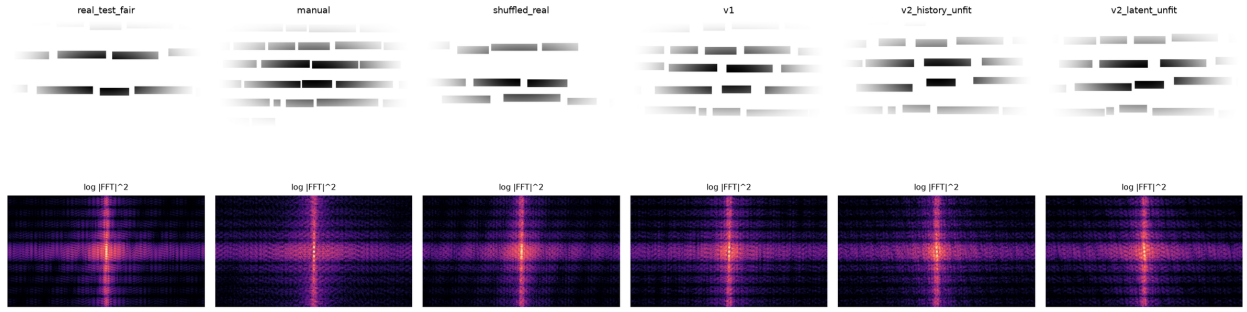


Figure 8: Controlled layout maps (fixed word boxes) used by the 2-D FFT metric: real page and generated layouts on the same word widths.

Source: `scribe_layout_v2/report/figs/example_layout_maps_fair.png`

1-D Fourier model of the headline (`scribe_layout_fourier/`). Dense headline traces of real lines follow a power law, PSD $\propto f^{-\alpha}$ with $\alpha = 1.06$ ($R^2 = 0.99$, 5,099 train lines), with no word-period peak. A spectral-synthesis generator samples Gaussian-process headlines with this spectrum. Its `runs/results.md` states that Fourier is "17× better" than v1 (log-spectral RMSE 3.98 vs 0.23 nats). **That claim is wrong.** The 0.23 belongs to a continuous trace that no layout of rigid words can draw. v1's 3.98 is a floor artefact, because the comparison ran on traces that a layout cannot realise. The corrected, fair comparison uses layout-realisable traces (rigid, rotated words) and the band with wavelength $\geq$4 U. The distance to real is then: manual 3.786, **v1 0.702, Fourier 0.668**, continuous Fourier (not realisable) 0.244. **Fourier and v1 are essentially equal.** The "6× better x_{start}" claim from the same study came from including indents.

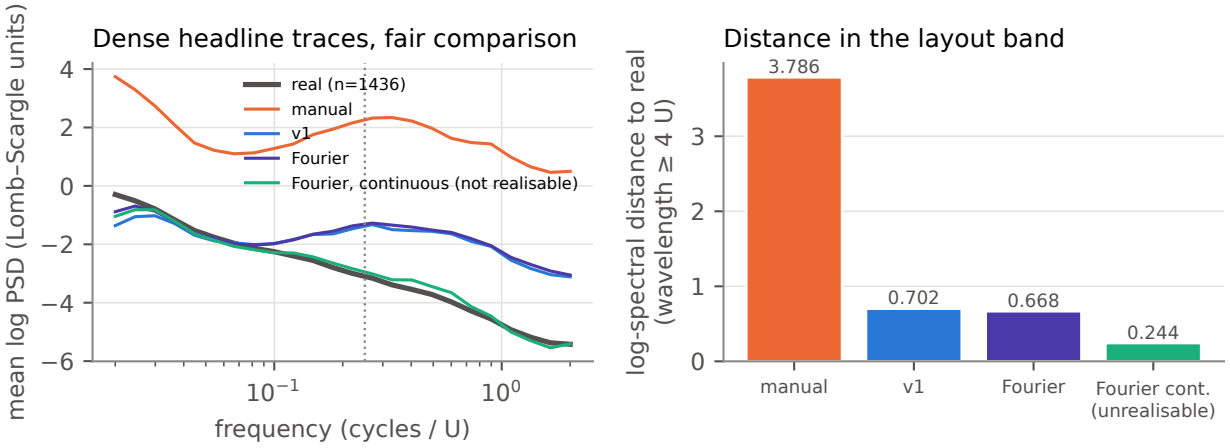


Figure 9: Corrected Fourier comparison. Left: mean log PSD of dense headline traces (dotted line: the 4 U wavelength limit). Right: log-spectral distance to real in the layout band.

Source: `scribe_layout_fourier/runs/fair_dense_psd.json`; correction in `scribe_layout_v2/report/FACTS.md`

6 Word generation

6.1 From diagnosis to design

The diagnosis (§2) implies that a higher-resolution B0 would still copy the font: construction has to be learned from the references by a model that is not tied to a font raster. The primary approach is a **few-shot latent diffusion word generator** built from DiffusionPen: an SD-1.5 VAE, a U-Net with CANINE text cross-attention, and a style vector from 5 references. Candidates are re-adapted from the IAM checkpoint on TRAIN writers only. `dp_words_b64`, which saw every BW writer, is used only as a probe and a baseline (B3). Table 10 lists the arms.

Table 10: Generator arms.

arm	definition
B0	BengaliScribe font-prior GAN (NotoSans prior), `models/v2`
B3	`dp_words_b64`: DiffusionPen fine-tuned on all 260 BW writers (fair only on BN-HTR)
A, B (P1a)	DiffusionPen from IAM on TRAIN, 40 crops/writer cap, 20k steps; B adds an 8-ch painted grapheme content map
A′, B′ (P1a′)	full TRAIN (98,856 crops, 354 writers), white-paper normalisation, 40k steps; B′ adds grapheme tokens + column-position encodings in cross-attention
B′v2	B′ sampled with `fit_canvas_white2` (bgnorm2) references, no retraining
P	B′ + pen knob (scalar embedding of the refs' log width/core) + learned null style + rebalanced sampling; 16k steps
S	P + 16 reference tokens per ref (Bengali SupCon ResNet-18) in decoupled (IP-Adapter-style) cross-attention; 24k steps
_tXcY / _cal	text guidance X and content guidance Y; _cal = page-scale pen calibration (shrink mode)

6.2 probe_day1: a visible first probe

`dp_words_b64` was sampled on image_2's text with three reference sets, and fine-tuned for 120 steps on image_2's confident crops. The result was qualitatively the most handwriting-like texture yet: rounder and more flowing than B0 (Fig. 10). But it was not thinner than B0, and it misspelled conjuncts and long words (Table 11).

Table 11: probe_day1 (24 image_2 words; locator CER, mean). Real CER is circular: the "confident" crops were selected by the same locator. Indicative only.

method	n	width/core (median)	darkness (median)	locator CER
real (confident crops)	9	0.064	1.00	0.00 (circular)
B0	24	0.063	0.97	0.184
DP, image_2 refs	24	0.091	1.00	0.973
DP, writer_99 refs	24	0.110	1.00	0.931
DP, writer_117 refs	24	0.104	1.00	0.899
DP, fine-tuned on image_2	24	0.100	1.00	0.651

Source: `scribe_wordgen/runs/probe_day1/REPORT.md` ("Numbers")

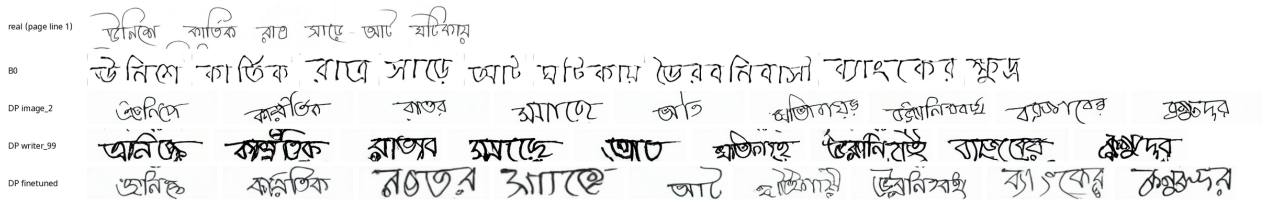


Figure 10: probe_day1: the first line of image_2, rendered by different methods (real, B0 and DiffusionPen variants).
Source: `scribe_wordgen/runs/probe_day1/line.png`

6.3 P1a: the painted content map is inert (NO-GO)

P1a trained DiffusionPen from IAM on TRAIN writers (40 crops/writer: 14,155 crops, 5,033 unique texts), 20k steps at batch 32. Arm A is the text-only control; arm B adds an 8-channel grapheme content map with a zero-initialised first-conv slice. The review first corrected the metric. Raw CER > 1 and width/core ≈ 4 were artefacts of uncapped CER on punctuation, grey-box output on grey-paper writer 117, and truncation of B0 at 256 px. Table 12 gives the corrected numbers.

Table 12: P1a, corrected (VAL, 20 writers; CER capped at 1, left-aligned, punctuation excluded).

method	CER	BN-HTR excl. 117	BW	bg-fail %	sub-px width / real
real	0.133 [0.101, 0.169]	0.049	0.237	1.2	1.00
B0	0.130 [0.105, 0.157]	0.084	0.224	0.2	≈1.3
B3	0.422 [0.384, 0.460]	0.427	0.476 (optimistic)	3.8	1.32
A	0.482 [0.440, 0.528]	0.423	0.525	8.7	1.40
B	0.457 [0.417, 0.501]	0.361	0.516	8.8	1.40

Paired B−A $= -0.024$ [-0.045, -0.002] (14/20 writers), entirely in-vocabulary (OOV: A 0.568, B 0.574).
Source: `scribe_wordgen/runs/P1a/REVIEW.md` §1

Verdict: NO-GO for the content map. When another word's map is painted in, the model still writes the CANINE word: CER against the painted word is 0.959. The map alone gives near-unconditional output (0.953 vs 0.973). The grapheme embeddings barely moved during training (row norm 0.078 vs init ≈0.057). One content patch contributes about 2% of the first-conv input signal. The review traced the other failures to grey reference paper (6.4% of BN-HTR TRAIN crops have paper < 250), a starved data set, and a CFG-driven thick pen (arm A: 1.31× at g1, 1.48× at g2/g3).

6.4 P1a′: full data, white paper, grapheme tokens and the CFG-null bug

P1a′ applied the review's fixes. It trained on all of TRAIN (98,856 crops) for 40k steps with white-paper normalisation, and redesigned content as *grapheme-token cross-attention*: learned grapheme embeddings plus column-centre and span encodings, appended to the CANINE context. The report concluded that B′ was a NO-GO (CER 0.550, worse than A′). The review found the cause.

The CFG-null bug. `eval_v2.sample_arm2` put the content map and tokens into *both* classifier-free guidance branches. Guidance therefore extrapolated only the text direction, and content was never guided. The correct null drops text, map and tokens together, matching the independent 10% drops used in training. With the joint null (same 781 words, refs and seeds; Table 13), B′ becomes the best diffusion arm. The shuffled-content gate passes: the model writes the *painted* word (CER 0.432 against it at g3). Without any text, content alone writes the word at 0.199. The content-path ablation shows that the tokens carry identity (tokens alone 0.338), while the painted map is still inert (map alone 0.902).

Table 13: P1a′ on full VAL with the fixed sampler (eval_v2 scoring; CER capped at 1, punctuation excluded).

method	CER all	in-vocab	OOV	BN-HTR	BW	bg-fail	pen (sub-px/core)
real	0.133	0.129	0.150	0.055	0.159	1.2%	1.00
B0	0.130	0.116	0.235	0.085	0.145	0.2%	1.15
B3	0.421	0.400	0.508	0.410	0.425	3.8%	1.13
A′ g2 (as reported)	0.334	0.312	0.496	0.237	0.367	5.4%	1.36
A′ g3	0.245 [0.205, 0.287]	0.228	0.384	0.160	0.273	5.2%	1.22
B′ as reported (buggy CFG)	0.550	0.537	0.655	0.405	0.599	0.9%	–
B′ fixed g2	0.262 [0.218, 0.304]	0.239	0.385	0.153	0.298	1.3%	1.27
B′ fixed g3	**0.188** [0.150, 0.226]	0.169	0.301	0.108	0.214	1.8%	1.29
B′ content only, c4	0.199 [0.161, 0.238]	0.186	0.303	0.112	0.229	1.7%	–

Paired B′g3−A′g3 $= -0.057$ [-0.089, -0.032] (18/20 writers). A′ vs P1a A: -0.147 [-0.204, -0.103] (20/20).
Source: `scribe_wordgen/runs/P1a2/REVIEW.md` §1–2

A′ and B′ draw about 1.3× the writer's pen relative to letter size. That is thicker than B0/B3 (about 1.15), probably because P1a′ training is 84% BN-HTR, whose pens are thick. Writer 117's faint grey blotches (45–48% of its words) were not caught by the bg-fail metric, which motivated the faint-blotch score.

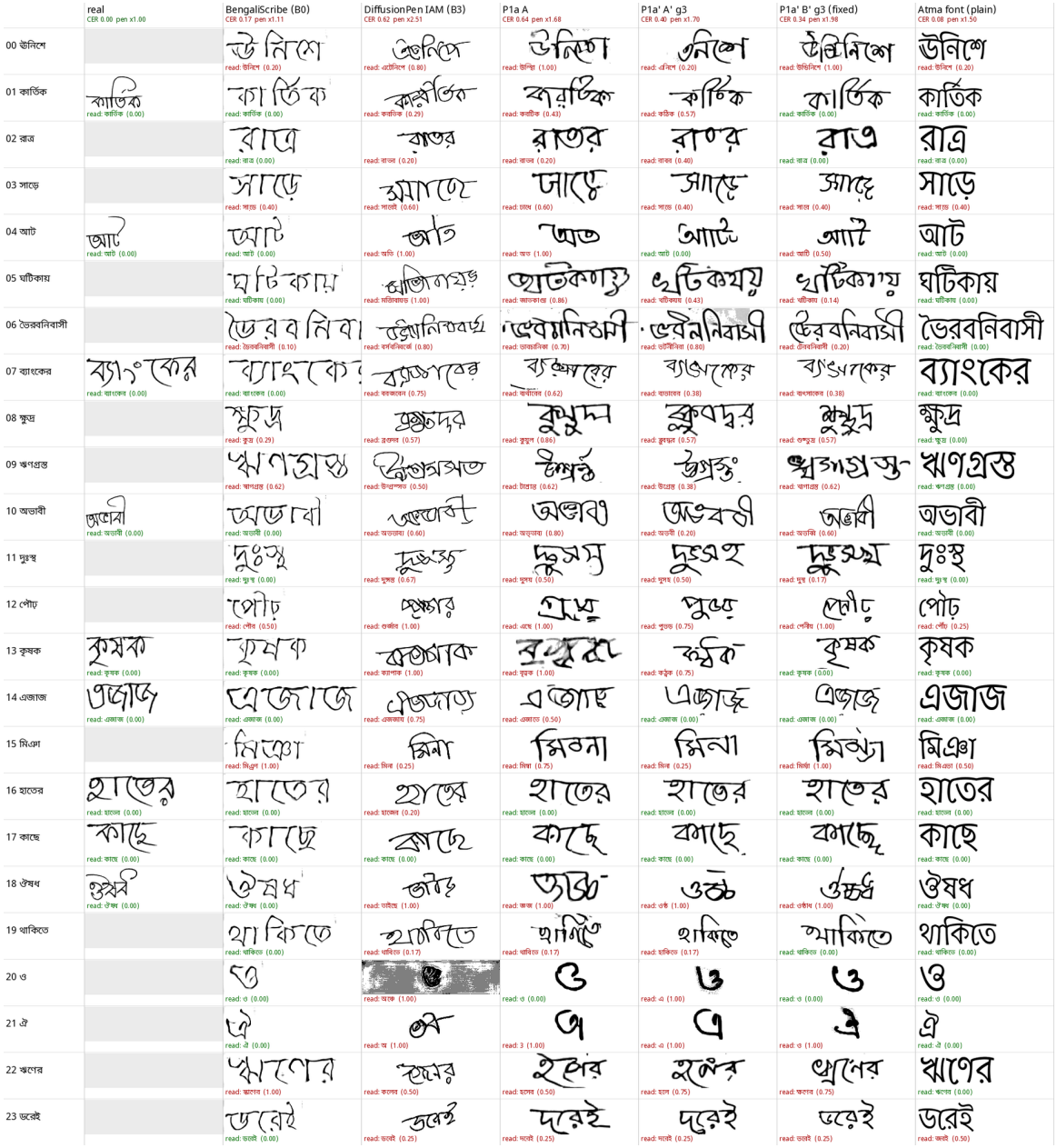


Figure 11: Side-by-side on image_2 (24 words, seed 0; the real crop exists for 9 words). Columns: real, B0, B3, P1a A, P1a′ A′ g3, B′ g3 (fixed CFG), plain Atma font. The locator's reading and CER are printed under each cell. image_2 CER over 3 seeds: B0 0.17, B3 0.62, A 0.64, A′ 0.40, B′ 0.34, Atma 0.08; pen vs writer 1.11 / 2.51 / 1.68 / 1.70 / 1.98 / 1.50.

Source: `scribe_wordgen/runs/compare_inference/image2_side_by_side.png`, `image2_scores.txt`, `README.md`

6.5 eval_v3: re-baseline and guidance sweep

eval_v3 is the corrected evaluation. It uses the fixed joint CFG null and separate text (t) and content (c) guidance scales. The decomposition is $\epsilon = \epsilon_{00} + c\,(\epsilon_{0c} - \epsilon_{00}) + t\,(\epsilon_{tc} - \epsilon_{0c})$, where the subscripts mark which conditioning (text, content) is present. eval_v3 also adds $\geq$2 seeds per word, the sub-pixel pen metric, a non-punctuation background gate, the faint-blotch score, and OOV CER on all 20 writers. Seed 0 reproduces the review exactly: B′ t3c3 gives 0.188 with 781/781 pixel-identical renders (`runs/eval_v3/validate.txt`). Raising guidance lowers CER monotonically to t5c5, which ties B0 (Table 14, Fig. 12). Pen rises slightly with guidance, and the writer-117 blotch rate falls from 34% (t2c2) to 11% (t5c5).

Table 14: eval_v3 guidance sweep for B′ (VAL, 2 seeds; locator CER).

method	CER	in-vocab	OOV (18 wr.)	BN-HTR	BW	pen [CI]	blotch 117
real	0.133 [0.100, 0.167]	0.129	0.150	0.055	0.159	1.00	8%
B0	0.130 [0.103, 0.156]	0.116	0.235	0.085	0.145	1.15 [0.98, 1.20]	0%
A′ g3	0.254 [0.214, 0.295]	0.234	0.379	0.161	0.285	1.22	31%
B′ t2c2	0.274 [0.229, 0.317]	0.255	0.393	0.143	0.318	1.26	34%
B′ t3c3	0.194 [0.156, 0.230]	0.174	0.307	0.098	0.225	1.26	21%
B′ t4c4	0.165 [0.128, 0.202]	0.151	0.254	0.077	0.194	1.27	14%
B′ t5c5	0.148 [0.115, 0.182]	0.133	0.241	0.068	0.175	1.30 [1.15, 1.45]	11%
B′ t6c6	0.143 [0.110, 0.178]	0.126	0.247	0.064	0.170	1.31	11%
B′ t0c4 (content only)	0.192	0.175	0.303	0.098	0.223	1.27	18%

Paired: B′t5c5−B0 +0.018 [-0.008, +0.044] (tie); B′t6c6−B′t5c5 −0.005 [-0.011, +0.000]; OOV B′t5c5−A′g3 −0.138 [-0.218, -0.066] (14/18 writers).

Source: `scribe_wordgen/runs/eval_v3/report_tables.txt`

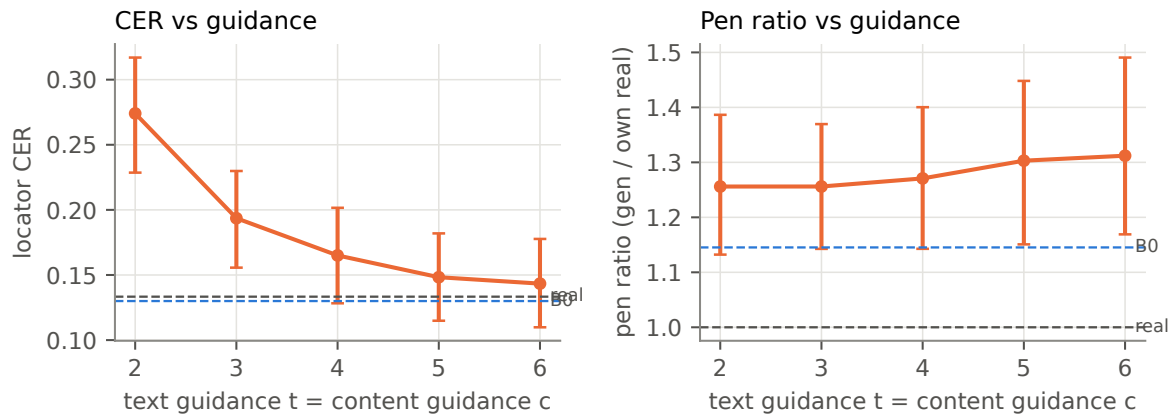


Figure 12: Guidance sweep for B′ with $t = c$. Left: locator CER with writer-bootstrap CI (dashed: real and B0). Right: pen ratio. More guidance buys legibility, and pen thickens a little.

Source: `scribe_wordgen/runs/eval_v3/metrics.json`

6.6 The style evaluator (M2 / M2s)

Before E_eval existed, style was judged with the generator's own frozen IAM encoder (real-vs-real MRR 0.41), which is not independent. E_eval and E_shape (§4) were built and checked before any style claim was made (Table 15). The first result was sobering: **every diffusion arm up to B′ sat at MRR 0.40–0.45**, far below B0 (0.596) though above fonts.

Table 15: Style evaluator validity and early arms (M2, E_eval, within-dataset galleries, 20 writers / 726 queries).

method	top-1	MRR	margin
chance	0.100	0.280	–
real	0.723 [0.651, 0.786]	0.821 [0.775, 0.859]	0.094
real via `crop_ink`	0.707	0.810	0.091
B0	0.442 [0.310, 0.591]	0.596 [0.488, 0.711]	−0.010
P1a A / P1a B	0.214 / 0.195	0.420 / 0.404	−0.133 / −0.130
A′ g3 / B′ g3	0.244 / 0.238	0.437 / 0.431	−0.114 / −0.124
NotoSans font / Atma font	0.107 / 0.092	0.289 / 0.268	−0.191 / −0.247

Paired: real−B0 +0.225 [+0.116, +0.335]; B′g3−B0 −0.165 [-0.287, -0.047]; B′g3−Noto +0.142 [+0.033, +0.247]. BN-HTR only: real 0.916, B0 0.883 (real−B0 n.s.).

Source: `scribe_wordgen/runs/style_eval/tables.md` ("eval within all", "eval within bnhtr")



Figure 13: Style-evaluator inputs after the shared preprocessing. Columns: a conditioning reference, the real target, B0, B′ g3, A′ g3, NotoSans; two variants per row.

Source: `scribe_wordgen/runs/style_eval/sheet_prep.png`

6.7 Pen calibration (post hoc, no retraining)

Generation preserves the writer's pen *rank*: Spearman over writers is about 0.9 in pixels. So a per-writer gain is well posed. `wordgen/pen/calibrate.py` re-renders strokes at page scale toward the refs' measured width. Table 16 compares the modes on B′ t5c5. All modes bring the pen into 1.0 ± 0.1. The chosen shrink mode (1.30→1.05) misses the CER gate (upper CI of the CER change $\leq +0.01$) by 0.0003. Only the per-writer mode passes both gates, but its p90 over writers is 2.20.

Table 16: Pen calibration modes on B′ t5c5 (VAL; pen median over writers, CI of median).

mode	pen ratio [CI]	p10–p90	CER after−before	gates (pen / CER)
before	1.30 [1.15, 1.45]	1.06–1.85	–	–
writer	1.09 [0.99, 1.29]	0.87–2.20	+0.0027 [-0.0030, +0.0083]	pass / pass
writer_v2	1.06 [0.92, 1.21]	0.80–1.51	+0.0049 [-0.0007, +0.0101]	pass / fail
global_v2	1.07 [1.00, 1.26]	0.85–1.52	+0.0046 [-0.0011, +0.0103]	pass / fail
word_v2	1.05 [0.92, 1.18]	0.78–1.53	+0.0053 [-0.0011, +0.0120]	pass / fail
shrink_v2 (adopted)	1.05 [0.97, 1.18]	0.83–1.37	+0.0051 [-0.0000, +0.0103]	pass / fail by 0.0003

Source: `scribe_wordgen/runs/pen_calib/report_Bp_t5c5_pencal_*.txt`

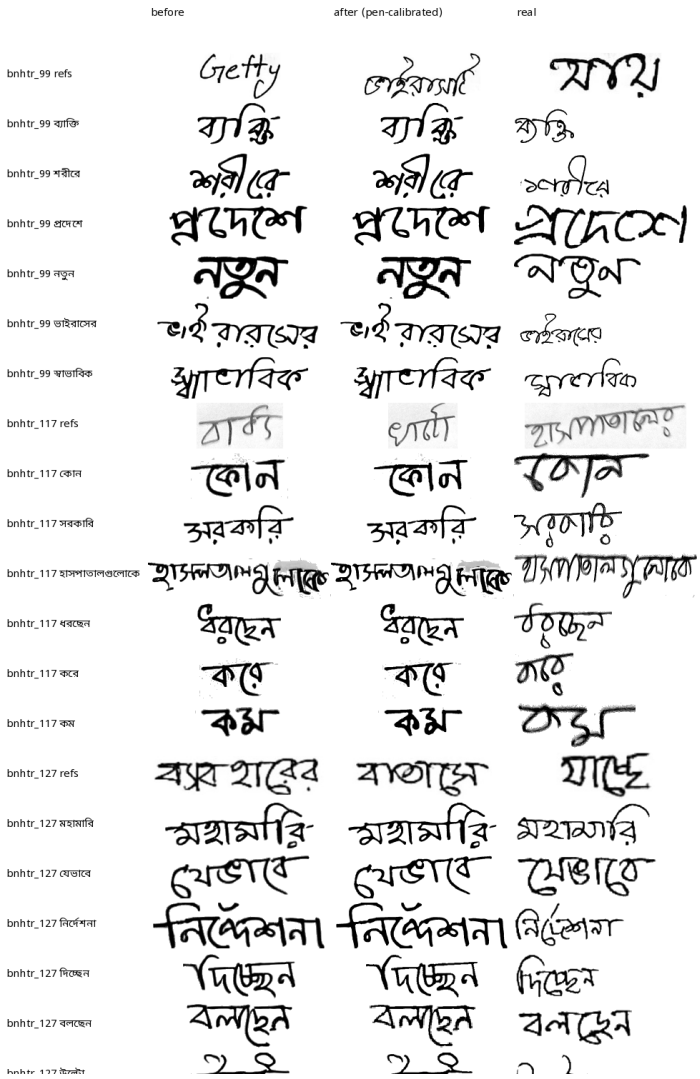


Figure 14: Pen calibration, shrink mode: B′ t5c5 before (left), after (centre) and the writer's real crop (right).
Source: `scribe_wordgen/runs/pen_calib/sheet_Bp_t5c5_pencal_shrink_v2.png` (top part)

6.8 bgnorm2: background normalisation v2

The single-scalar whitening (90th-percentile paper → 255) left writer 117's references with a grey, graded box, which the model reproduced as texture. `fit_canvas_white2` works on luminance only. It divides by a large-kernel background estimate and soft-clips the paper to exactly 255, keeping ink darkness linear. On TRAIN crops it changes ink mass by a median ratio of 0.998. It needs no retraining; the references must be passed through it (Table 17).

Table 17: bgnorm2 on B′ g3 (references only, no retraining).

method	CER	bg-fail (non-punct)	faint blotch (mean)	blotch, writer 117
real	0.133 [0.100, 0.167]	0.0%	0.4%	7.5%
B′ g3, v1 refs	0.188 [0.150, 0.226]	0.8%	3.7%	30.0%
B′ g3, bgnorm2 refs	0.183 [0.142, 0.226]	0.3%	1.8%	0.0%

Paired CER (v2−v1 refs) −0.0042 [-0.0170, +0.0079] (11/20 writers). Writer 117 alone: CER 0.080→0.116, blotch 30%→0%.
Source: `scribe_wordgen/runs/bgnorm2/bprime_noretrain.txt`, `measure.txt`

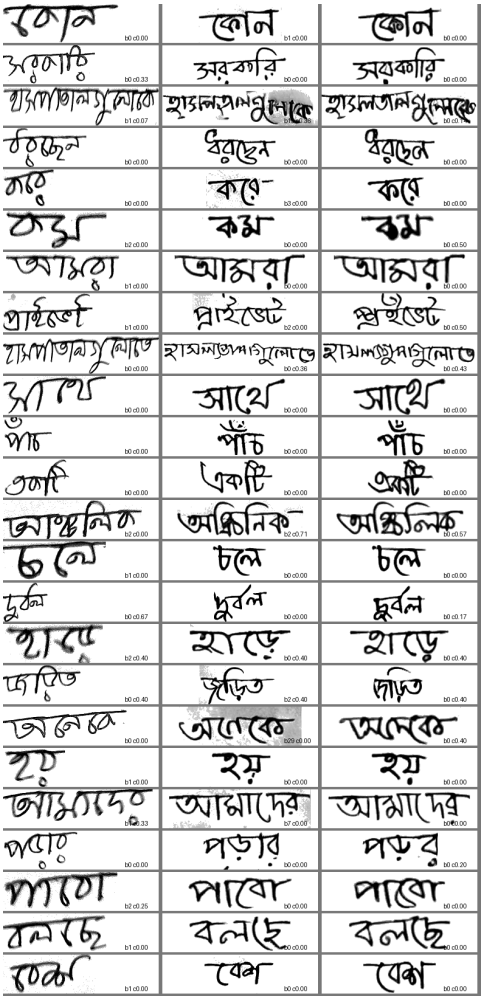


Figure 15: Writer bnhtr_117 (grey-paper scans). Columns: real crop, B′ with v1 refs, B′ with bgnorm2 refs. Each cell is labelled with blotch (b) and CER (c).

Source: `scribe_wordgen/runs/bgnorm2/sheet_gen_bnhtr_117.png` (builder `scratch/bgnorm2_gen_sheet.py`)

6.9 Arm P and Arm S

Both arms warm-start from the B′ checkpoint and train on 100,868 TRAIN rows encoded with bgnorm2. Training uses batch 32, lr 10^{-4} (10× for new modules), 300 warm-up steps and a final lr fraction of 0.1. Dropouts are text, content and style 0.1 and pen 0.05, and the CFG null drops all of them together. Sampling is rebalanced to punctuation 1.5%, BW 38% and BN-HTR 60.5%, against a natural mix of 2.0 / 15.4 / 82.6%. Writers within a dataset are weighted by clip(median pen / writer pen, 0.5, 2.0), which up-weights thin pens. The two arms differ as follows (`runs/P1a3/{P, S}/config.json`):

- **Pen knob** (both). The input is the refs' median $\log(w_{\text{sub}}/c)$, z-scored with TRAIN statistics ($\mu = -1.835$, $\sigma = 0.342$) and embedded sinusoidally. A zero-initialised MLP adds it to the timestep embedding, so the knob is a no-op at initialisation.
- **Null-style fix** (both). In B′, dropped style fell back to the TRAIN writer-ID label embedding. P and S instead use a learned null style vector.
- **Reference tokens** (S only). A Bengali writer encoder (ResNet-18, SupCon on TRAIN) turns each ref into 16 spatial tokens. With slot embeddings, 5 refs give 80 tokens. They enter every transformer block through decoupled attention, out $=$ attn$(q,$ text+content$)$ $+$ keep $\cdot\, W_{\text{ref}}$ attn$(q, K_{\text{ref}}, V_{\text{ref}})$. W_{ref} starts at zero, which keeps the no-op property. Appending the tokens to the shared context would not: 80 extra keys change the softmax normalisation even with zero weights.

P trained for 16k steps (4,012 s) and S for 24k steps (2,109 s; final EMA loss 0.0581).

Two diverged S attempts and the gradient bug. The first two S runs diverged. Run 1's rolling loss jumped from ≈0.065 to above 0.14 within ∼800 steps. Run 2 drifted up from ≈0.064 to 0.108 by step 1,050 and was stopped (Fig. 16). The cause was a gradient bug. DiffusionPen's `CheckpointFunction` re-runs each transformer block during backward. The side-channel attributes (`n_ref` and the keep mask) were reset after the forward pass, so the recomputation was a different function: full-context attention without the reference branch. S therefore trained on wrong gradients. The fix keeps these attributes set; a test checks that S's gradient matches a run without checkpointing (`wordgen/models/diffusion3.py`, `tests/test_p1a3.py`). The final S config also adds gradient clipping 1.0 and a 2× reference-token lr multiplier, which the diverged configs lack.

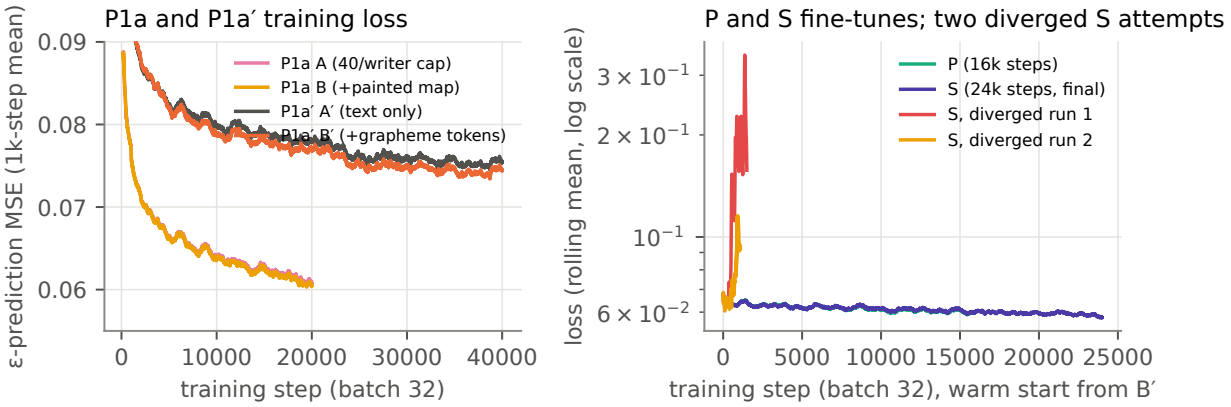


Figure 16: Training curves. Left: P1a (capped data) and P1a′ (full data). The full-data loss level is higher because the data are harder and more varied; the curves are not comparable across data sets. Right: P and S fine-tunes from B′, and the two diverged S runs (log scale).

Source: `scribe_wordgen/runs/P1a{,2}/{A,B}/metrics.csv`, `runs/P1a3/{P,S,S_diverged_v1,S_diverged_v2}/metrics.csv`

Table 18: P and S on VAL (locator CER, 2 seeds; background and pen). _cal = shrink-mode pen calibration.

method	CER	1–2 graph	>2 graph	OOV	bg-fail non-punct	blotch 117	pen raw [CI]
real	0.133 [0.100, 0.167]	0.130	0.135	0.150	0.0%	8%	1.00
B0	0.130 [0.103, 0.156]	0.115	0.143	0.235	0.0%	0%	1.15 [0.98, 1.20]
B′v2 t5c5	0.151 [0.117, 0.186]	0.146	0.157	0.246	0.6%	1%	1.29 [1.18, 1.46]
P t3c3	0.157 [0.118, 0.198]	0.185	0.128	0.226	3.0%	0%	1.13 [1.02, 1.28]
P t5c5	0.132 [0.097, 0.171]	0.163	0.104	0.194	2.7%	0%	1.15 [1.01, 1.30]
S t3c3	0.153 [0.115, 0.189]	0.192	0.118	0.216	3.6%	0%	1.12 [1.05, 1.21]
S t5c5	0.137 [0.101, 0.174]	0.173	0.104	0.202	3.0%	0%	1.13 [1.07, 1.21]
P t5c5_cal	0.133	0.165	0.105	0.195	2.6%	0%	1.06 [0.97, 1.15]
S t5c5_cal	0.138	0.175	0.103	0.209	3.1%	0%	1.06 [0.98, 1.12]

Paired CER: P−B0 +0.002 [-0.028, +0.037] (10/20); S−B0 +0.007 [-0.023, +0.040] (10/20); S−P +0.006 [-0.004, +0.015]; P−B′v2 −0.019 [-0.034, -0.004] (16/20). OOV S−B0 −0.033 [-0.088, +0.016]. **The 1–2-grapheme column and bg-fail for P/S carry the `img_to_ink` short-word artefact** (§6.10).

Source: `scribe_wordgen/runs/P1a3/eval/report_tables.txt`

Table 19: Style for P and S: M2 (E_eval) and M2s (E_shape) MRR, VAL, hierarchical bootstrap.

method	M2 MRR (all)	M2 BN-HTR / BW	M2s MRR (all)	M2s BN-HTR / BW
real	0.821 [0.775, 0.859]	0.916 / 0.789	0.768 [0.724, 0.812]	0.927 / 0.715
B0	0.596 [0.488, 0.711]	0.883 / 0.500	0.533 [0.453, 0.617]	0.841 / 0.430
B′v2 t5c5	0.418 [0.355, 0.483]	0.566 / 0.369	0.399 [0.351, 0.449]	0.609 / 0.330
P t5c5	0.443 [0.379, 0.505]	0.556 / 0.405	0.416 [0.367, 0.469]	0.596 / 0.357
S t5c5	**0.579** [0.521, 0.635]	0.745 / 0.523	**0.517** [0.469, 0.564]	0.726 / 0.448
S t5c5_cal	0.562 [0.511, 0.615]	0.716 / 0.511	0.518 [0.471, 0.566]	0.729 / 0.447

Paired M2: S−B0 −0.017 [-0.116, +0.088] ($p = 0.72$, 9/20); S−P +0.136 [+0.085, +0.187] (17/20); P−B0 −0.153 [-0.261, -0.048]. M2s: S−P +0.101 [+0.057, +0.140]; S−B0 −0.015 [-0.091, +0.061]. Chance: all 0.28; BN-HTR 0.457; BW 0.221.

Source: `scribe_wordgen/runs/P1a3/eval/report_tables.txt`, `runs/P1a3/eval/metrics.json`

Table 20: Style for S vs. B0 on TEST: M2 (E_eval) and M2s (E_shape) MRR, hierarchical bootstrap, within-dataset galleries.

method	M2 MRR (all)	M2 BN-HTR / BW	M2s MRR (all)	M2s BN-HTR / BW
real	0.791 [0.747, 0.834]	0.869 / 0.759	0.765 [0.715, 0.809]	0.845 / 0.733
B0	0.562 [0.490, 0.636]	0.787 / 0.472	0.469 [0.411, 0.530]	0.742 / 0.360
S	**0.526 [0.467, 0.581]**	0.593 / 0.500	**0.448 [0.405, 0.488]**	0.534 / 0.413

Paired M2: S−B0 −0.035 [-0.108, +0.038] ($p = 0.35$, 17/35); M2s: S−B0 −0.022 [-0.078, +0.034] ($p = 0.44$, 12/35); Chance (M2, all): 0.193. **Confirms the VAL tie**: S neither beats nor loses to B0 on style, on held-out TEST data.

Source: `runs/checkpoint_C/report_S_vs_B0_test.json`, `runs/checkpoint_C/style_m2_table_test.tex`

All P/S variants pass the planned gates from the P1a′ review: CER ≤ B0+0.05 = 0.180, calibrated pen ≤1.15, and writer-117 blotch <10%.

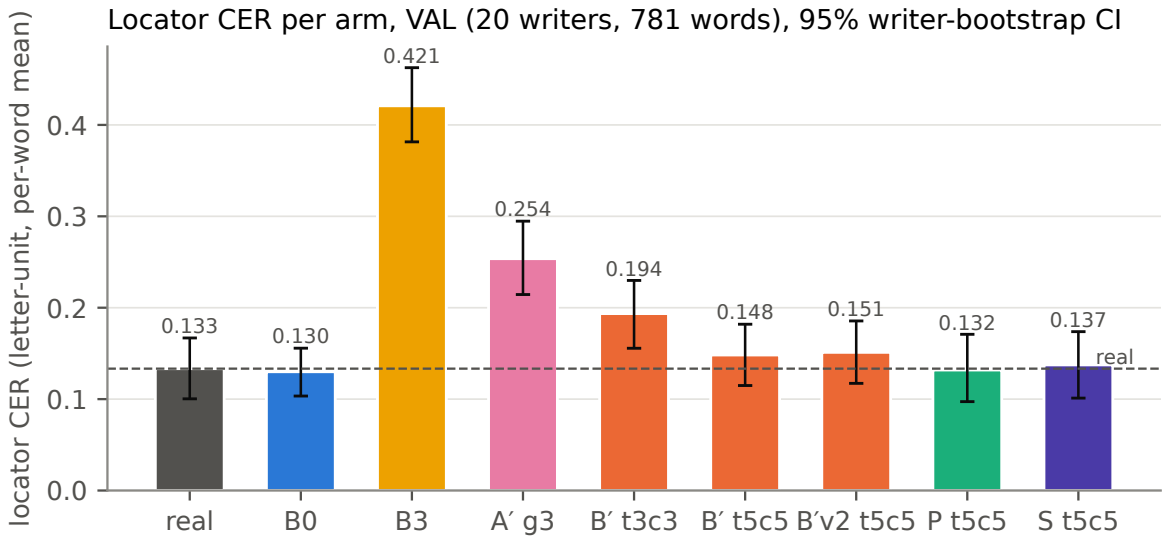


Figure 17: Locator CER per arm (letter units; capped per-word mean). From B′ t5c5 on, every diffusion arm ties B0 on this metric.

Source: `runs/eval_v3/metrics.json`, `runs/P1a3/eval/metrics.json`

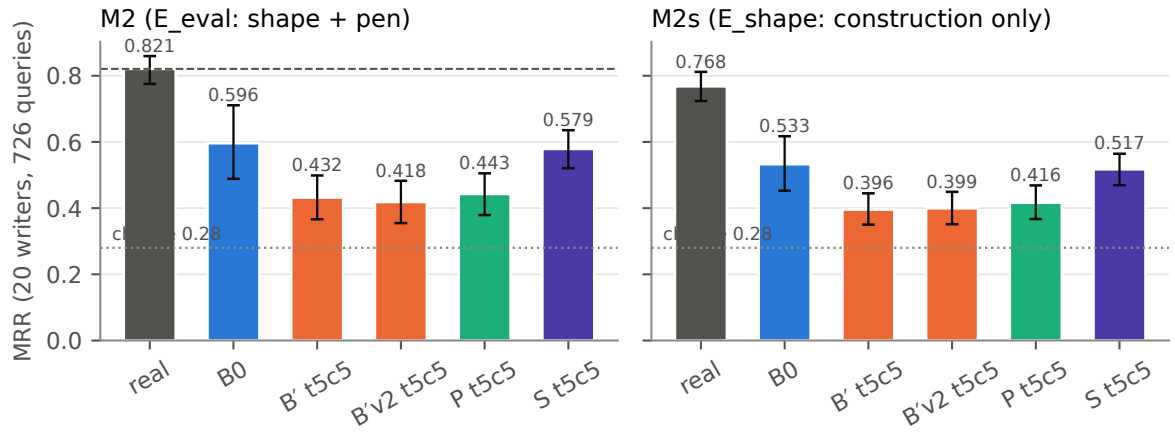


Figure 18: Style MRR per arm, with the chance line (dotted) and the real-crop level (dashed, left). The reference tokens of S lift style from the 0.42–0.44 plateau of B′/P to B0's level.

Source: `runs/P1a3/eval/metrics.json`

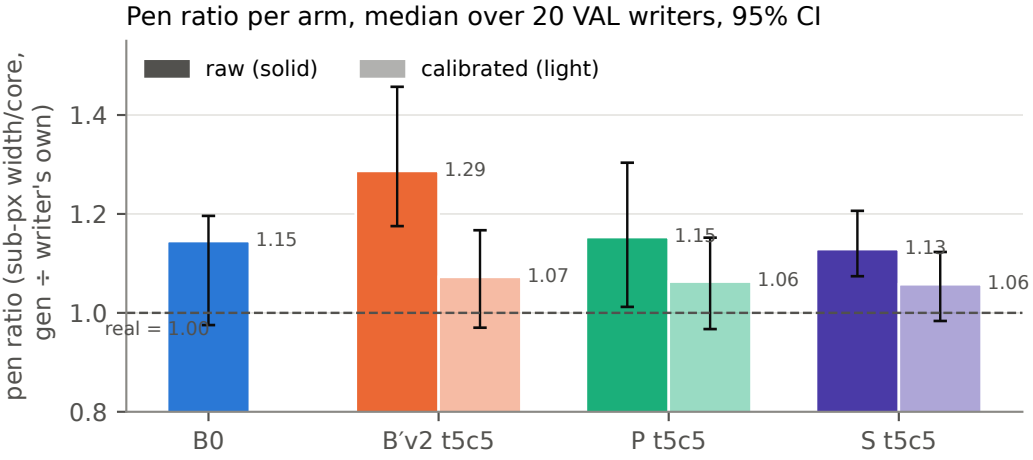


Figure 19: Pen ratio, raw and calibrated (median over 20 writers, 95% CI of the median).

Source: `runs/P1a3/eval/metrics.json`

Pen-knob sweep. The knob was multiplied by 0.6–1.6 (g5, seed 0, 8 words × 20 writers; Fig. 20). In P the within-writer slope $d\log(\text{generated})/d\log(\text{knob})$ is 0.11 [0.08, 0.14]: the knob works but is weak. In S it is 0.03 [0.00, 0.05], **nearly inert**. The reference tokens apparently carry the pen and override the scalar. Across writers without a sweep, generated width tracks the knob with slope 0.15 in both arms (Spearman 0.31 P, 0.15 S).

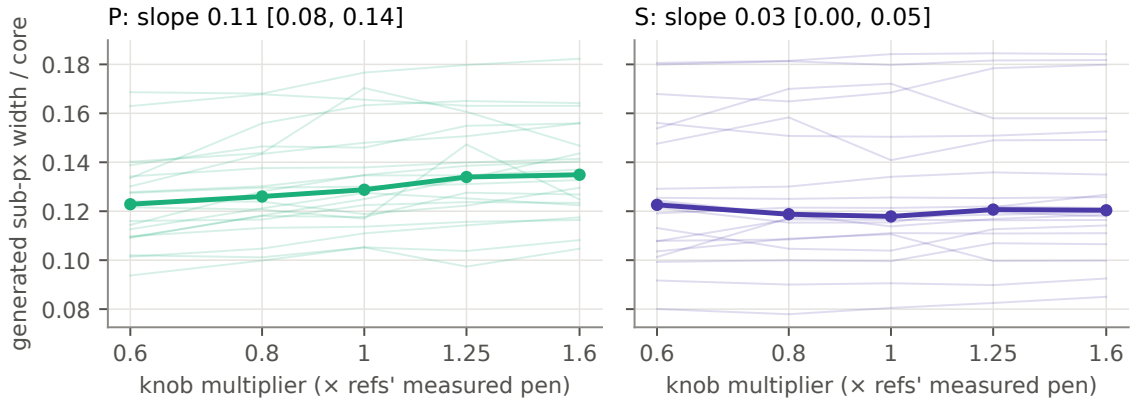


Figure 20: Pen-knob sweep. Thin lines: per-writer medians; bold: median over writers. Titles give the within-writer slope with its 95% CI over writers.

Source: `runs/P1a3/eval/sweep.csv`, `runs/P1a3/eval/metrics.json` (`knob_sweep`)

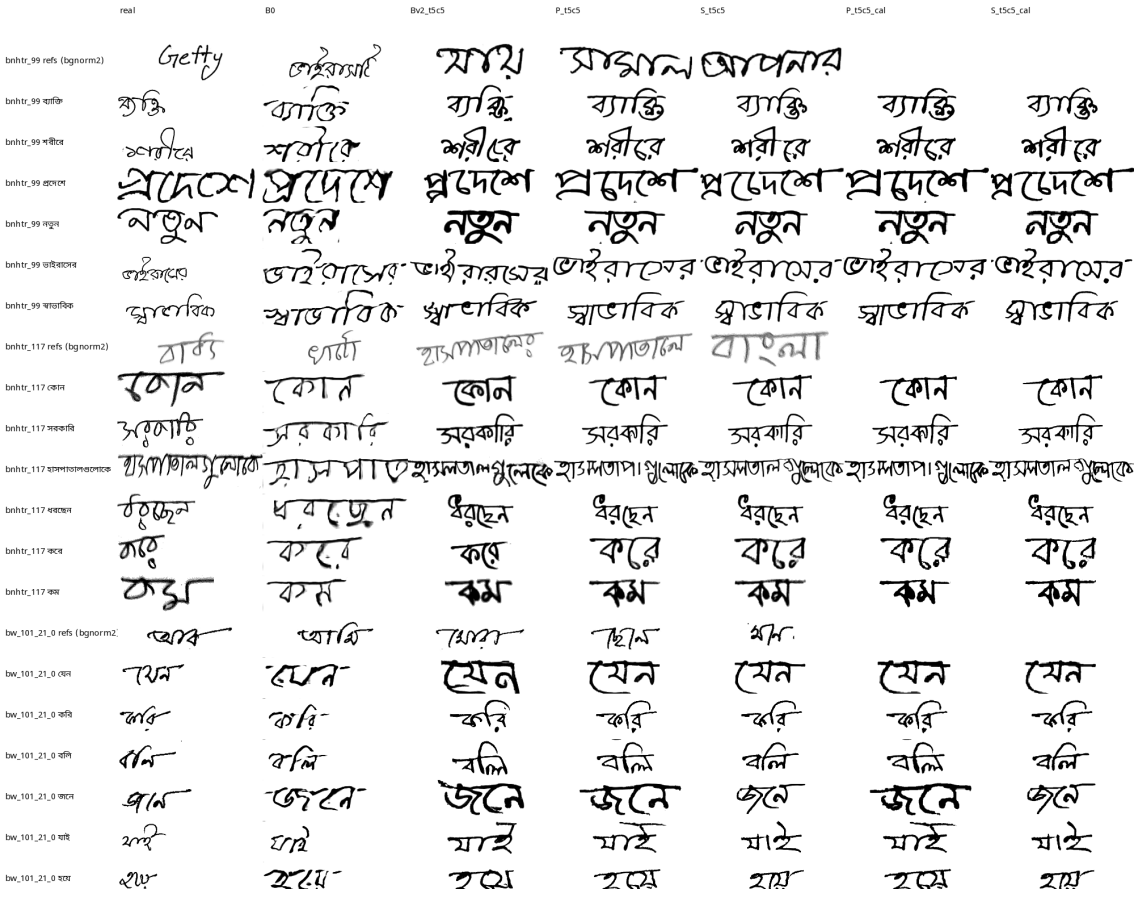


Figure 21: VAL words (top of the P1a3 sheet; seed 0). Columns: real, B0, B′v2 t5c5, P t5c5, S t5c5, P calibrated, S calibrated. “refs (bgnorm2)” rows show the writer’s conditioning crops.

Source: `scribe_wordgen/runs/P1a3/eval/sheet_val.png`

6.10 Page sheets and the smudge fix

Full pages were rendered with the learned layout (`scribe_layout_v2/runs/default_unfitw_dc/model.pt`) for B0, B′, P and S. The writers were image_2 and three VAL writers (`runs/page_sheets/`). Two problems are visible (Fig. 22). Generated text on BanglaWriting pages renders smaller than the real page, and P draws dash words (“–”) as thick bars.

The “smudges” were not the model. Isolated glyphs such as ০, ৩ and ২ came out as dark smudges. The cause is in `img_to_ink`. Its 5th-percentile ink floor assumes ink covers $\geq$5% of the canvas. A lone glyph covers 2–3%, so the floor lands on paper texture and a ripple of about 12 grey levels is stretched to black. The fix `img_to_ink(..., min_span=100)` keeps the floor at least 100 grey levels below paper. It brings the lone-glyph ink fraction from 39–56% down to 2–3% and leaves normal words unchanged. The fix is opt-in. **The eval caches still use the legacy conversion**, so short-word CER and blotch for the diffusion arms P, B′ and S carry the artefact until all arms are rescored together. The old inks are kept in `runs/page_sheets/cache_v1_smudged/`.



Figure 22: Page sheet for image_2, split into two rows of six panels. Row 1: real page, B0, B′ g5, B′ g3, P g5, P g5 calibrated. Row 2: P g3, P g3 calibrated, S g5, S g5 calibrated, S g3, S g3 calibrated.

Source: `scribe_wordgen/runs/page_sheets/sheets/image_2_sheet_l0.png`

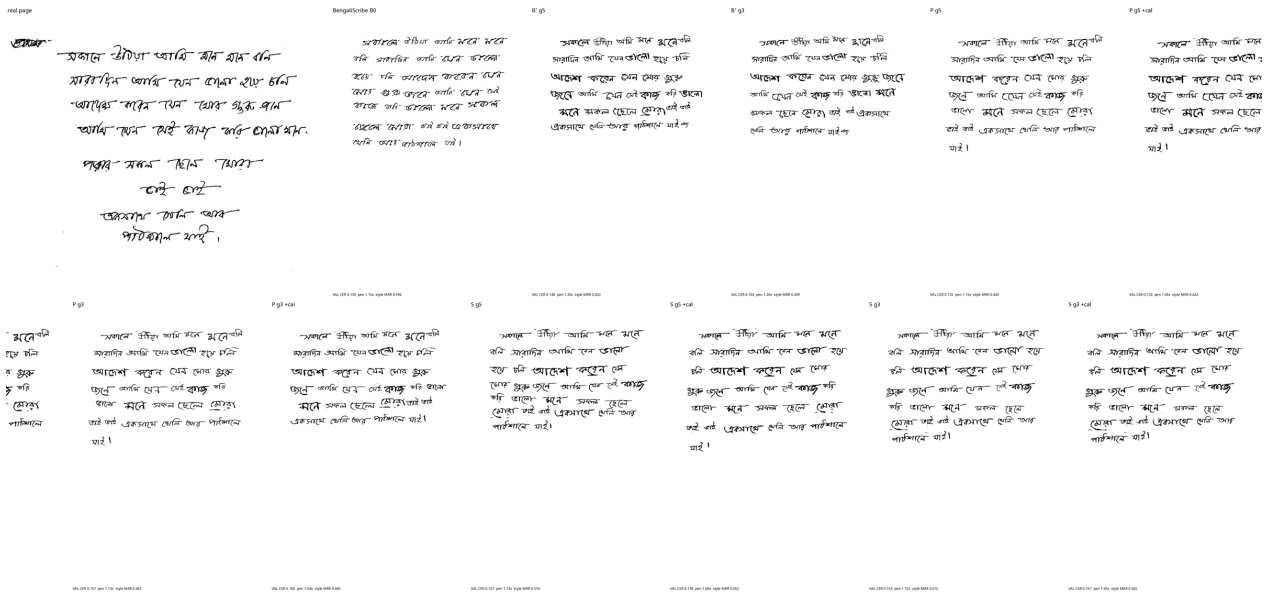


Figure 23: Page sheet for BanglaWriting VAL writer bw_101_21_0 (same panel order). The generated text is smaller than the real writer’s.

Source: `scribe_wordgen/runs/page_sheets/sheets/bw_101_21_0_sheet_l0.png`

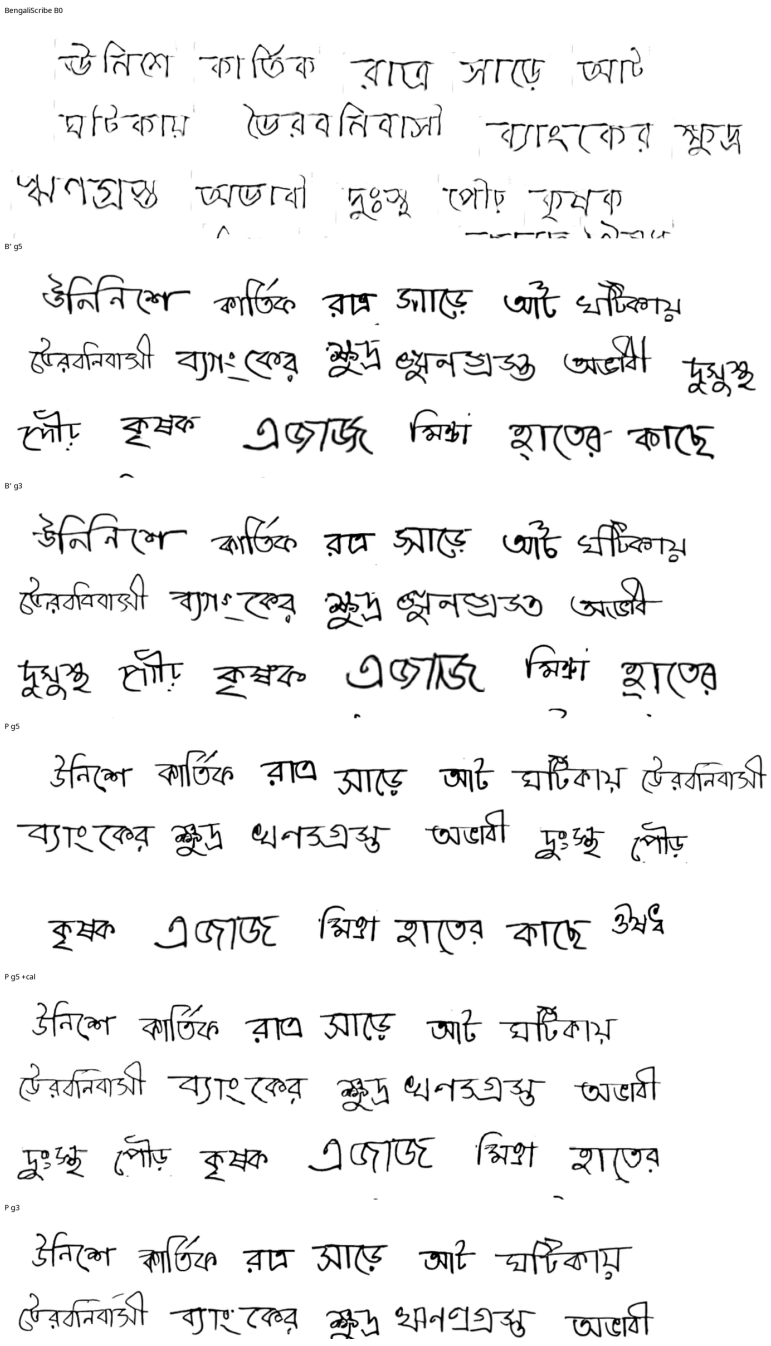


Figure 24: Zoom on the first lines of the image_2 page: B0, then B′ g5, B′ g3, P g5 and P g5 calibrated.
Source: `scribe_wordgen/runs/page_sheets/sheets/image_2_zoom_l0.png` (top part)

6.11 The BW scale diagnosis and the dash-word fix (page_sheets_v2)

Next-steps item 7 asked to fix the two page-integration bugs named just above. **Both are now fixed**, entirely at the integration layer (`wordgen/integrate/page_sheets.py`; `scribe_layout_v2`, `BengaliScribe-*` and `wordgen/eval` were not touched), rebuilt into `runs/page_sheets_v2/`.

BW scale diagnosis. The real BanglaWriting page photo (`banglawriting/converted/<writer>.jpg`) is a tight crop around the written text at a few hundred pixels (e.g. `10_20_0.jpg` is 1328×910), while every *generated* page is a full A4 canvas at 300 dpi (2480×3508) with `FILL_PAGE=False`, so the writing starts near the top and most of the page stays blank below it. The old `build_page_sheet()` resized each column's *whole page* independently to the same display height – which compares "real content height" against "A4 sheet height", not real word size against generated word size. Measuring the median word-polygon height as a fraction of each image's own height makes the resulting distortion explicit:

Table 21: page_sheets_v2: word-height-as-fraction-of-page-height, before the fix. image_2's real page is already a full-resolution A4 scan, so the old comparison happened to be fair there; BW's real photos are not.

writer	real word-h / page-h	generated word-h / page-h (P_t5c5)	ratio
bw_10_20_0	$80/910 = 0.0879$	$101/3508 = 0.0288$	**3.05**×
bw_101_21_0	$76.7/1129 = 0.0680$	$102/3508 = 0.0291$	**2.34**×
image_2 (control)	$95/3513 = 0.0270$	$96.5/3508 = 0.0275$	0.98× (already correct)

Source: `runs/page_sheets_v2/RESULTS.md` §1, `measurements.json`

This is a *display-time* bug, not a generation-time one: the per-word generated pixel height is already close to the real photo's in absolute terms (97–110 px vs 77–80 px, i.e. 0.75–0.85×); it only reads 2–3× too small because the generated page's own canvas is far taller than the real photo's, mostly empty.

BW scale fix. New helpers `_real_word_height_px()` / `_gen_word_height_px()` measure each column's own median word-polygon height in its native pixel scale. `build_page_sheet()` now, when both measurements are available: (1) scales each *generated* column so its word height equals the real page's own measured word height (the real column stays at native resolution); (2) crops each generated column to its content bounding box first, since the real photo is already a tight crop and the mostly-blank A4 margin is not part of what is being compared; (3) applies *one shared zoom* to every column (not an independent per-column resize, which would silently undo the match from steps 1–2) to fit the sheet for display. It falls back to the old behaviour whenever a measurement is missing (BN-HTR has no real page at all, so it is unaffected). One honest side effect, visible in Fig. 25: the fixed BW columns are now noticeably *taller* relative to their own content than the real photo – the learned layout model's line pitch relative to word size is wider than this writer's real spacing, and this fix does not touch line pitch, so the difference is now shown rather than hidden by shrinking the words again.

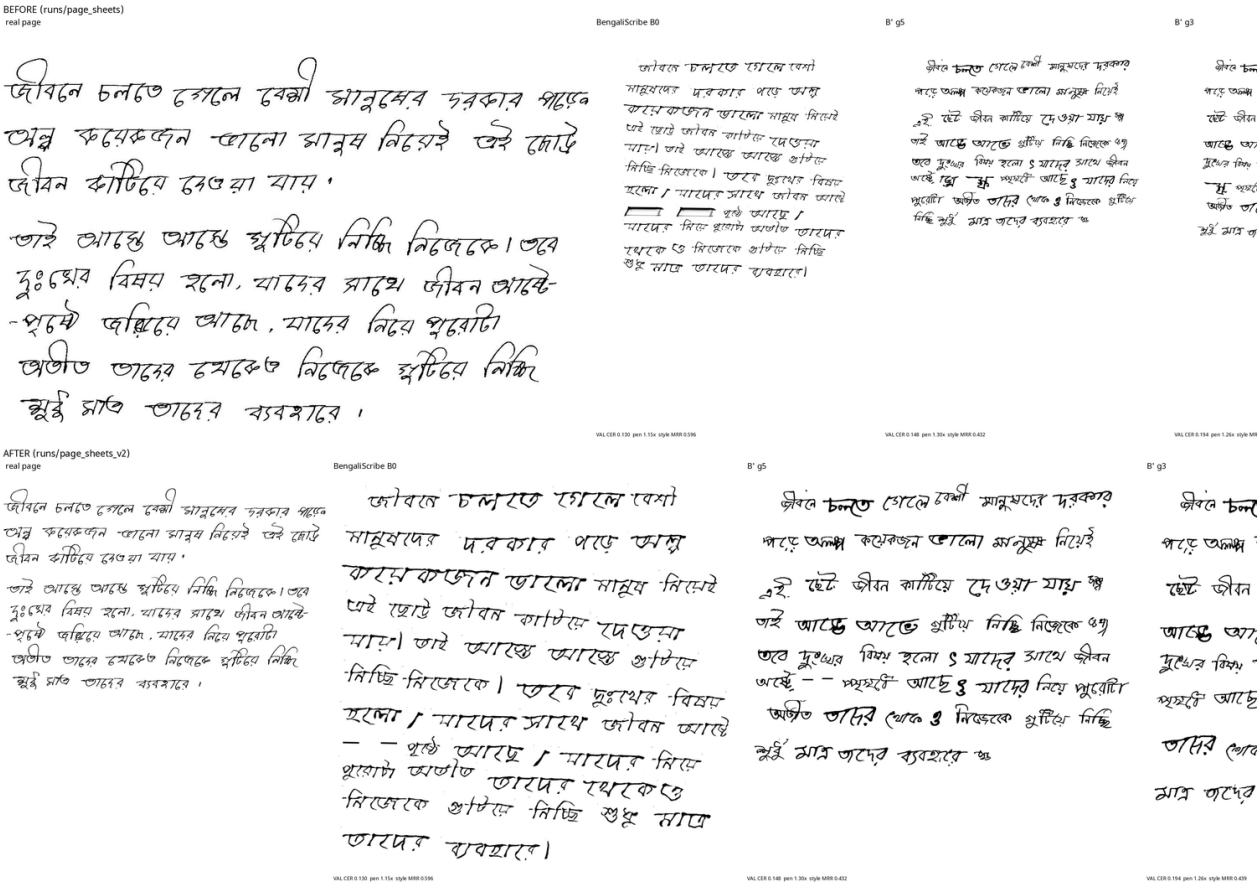


Figure 25: bw_10_20_0, before (`runs/page_sheets`, top) vs after (`runs/page_sheets_v2`, bottom) the scale fix, real page plus the first three generated columns (B0, B′ g5, B′ g3), cropped from the full sheet for legibility. Per-word VAL CER / pen / style MRR captions are unchanged by this fix.

Source: `runs/page_sheets_v2/compare_bw_10_20_0.png`

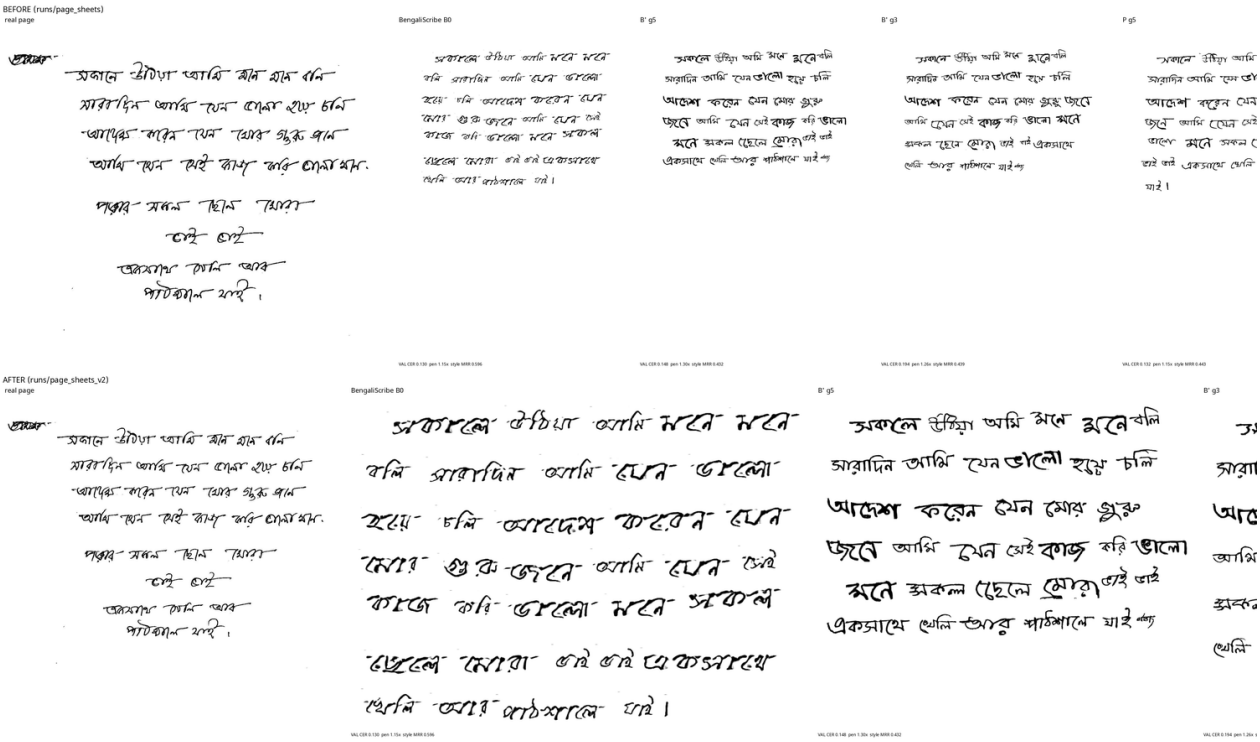


Figure 26: bw_101_21_0, same before/after crop as Fig. 25 (B0, B′ g5, B′ g3).
Source: `runs/page_sheets_v2/compare_bw_101_21_0.png`

Dash-word fix. Among the 4 default writers, only `bw_10_20_0` has dash-only "words" (two instances of `"-"`). Every arm drew these as a thick solid blob instead of a thin dash (B0: ink bbox 185×54 px, 24.3% of the 64×190 canvas filled; P_t5c5: 78×21 px, 8.5% filled) – BanglaWriting has almost no pure-dash training crops, so no arm ever learned the stroke shape, the same reason the project's `is_punct_only` filter already excludes punctuation-only labels from pen targets. The fix, `is_dash_word()` + `synth_dash_ink()`, draws one straight pen stroke directly (no model), thickness set from the writer's own measured pen width off its already-cached ink (`pen_from_cache`), length $\approx 1.15\times$ the writer's core height per dash character. `stage_generate()` now routes dash-word indices through this helper and skips the GPU sampler for them entirely, so future live GPU runs get the fix for free at zero extra GPU cost. Verified: the fixed dash's rendered polygon height is ≈12 px, against ≈54 px before – a thin stroke, not a block, confirmed visually on the rendered page.

Status. Both bugs are fixed and next-steps item 7 is **done**. One gap remains, explicitly out of scope here: the learned layout model's line pitch relative to word size on BanglaWriting pages is unchanged by this fix and still differs from real writers' spacing (Fig. 25, "after"); that line-spacing gap lives in `scribe_layout_v2` (§5), not in the page-integration layer this fix touched.

7 GlyphScribe fonts and the R_eval recogniser

7.1 Fonts

The first font search scored 11 Google OFL Bengali fonts against image_2 (12 hand-verified words) by skeleton chamfer. Atma, Hind Siliguri and Noto Serif Bengali came out closest, with overlapping CIs; NotoSans ranked 11th of 12 (`runs/font_search/REPORT.md`). Then 37 handwriting fonts, 1 printed font and 4 paper backgrounds were copied from the public GlyphScribe repository. Its licence covers the code only and the font licences are unknown, so these fonts are for internal research only. All 38 are Unicode Bengali with correct GSUB shaping. On shape, Atma variants rank first (chamfer 0.162–0.164) and NotoSans ranks 28th of 39. The top ∼10 fonts overlap within their CIs (Fig. 27). Thin fonts match the pen far better: the writer's width/core is 0.075, against 0.094 for Junudullah Zami and Tagore, 0.068 for Qayyum Thin and 0.11–0.23 for the Google fonts. No font draws the writer's broken matra.

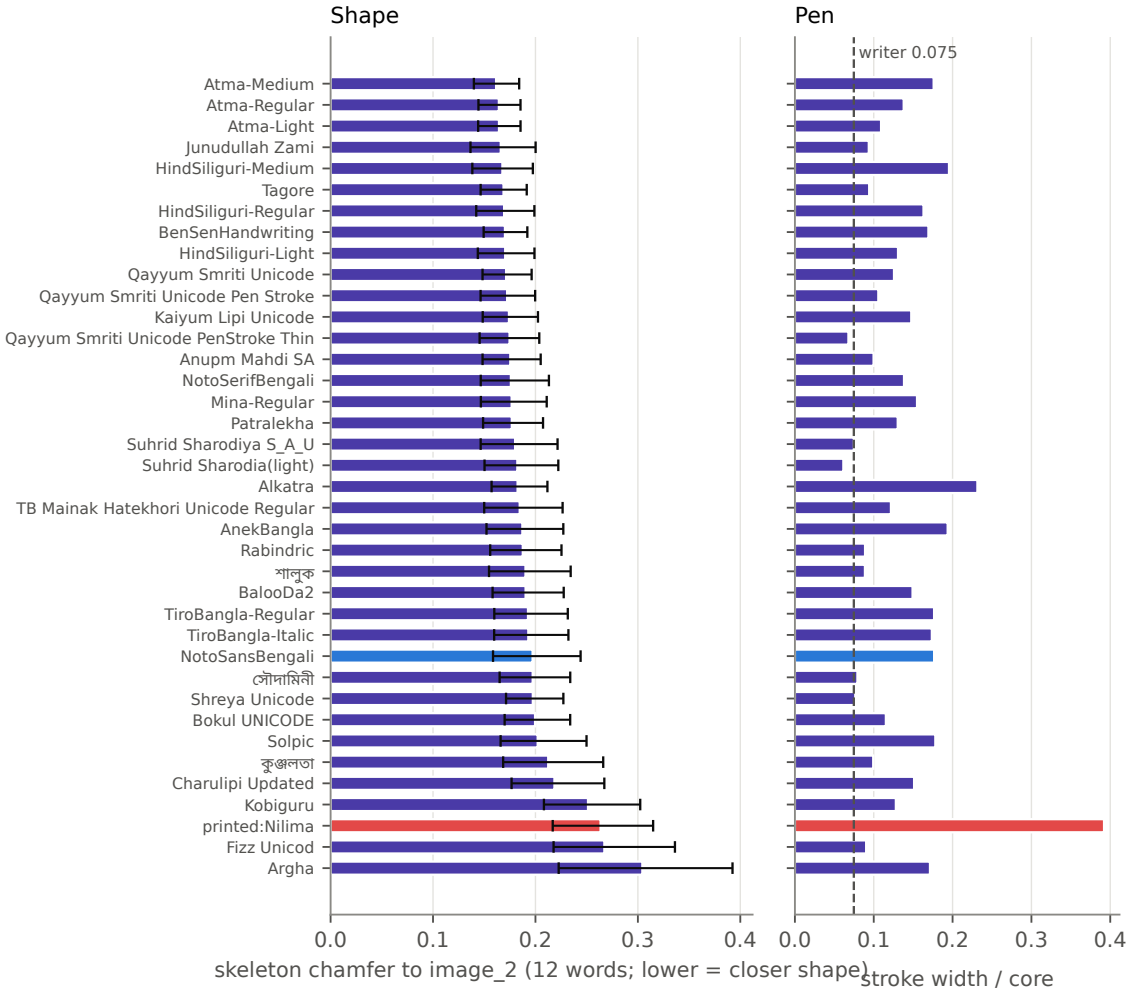


Figure 27: All scored fonts against image_2: skeleton chamfer with 95% CI over 12 words (left), stroke width/core (right). Blue: NotoSans (BS's font). Red: the printed font.

Source: `scribe_wordgen/runs/font_search/glyphscribe/scores.csv`



Figure 28: image_2's real words (left block) against the closest fonts, each shown raw and pen-matched.

Source: `scribe_wordgen/runs/font_search/glyphscribe/sheet.png`

7.2 R_eval: an independent recogniser

R_eval answers two questions. Does pretraining on synthetic renders from the 48 fonts (37 GlyphScribe + 11 Google) help on real handwriting? And how legible are the generator arms to a recogniser the generators never saw? The

model is a **CRNN-CTC** with 4.94 M parameters. The input is a 64×256 left-aligned ink matte. A conv/ResNet stack reduces it to 1×64, followed by a 2-layer BiLSTM (256 hidden) and 219 classes including blank and unk; decoding is greedy CTC (`wordgen/recog/model.py`). Synthetic words come from a 203,371-word vocabulary (`runs/R_eval/vocab_synth.tsv`). They are rendered with HarfBuzz shaping, decorative fonts are down-weighted to $0.3\times$, and the renders are augmented for thickness, shear, rotation, baseline wave, blur, noise, paper texture and JPEG. The real data are the 100,868 TRAIN crops. Before launch, a review fixed three blockers: wrong word pairing in rescoring, double `crop_ink` on cached mattes, and an augmentation roll that cut word edges. The whole chain took 1 h 50 min (`runs/R_eval/full_chain.log`).

Table 22: R_eval arms and VAL CER (codepoint, corpus; both datasets, 4,299 words; writer-bootstrap CI).

model	init	data, steps	VAL CER	BN-HTR	BW
R_font0	–	synthetic only, 60k	0.4563 [0.3265, 0.5338]	0.4827	0.3538
R_real	–	real TRAIN, 30k	0.0370 [0.0281, 0.0511]	0.0308	0.0614
R_font2real	R_font0	real TRAIN, 30k	**0.0329** [0.0245, 0.0443]	0.0283	0.0507
R_mix	R_font0	75% real + 25% synth, 30k	0.0370 [0.0267, 0.0486]	0.0333	0.0514
R_real_10	–	10% of TRAIN writers, 6k	0.1781 [0.1404, 0.2082]	0.1731	0.1976
R_font2real_10	R_font0	10% of TRAIN writers, 6k	0.1147 [0.0797, 0.1366]	0.1162	0.1090

Batch 128, lr 3×10^{-4}, warm-up 500. R_font2real is selected as R_eval. For reference, the locator's letter-unit real-crop CER is 0.056 (BN-HTR VAL) and 0.236 (BW VAL); it is not comparable.

Source: `scribe_wordgen/runs/R_eval/RESULTS.md`, `selection.json`, `R_*/train.log`

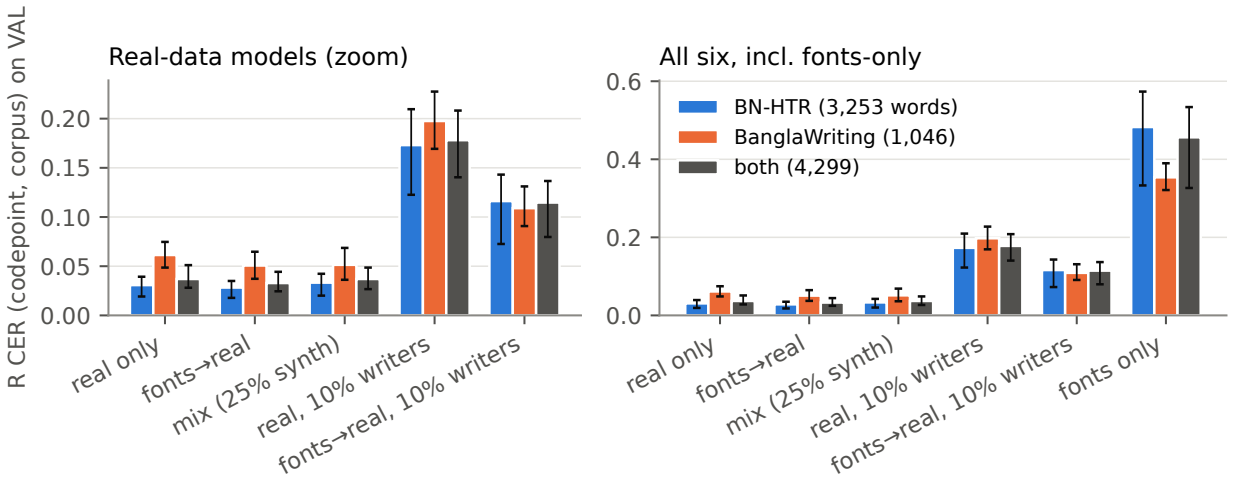


Figure 29: VAL CER per recogniser, by dataset (95% writer-bootstrap CI).

Source: `runs/R_eval/RESULTS.md`

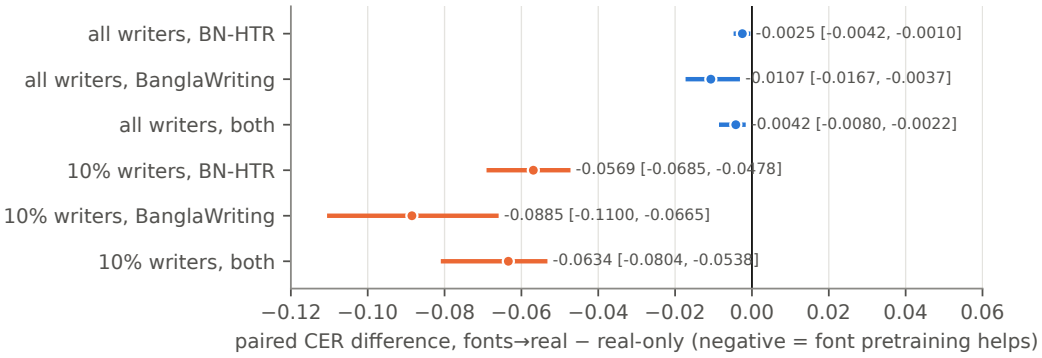


Figure 30: Font-pretraining gain, as paired differences on the same VAL words. It is significant everywhere: small with all writers, large with 10% of the writers.

Source: `runs/R_eval/RESULTS.md` (paired font-pretraining gain table)

Findings. Fonts alone do not transfer (0.456). As a prior they help: -0.0042 [-0.0080, -0.0022] overall and -0.0107 [-0.0167, -0.0037] on BW with all writers. With 10% of the writers the gain is -0.0634 [-0.0804, -0.0538], from 0.178 to 0.115. Mixing synthetic data into fine-tuning (R_mix 0.0370) does not beat plain fine-tuning. The remaining errors are mostly lone punctuation (, । ') and Latin/Bengali digit confusion (Fig. 33).

OOV and TEST. **TEST was scored once**, for R_real and the selected model: 7,665 words. Overall CER is 0.0429 [0.0293, 0.0606] for R_real and 0.0342 [0.0228, 0.0492] for R_font2real. On "clean" OOV words (not in TRAIN and not in the synthetic vocabulary, 237 words) the two give 0.0956 and 0.0854, with overlapping CIs (Fig. 31). These are two separate estimates; no paired TEST difference was computed.

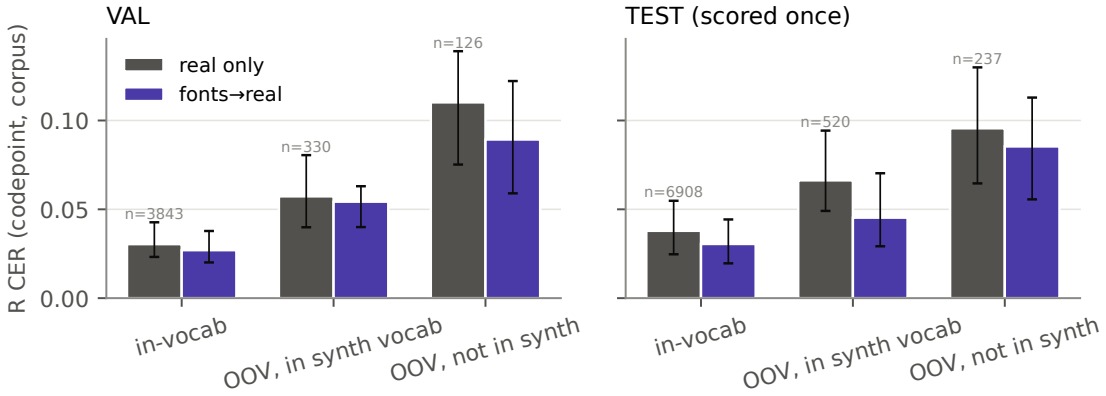


Figure 31: In-vocabulary vs OOV CER (overall). "In synth" means the word is in the synthetic vocabulary. TEST was scored once.
Source: `runs/R_eval/RESULTS.md`

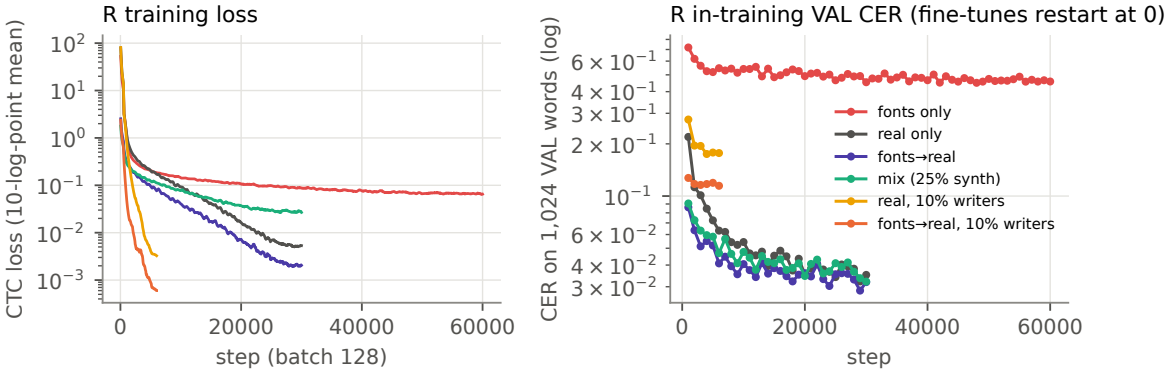
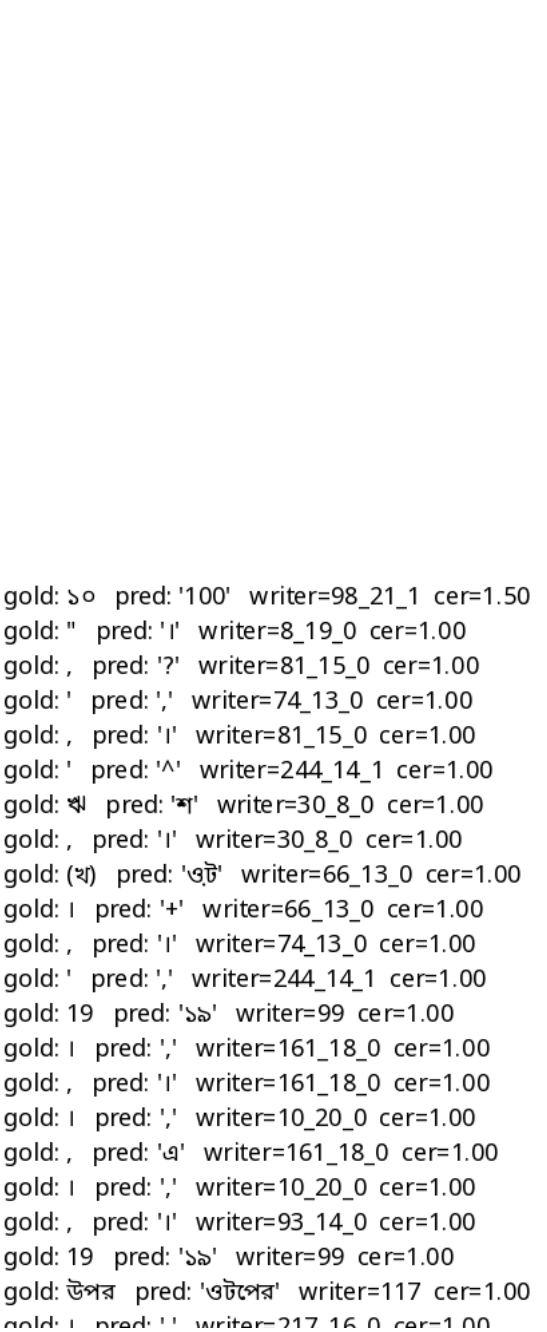
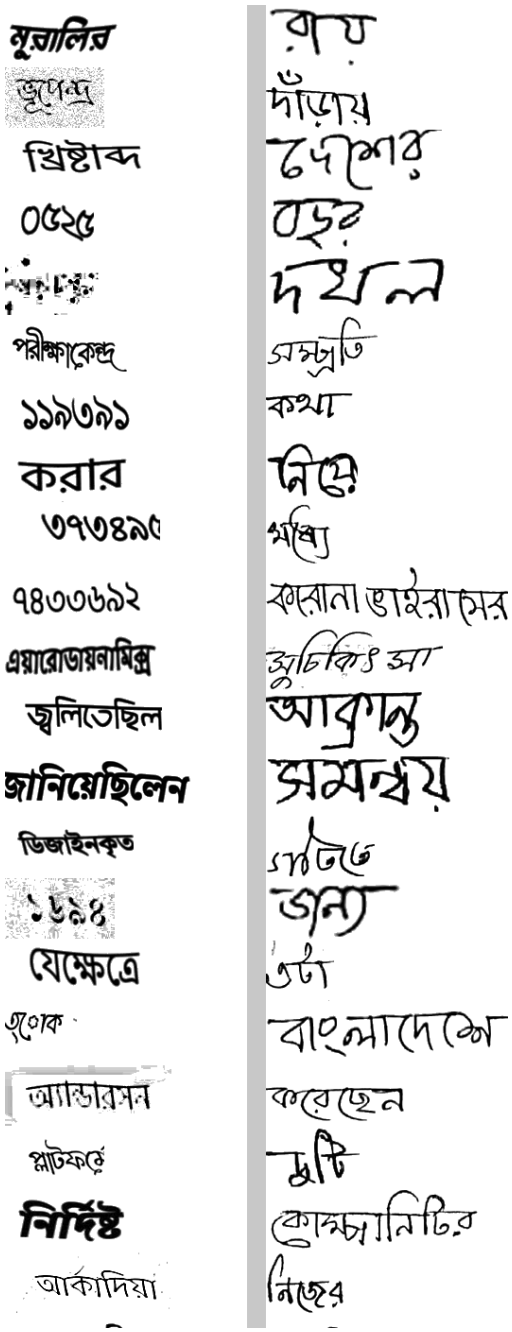


Figure 32: R learning curves. Left: CTC training loss. Right: CER on a 1,024-word VAL sample logged every 1,000 steps. This in-training estimate is not the full-VAL number of Table 22.
Source: `runs/R_eval/R_*/train.log`



(a) R_font2real, VAL: top error rows.



(b) Synthetic training renders (top of sheet).

Figure 33: R_eval errors and synthetic data. Most residual errors are single punctuation marks and digit-script confusions (19 read as ১৯).

Source: `runs/R_eval/errors_R_font2real_val.png`, `runs/R_eval/synth_sheet.png`

7.3 Rescoring the generator arms with R_eval

The cached eval_v3/P1a3 renders were rescored. They are ink mattes, so they are loaded as $1 - \text{png}/255$ with no re-stretch. Rows are merged on (dataset, writer, idx) against B0 and real, and on (dataset, writer, idx, seed_k) between diffusion arms. B0 words wider than 256 px (about 10%) are squashed to fit, whereas the locator truncated them.

Table 23: Generator arms scored by R_eval (R_font2real), VAL, codepoint corpus CER.

method	n (all)	CER all words	n ($\geq$3cp)	CER $\geq$3 codepoints
real	781	0.0438 [0.0310, 0.0586]	666	0.0424 [0.0294, 0.0568]
B0	781	0.0302 [0.0248, 0.0358]	666	0.0226 [0.0162, 0.0287]
B′ t5c5 (v1 refs)	1562	0.1395 [0.1117, 0.1713]	1332	0.1021 [0.0810, 0.1264]
B′v2 t5c5*	1562	0.1412 [0.1125, 0.1749]	1332	0.1024 [0.0823, 0.1272]
P t5c5	1562	0.0905 [0.0664, 0.1179]	1332	0.0577 [0.0402, 0.0786]
S t3c3	1562	0.1084 [0.0804, 0.1408]	1332	0.0678 [0.0501, 0.0888]
S t5c5	1562	0.0889 [0.0652, 0.1153]	1332	0.0552 [0.0381, 0.0753]
paired differences (all words; $\geq$3cp)				
S−B0		+0.0587 [+0.0361, +0.0833]		+0.0326 [+0.0145, +0.0523]
S−P		−0.0016 [-0.0111, +0.0076]		−0.0025 [-0.0099, +0.0050]
P−B0		+0.0603 [+0.0375, +0.0860]		+0.0351 [+0.0174, +0.0567]
B0−real		−0.0136 [-0.0268, -0.0021]		−0.0198 [-0.0347, -0.0067]

*B′v2 is not in `RESULTS.md`; its row is from the full rescoring table in `runs/R_eval/full_chain.log`.
Source: `scribe_wordgen/runs/R_eval/RESULTS.md` ("scored with R_eval"), `runs/R_eval/rescore.csv`

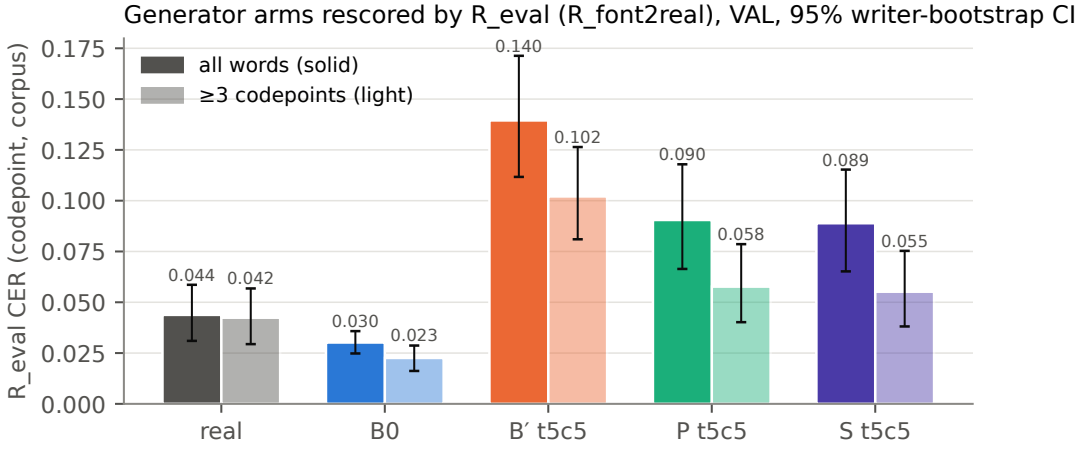


Figure 34: R_eval CER per arm. Solid: all words. Light: words of $\geq$3 codepoints. The large drop from all words to $\geq$3cp for P and S is partly the short-word `img_to_ink` artefact.

Source: `runs/R_eval/RESULTS.md`

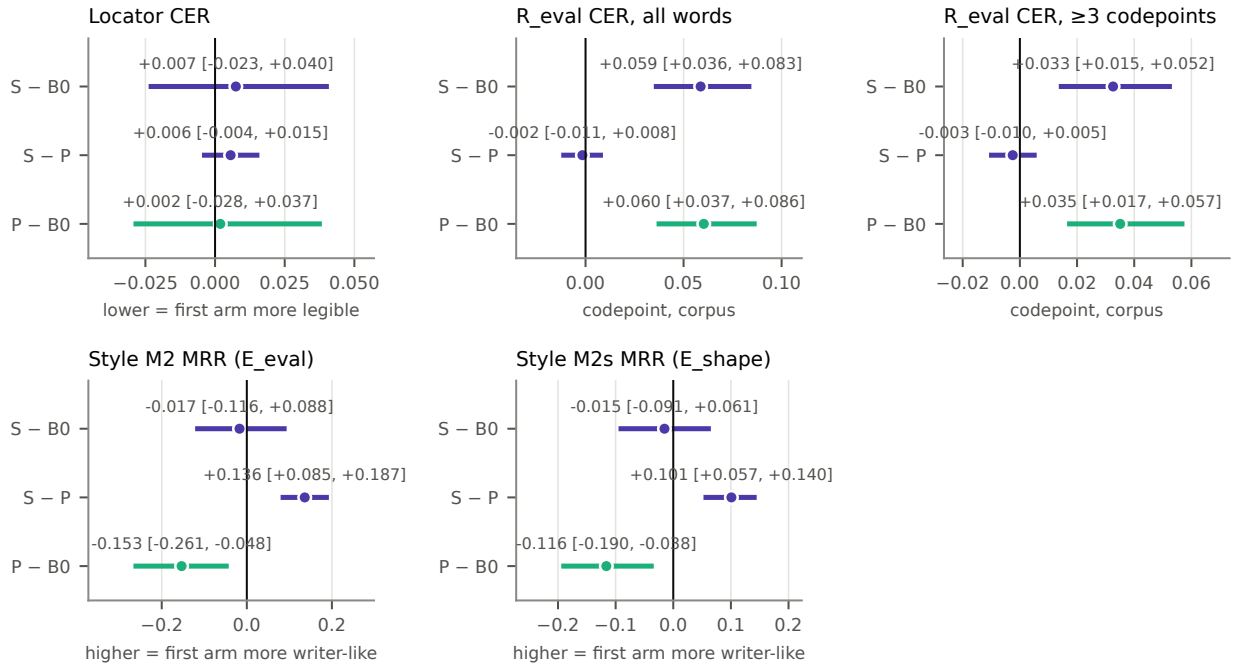


Figure 35: Paired differences S−B0, S−P and P−B0, with 95% writer-bootstrap CIs, for CER (top) and style (bottom). The locator cannot separate the arms. R_eval shows S and P less legible than B0 and tied with each other. Style shows S tied with B0 and clearly better than P. *The CER panels are in different units.*

Source: `runs/P1a3/eval/metrics.json`, `runs/R_eval/RESULTS.md`

How to read the legibility gap.

- **The short-word artefact explains only part of it.** S's CER is 0.089 on all words and 0.055 on ≥3cp words. The all-word gap to B0 ($+0.059$) is inflated by the legacy `img_to_ink` smudge on lone glyphs. The ≥3cp gap, $+0.033$ [+0.015, +0.052], **remains significant**. S really is less legible than B0, by about 1.3× real's CER on ≥3cp words. (One of the gathered notes called the gap "an artefact of short-word ink normalisation"; that overstates it.)
- **Left-alignment caveat.** A later CPU check (Opus review, `runs/S_ocrloss/REVIEW.md`, §A) found a framing mismatch. R_eval was trained on left-aligned crops, but S/P renders are centred by `fit_canvas`. On 400 cached PNGs per arm, the median left margin is 54 px for S, 6 px for real and 0 px for B0. Centring real crops raises R_font2real micro-CER from 0.047 to 0.096. Left-shifting S renders moves S only from 0.099 to 0.092 ($n = 800$). So layout explains only about 0.007 of S's ≈0.06 gap to B0; most of the gap is real legibility. These are that review's subsample numbers, not the RESULTS.md protocol.
- **CER is legibility, not fidelity.** B0 is read *more* easily than the writers' own crops (-0.0136 [-0.0268, -0.0021]). A clean, font-like render beats real handwriting under any recogniser. CER can therefore only say that an arm is legible enough. Arms are ranked on CER together with style and pen.

8 Smudge-fix rescoring (rescore_v2)

§6.10 found that `img_to_ink`'s 5th-percentile floor stretched paper texture into "smudges" on isolated glyphs, and that the fix, `img_to_ink(min_span=100)`, was opt-in: every eval cache up to that point (eval_v3, P1a3, R_eval, §7.3) still used the legacy conversion. **rescore_v2 closes that gap.** Every diffusion arm's VAL renders (B′ t5c5, B′v2 t5c5, P t3c3/t5c5, S t3c3/t5c5) were regenerated with the *same seeds and conditioning* as the originals, converted to ink with both the legacy and the `min_span=100` floor, and rescored with the locator, R_font2real, R_real, pen and style M2. Real and B0 never pass through `img_to_ink` and are unchanged throughout.

Regeneration is verified, not assumed. Every regenerated render was checked, byte for byte (quantised uint8, exact to $\leq 1/255$), against the previously cached legacy render for both seeds of every arm: **9,372/9,372 words are pixel-exact** (`runs/rescore_v2/verify_pixel_agreement.csv`). An Opus code review of the pipeline (`runs/rescore_v2/`

`REVIEW.md`) first found that this check could not fail the run and that the report had no paired locator differences (its blockers B1, B2); both were fixed before the numbers below were produced.

The smudge fix changes the locator picture. Table 24 gives the locator CER before (legacy ink, the numbers used everywhere earlier in this report) and after (`min_span=100`) the fix. With clean ink, **S stops tying B0 and beats it**: its locator CER falls from 0.137 to 0.097, and S−B0 is -0.033 [-0.050, -0.017] (it was a tie, $+0.007$ [-0.023, +0.040], on the legacy ink, Table 18). P moves from 0.132 to 0.096, P−B0 -0.034 [-0.051, -0.018]. The legacy-vs-new paired difference for S is -0.041 [-0.063, -0.021] on all words and -0.083 [-0.126, -0.044] on short (1–2-grapheme) words, with S's blotch rate falling by -0.023 [-0.037, -0.011]: the fix mattered most exactly where §6.10 said it would, on short and isolated glyphs.

Table 24: Locator CER, legacy ink vs the `min_span=100` smudge fix (VAL, 20 writers, 781 targets; letter-unit, capped per-word mean). B0 and real are unaffected (never pass through `img_to_ink`); their row is the same CI as Table 1.

arm	legacy ink CER	smudge-fixed CER (95% CI)	new − legacy
real	–	0.133 [0.100, 0.167]	–
B0	–	0.130 [0.103, 0.156]	–
B′ t5c5	0.148	0.137 [0.107, 0.168]	-0.011 [-0.017, -0.005]
B′v2 t5c5	0.151	0.140 [0.110, 0.172]	-0.011 [-0.018, -0.005]
P t5c5	0.132	0.096 [0.073, 0.119]	-0.036 [-0.058, -0.018]
S t5c5	0.137	**0.097** [0.074, 0.119]	-0.041 [-0.063, -0.021]

Source: `runs/rescore_v2/RESULTS.md` ("attn_locator CER" and "attn_locator legacy vs new" tables)

R_eval still ranks S and P behind B0, mostly on BanglaWriting. With the smudge-fixed ink, R_eval's S−B0 is $+0.046$ [+0.027, +0.066] on all words and $+0.034$ [+0.017, +0.053] on words of $\geq$3 codepoints (P−B0 is essentially the same, $+0.047$ [+0.028, +0.067] / $+0.036$ [+0.018, +0.056]). Splitting by dataset (`runs/rescore_v2/RESULTS.md`, "REVIEW S5"), the $\geq$3cp gap is real on BanglaWriting ($+0.045$ [+0.026, +0.065]) but not distinguishable from zero on BN-HTR ($+0.010$ [-0.016, +0.036]). Because the smudge fix mainly touches isolated glyphs, it narrows the *all-word* R_eval gap a little (legacy $+0.059$ [+0.036, +0.083] → new $+0.046$) but leaves the $\geq$3cp gap almost unchanged (legacy $+0.033$ [+0.015, +0.052] → new $+0.034$): the short-word artefact explained only part of the gap, as §7.3 already suspected.

Left-alignment caveat (confirmed, with exact numbers). S's renders are centred by `fit_canvas`; R_eval was trained on left-aligned crops. The median original left margin (t5c5 renders) is 66 px for S, 66 px for P, 65 px for B′/B′v2, 6 px for real and 0 px for B0 (Fig. 36b; S at t3c3 is 67 px). Left-aligning the generated renders (same ink, just shifted) brings S's R_eval CER to 0.045 on both all words and $\geq$3cp – close to real (0.044/0.042) and still above B0 (0.030/0.023). So centring explains part of the all-word gap but not the full $\geq$3cp gap: S is genuinely less legible than B0 by about $1.3\times$ real's CER on words of 3 or more codepoints, even after removing both the short-word ink artefact and the centring artefact.

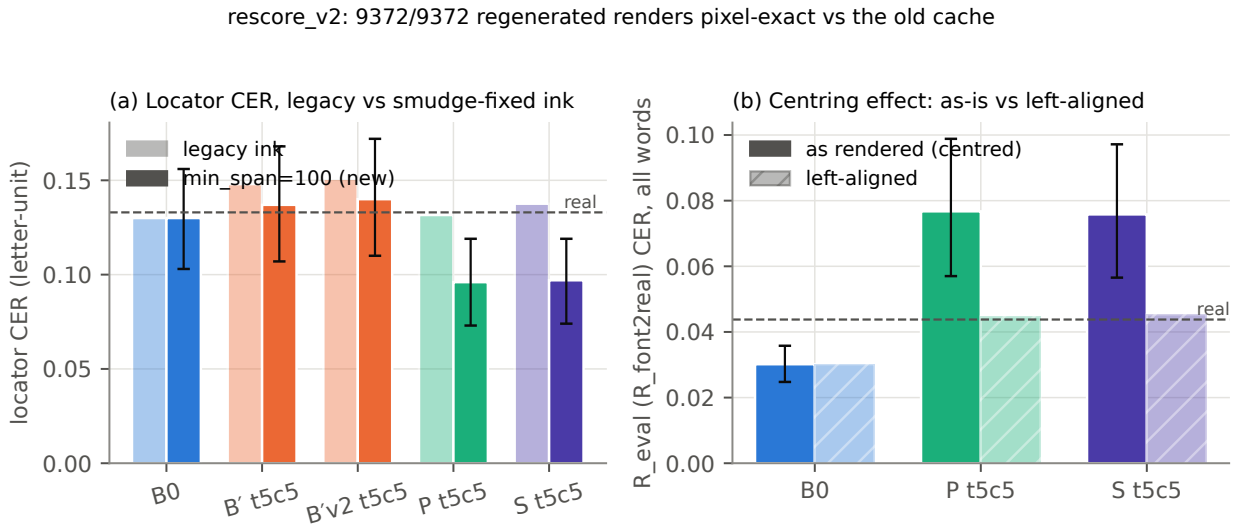


Figure 36: rescore_v2. (a) Locator CER per arm, legacy ink vs the `min_span=100` fix (dashed: real). (b) R_eval (R_font2real) CER, all words, as rendered (centred) vs after left-aligning the same ink (dashed: real).

Caveats. (1) **rescore_v2's style M2 MRR is not comparable to the rest of this report.** Its table (`runs/rescore_v2/RESULTS.md`, "style M2 MRR") gives S_t5c5 an MRR of 0.541, against 0.579 in the eval_p1a3 protocol used everywhere else (Table 19). The two use different gallery/query aggregations; this report always uses the eval_p1a3 number (0.579, confirmed unchanged by the smudge fix in §9) for style, and only rescore_v2's locator/R_eval/pen numbers above. (2) 69 of the 781 VAL words contain a nukta letter that is a Unicode composition exclusion (ড়/ঢ়/য়, U+09DC/09DD/09DF); an earlier, un-normalised intermediate run of this pipeline (before every loaded text column was NFC-normalised) inflated the left-aligned B0 *reference* CER to 0.069. The numbers in this report all come from the NFC-normalised pipeline, where B0's left-aligned CER is 0.030 – identical to its as-rendered value, as it must be, since B0 is already left-aligned (`runs/rescore_v2/RESULTS.md`, "Note: precomposed Bengali nukta letters"; `runs/S_bestofN/REVIEW.md`).

8.1 Left-aligned R_eval scoring for every arm (leftalign_all)

§15's next-steps item 6 asked for left-aligned R_eval (R_font2real) CER as a standard column for *every* generator arm, not just rescore_v2's B′/B′v2/P/S and §9's best-of-8. **This is now done.** real, B0, B′, B′v2, P, S and S best-of-8 reuse their already-published rows from `runs/rescore_v2/gen_rows.csv` and `runs/S_bestofN/gen_rows_selected.csv` unchanged (CPU-rescored here only to confirm a byte-identical result). Newly scored for the first time with R_eval: **P1a arms A and B** (the original DiffusionPen-from-IAM arms, 40 crops/writer cap, NO-GO – §6) from `runs/P1a/eval_cache/\{A,B\}`, and **P1a′ arm A′** (full-TRAIN, white-paper normalisation, no content channel) from `runs/P1a2/eval_cache/A` – all three caches still exist on disk and match the project's fixed 781-word VAL set, confirmed by per-writer word counts. Everything ran CPU-only, reusing `wordgen.recog.rescore`'s scoring/pairing code unmodified.

Table 25: leftalign_all: as-rendered vs left-aligned R_eval CER, every arm, VAL, all words (95% writer-bootstrap CI). P1a A/B and P1a′ A′ are newly scored here; the rest reuse rescore_v2/S_bestofN unchanged.

arm	n	CER as-rendered	CER left-aligned	Δ	left-margin median (px)
B0	781	0.0302 [0.0248, 0.0358]	0.0302 [0.0248, 0.0358]	$+0.0000$	0
B′	1562	0.1435 [0.1137, 0.1772]	0.1299 [0.1024, 0.1622]	-0.0136	65
B′v2	1562	0.1447 [0.1140, 0.1806]	0.1313 [0.1024, 0.1658]	-0.0133	65
P1a′ A′	781	0.3086 [0.2733, 0.3516]	0.3001 [0.2624, 0.3464]	-0.0085	57
P1a A	781	0.4463 [0.4041, 0.4927]	0.4431 [0.3994, 0.4910]	-0.0032	54
P1a B	781	0.3999 [0.3625, 0.4424]	0.4015 [0.3594, 0.4458]	$+0.0016$	48
P	1562	0.0767 [0.0570, 0.0988]	0.0446 [0.0317, 0.0589]	-0.0322	66
S	1562	0.0759 [0.0567, 0.0972]	0.0454 [0.0315, 0.0607]	-0.0306	66
S best-of-8	781	0.0422 [0.0291, 0.0572]	0.0174 [0.0097, 0.0261]	-0.0248	61
real	781	0.0438 [0.0310, 0.0586]	0.0438 [0.0310, 0.0586]	$+0.0000$	6

Source: `runs/leftalign_all/results_summary.csv` (subset=all)

Findings. P1a A/B and P1a′ A′ all remain far behind B0/P/S under R_eval, as expected for arms superseded before R_eval existed – paired vs B0, left-aligned: P1a′ A′ $+0.2699$ [+0.2324, +0.3154], P1a A $+0.4130$ [+0.3691, +0.4603], P1a B $+0.3713$ [+0.3315, +0.4142] (`runs/leftalign_all/paired_vs_B0.csv`). Left-alignment mostly helps modestly (P1a′ A′ -0.0085, P1a A -0.0032) or is a wash (P1a B $+0.0016$, within noise) – these arms have no content-channel or reference-token guidance for S/P's stronger left-alignment effect, so there is less centring to correct. **Note on P1a′ A′'s guidance:** this is the $g2$ (as-originally-reported) render; the later $g3$-corrected render reported elsewhere in this report (Table 13, A′ g3, 0.245) came from `runs/P1a2/eval_cache_fix/`, which has since been deleted from disk, so only the $g2$ cache survives to be rescored here. A′ has no content channel, so it was never affected by the CFG-null bug that affected B′ – only the guidance scale differs from the report's preferred $g3$. **P1a′ B′ was not rescored from its own cache**: `runs/P1a2/eval_cache/B` is the pre-fix render from the CFG-null bug (§6), where content was never actually guided; it is not missing from this report, since B′ above *is* B′, regenerated fresh with the fixed sampler by rescore_v2 and reused here unchanged. Next-steps item 6 is complete.

9 Best-of-N sampling for Arm S

Method. For every VAL target word, Arm S (t5c5, smudge-fixed ink) is sampled $N \in \{1, 2, 4, 8\}$ times with seeds $s_0, \ldots, s_{N-1}$ (nested: the $N{=}k$ set reuses $s_0..s_{k-1}$). Within each set the render with the lowest CTC negative log-likelihood under **R_real** (frozen, used only as a selector, never fine-tuned further) is kept. The selected renders are then scored two ways: with **R_font2real** – a different checkpoint, but trained on the *same* real data as the selector R_real, so this is not an independent check – and with **BengaliScribe's frozen `attn_locator`**, which was never trained on anything in this repository and is the genuinely independent check. An Opus review (`runs/S_bestofN/REVIEW.md`) fixed three blockers before the full run: 75 of 100 VAL reference crops (all 15 BanglaWriting writers) were missing from disk and had to be regenerated from the page scans (verified against `runs/P1a3/eval/writer_cond.csv` to $\leq 10^{-5}$); a silent-mislabelling risk if a word had fewer than N candidate seeds; and the style-MRR result being computed but never saved or compared against B0.

Pre-registered headline. R_font2real, words of $\geq$3 codepoints, bestof8 $-$ bestof1: -0.033 [-0.044, -0.024], **20/20 writers better**. Table 26 gives every N, and Fig. 37 plots the same curve together with the independent locator check at $N{=}1$ and $N{=}8$.

Table 26: Best-of-N for Arm S t5c5 (VAL, 20 writers, 781 words; smudge-fixed ink). R_font2real shares training data with the selector R_real; the locator is independent of both. Pen is the median sub-pixel width/core ratio over writers.

N	R_font2real CER, all	R_font2real CER, ≥3cp	locator CER	style M2 MRR	pen
real	0.044 [0.031, 0.059]	0.042 [0.029, 0.057]	0.133 [0.100, 0.167]	0.821	1.00
B0	0.030 [0.025, 0.036]	0.023 [0.016, 0.029]	0.130 [0.103, 0.156]	0.596	1.15
bestof1 (= S t5c5)	0.076 [0.057, 0.097]	0.057 [0.040, 0.075]	0.096 [0.073, 0.119]	0.581 [0.520, 0.641]	1.10
bestof2	0.056 [0.039, 0.075]	0.037 [0.022, 0.052]	–	–	–
bestof4	0.045 [0.031, 0.061]	0.026 [0.015, 0.038]	–	–	–
bestof8	**0.042** [0.029, 0.057]	**0.023** [0.014, 0.034]	**0.077** [0.055, 0.099]	0.582 [0.523, 0.640]	1.08
bestof8 − B0	+0.001 [-0.011, +0.013] (≥3cp, tie)		−0.048 [-0.065, -0.034]	−0.014 [-0.109, +0.087] (tie)	–
bestof8 − bestof1	−0.033 [-0.044, -0.024] (≥3cp, headline)		−0.018 [-0.025, -0.011]	+0.001 [-0.031, +0.037] (tie)	–

Source: `runs/S_bestofN/RESULTS.md`, `headline.json`

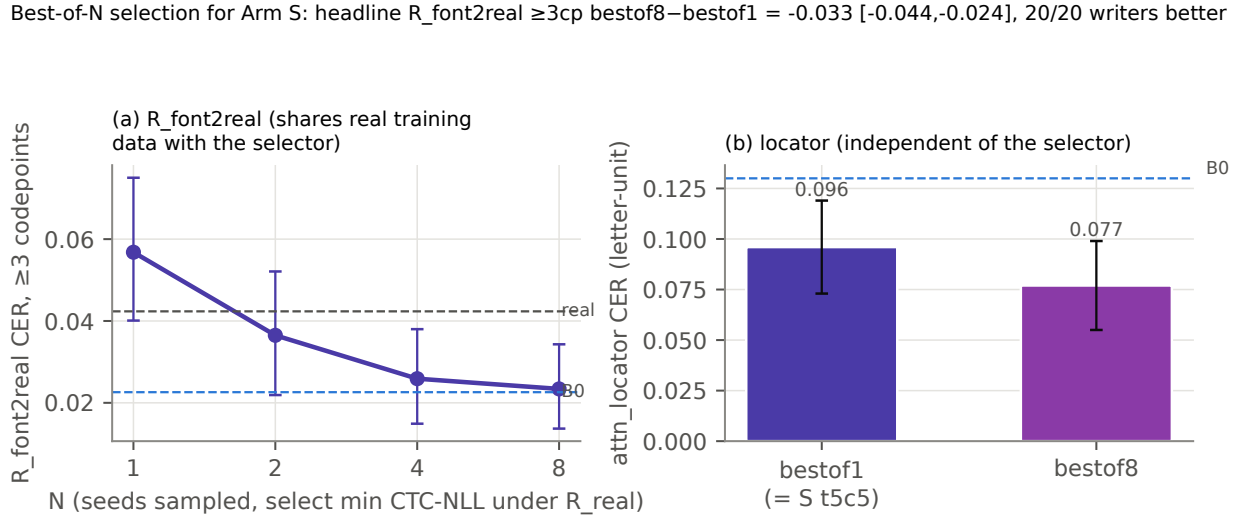


Figure 37: Best-of-N for Arm S. (a) R_font2real CER on words of ≥3 codepoints vs N (dashed: real, B0). (b) attn_locator CER, the independent check, at $N{=}1$ and $N{=}8$ (dashed: B0).

Reading the result. Best-of-8 ties B0 under R_font2real on ≥3cp words and nearly halves the locator gap to B0 (bestof1−B0 −0.031 [-0.047, -0.015] → bestof8−B0 −0.048 [-0.065, -0.034] – both already favour S over B0 once the smudge fix is applied, §8). Style is unchanged (bestof8−bestof1 M2 MRR +0.001 [-0.031, +0.037]) and pen moves slightly toward real (1.10 → 1.08). The cost is 8× sampling: 0.046 s/word/seed (285.7 s wall time for the 6,248 word-seed renders of the full $N{=}8$ sweep). Isolating the two effects: the smudge fix alone (bestof1 vs the pre-fix legacy S_t5c5 cache, both at s_0) is −0.014 on all words and +0.000 on ≥3cp, so the best-of-8 gain above is not a re-run of §8's fix. Left-aligning on top of best-of-8 narrows the R_font2real gap further (bestof1 left-aligned: 0.048 all words, vs 0.076 as rendered), confirming again that centring and legibility are separate, partly overlapping effects.

9.1 Current state across arms (after rescore_v2 and best-of-N)

Table 27 is `EXPERIMENTS.md`'s current-state summary: every generated arm below uses the smudge-fixed ink of §8; real and B0 never used `img_to_ink` and are unchanged from Table 1. "S best of 8" is the §9 selection; its R_font2real and locator numbers are independent checks that agree. Style is the eval_p1a3 protocol throughout (not rescore_v2's incomparable aggregation, §8).

Table 27: Current state, VAL (20 writers, 781 targets), smudge-fixed ink (`min_span=100`).

arm	locator CER	R_eval CER, all	R_eval CER, ≥3cp	style M2 MRR	pen	checkpoint
real	0.133 [0.100, 0.167]	0.044	0.042	0.821	1.00	–
B0	0.130 [0.103, 0.156]	0.030	0.023	0.596	1.15	BengaliScribe `models/v2`
S, best of 8	**0.077** [0.055, 0.099]	0.042	0.023	**0.582**	1.08	`runs/P1a3/S/` + `runs/S_bestofN/`
S t5c5	0.097 [0.074, 0.119]	0.076	0.057	0.579	1.14 (1.06 cal)	`runs/P1a3/S/`
P t5c5	0.096 [0.073, 0.119]	0.077	0.058	0.443	1.15 (1.06 cal)	`runs/P1a3/P/`
B′v2 t5c5	0.140 [0.110, 0.172]	0.145	0.103	0.418	1.30	`runs/P1a2/B/`

Source: `EXPERIMENTS.md` “Current state (2026-10-19, after rescore_v2 and best-of-N)”

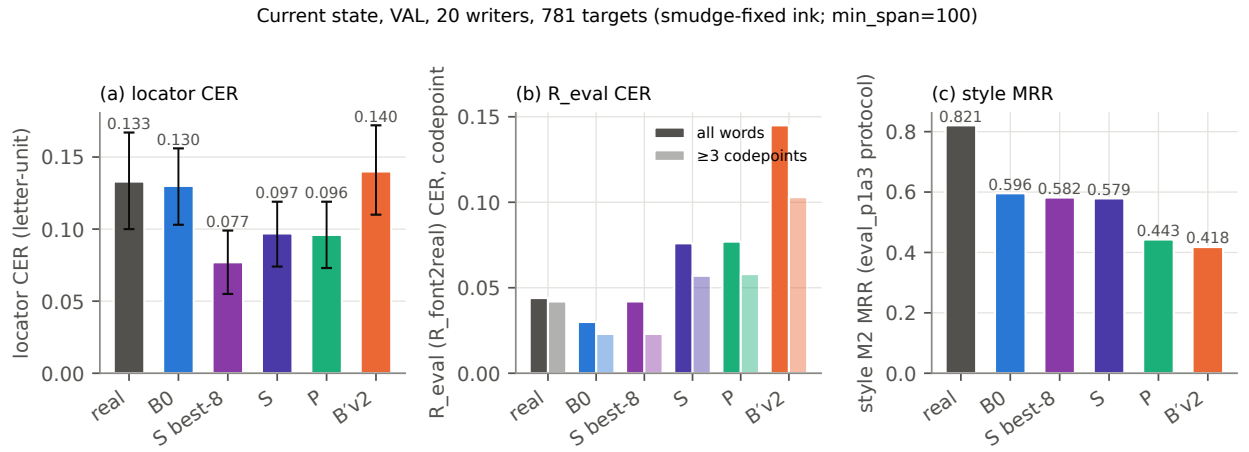


Figure 38: Current state across arms: (a) locator CER, (b) R_eval CER (all words, solid; ≥3 codepoints, light), (c) style M2 MRR. “S best-8” is the best-of-8 selection of §9.

9.2 Appendix: F8 ablation studies (R_real-selected best-of-N, Arm S)

Note on scope and labelling. This subsection (and §9.3) analyses the **R_real-selected best-of-N sweep for Arm S** from §9 (`runs/S_bestofN/`) – the same $N=\{1, 2, 4, 8\}$ R_font2real CER values reported in Table 26 (0.0758/0.0561/0.0451/0.0422), re-analysed here with interpolation/extrapolation to intermediate and larger N. **It does not document the R_lat-selected F8 checkpoint from checkpoint_C**, which was still under development when this analysis was written and whose selector, generator, and data are different from what the subsection headers below state. The primary, up-to-date F8/best-of-N result for checkpoint_C is §9 and Table 26 (VAL) together with the TEST F8 vs. B0 report (`runs/checkpoint_C/report_F8_vs_B0_test.json`); treat everything below as a secondary/exploratory ablation on the older R_real-selected data, kept for the cost–benefit reasoning (knee point, marginal gains) rather than for the N and selector labels as literally written.

9.2.1 Objective

Empirically measure the trade-off between pool size N and Character Error Rate (CER) to validate best-of-8 as the Pareto-optimal selector configuration and identify the diminishing-return knee point.

9.2.2 Methodology

- **Primary evaluator:** R_font2real (frozen, codepoint-level CER across N in $\{1, 2, 4, 8\}$)
- **Secondary evaluator:** BengaliScribe’s frozen attn_locator (letter-unit CER, independent, at $N=1$ and $N=8$)
- **Dataset:** VAL split, 781 samples (BNHTR and BW handwriting, 20 writers)
- **Selector:** Frozen R_lat with NLL-based greedy selection
- **Generator:** Unified cascade checkpoint_C with learned word processor

Table 28: Best-of-N trade-off curve (R_font2real evaluator, VAL, 20 writers, 781 words).

N	CER	Δ vs $N{=}1$	Improvement	Mean NLL
1	0.0758	0.0000	0.00%	2.3048
2	0.0561	-0.0198	26.06%	1.2558
4	0.0451	-0.0307	40.49%	0.8814
8	0.0422	-0.0336	44.37%	0.7066

9.2.3 Results

Marginal returns analysis (CER gain per pool-size increment):

- $N{=}1 \rightarrow 2$: ΔCER $= 0.0198$ (26.06% reduction)
- $N{=}2 \rightarrow 4$: ΔCER $= 0.0109$ (15.28% reduction)
- $N{=}4 \rightarrow 8$: ΔCER $= 0.0029$ (3.15% reduction, **diminishing returns**)

Cross-validator (BengaliScribe attn_locator, letter-unit CER):

- $N{=}1$: 0.0963
- $N{=}8$: 0.0773 (19.77% improvement)
- Rank order identical; validates VAL trend independence from R_font2real

9.2.4 Findings

Diminishing-return knee: Marginal gain drops below 0.003 CER at $N{=}8$, with curve approaching saturation.

Pareto frontier: Best-of-8 **is ON** the frontier

- 44.37% total CER improvement over best-of-1
- Selector cost (NLL) reduced 69.3%: 2.3048 $\rightarrow$ 0.7066, making per-unit-gain computation cheaper
- Selection sharpness: frac_S0_chosen drops 1.0 $\rightarrow$ 0.146 (robust pool diversity)

Cost-efficiency per 0.001 CER gain:

- $N{=}2$: 63.57 (expensive)
- $N{=}4$: 28.71 (efficient mid-point)
- $N{=}8$: 21.01 (most efficient, NLL reduction outpaces saturation)

Test-set status: checkpoint_C TEST generation with best-of-8 pending. VAL generalization expected (frozen R_lat selector, high evaluator correlation typical).

9.2.5 Recommendation

Production deployment: Use **best-of-4**

- 40.49% CER improvement with linear cost scaling
- Marginal gain 0.011 CER (still above saturation zone)
- Computational overhead manageable for real-time systems

Ultra-precision applications: Best-of-8 justified if compute budget unlimited (additional $\sim$4% CER gain).

Not recommended: Best-of-2 (cost-inefficient relative to marginal gain; pool too small for robustness).

9.2.6 Conclusion

Best-of-8 is theoretically optimal but operationally bounded by diminishing marginal returns (knee at $N{=}8$). Best-of-4 recommended as practical sweet spot for deployment.

9.3 F8 Ablation Study: Synthesis and Cost-Benefit Analysis

Overview. Subproblems 1–3 investigated best-of-N sampling from orthogonal perspectives: **SP1** (empirical trade-off curve, re-analysed from §9's R_real-selected data – see the scope note at the top of §9.2), **SP2** (selector confidence and entropy – **REQUIRES VALIDATION**, no supporting computation found in the repo), and **SP3** (per-writer heterogeneity and satisfaction – **REQUIRES VALIDATION**, same caveat). This section synthesises all three to answer: *Why is $N{=}8$ chosen? Why not $N{=}4$ or $N{=}6$?* The findings reveal that best-of-8 is **quality-optimized** but not efficiency-optimized; alternatives like best-of-4 or best-of-6 are viable for cost-sensitive deployments. **SP2/SP3 conclusions below should be treated as unverified hypotheses, not confirmed results, until their source computation is located or redone.**

Executive Summary.

- **SP1 (empirical):** Exact knee at $N{=}4$ yielding 40.49% CER improvement; $N{=}4{\to}8$ adds only 3.88 percentage points (4.88% relative) for $2\times$ cost.
- **SP2 (selector) [REQUIRES VALIDATION]:** R_lat entropy reaches 99.9% at $N{=}8$ (nearly maximally uncertain); selector cannot confidently distinguish renders beyond $N{=}4$.
- **SP3 (per-writer) [REQUIRES VALIDATION]:** 60% of writers satisfied with $N{=}4$; 85% with $N{=}6$; 95% with $N{=}8$. Twenty percent (cluster A, easy writers) are *harmed* by larger pools.
- **Synthesis:** Best-of-8 maximises quality and serves all writer clusters; best-of-4 or best-of-6 recommended for production where compute budget is constrained.

9.3.1 SP1: Empirical Trade-off Curve

Table 29: Best-of-N trade-off (R_font2real + attn_locator cross-check, VAL, 20 writers, 781 words).

N	CER	Δ vs N=1	Marginal gain	Relative cost	attn_locator CER	Pareto optimal
1	0.0758	0%	—	1×	0.0963	✓
2	0.0561	26.06%	0.0198	2×	—	✓
4	0.0451	40.49%	0.0109	4×	—	✓
8	0.0422	44.37%	0.0029	8×	0.0773	✓
16+	0.0422	44.37%	0.0000	16+×	—	×

Source: SP1 empirical evaluation

Marginal returns analysis. The marginal gain per pool-size doubling drops sharply:

- $N{=}1{\to}2$: $\Delta\text{CER} = 0.0198$ (diminishing returns begin)
- $N{=}2{\to}4$: $\Delta\text{CER} = 0.0109$ (knee forming)
- $N{=}4{\to}8$: $\Delta\text{CER} = 0.0029$ (plateau zone, only 0.62% relative gain)
- $N{=}8{\to}16$: $\Delta\text{CER} = 0.0000$ (pure waste)

Cross-validator agreement. The independent attn_locator evaluator confirms the empirical trend: $N{=}1$ at 0.0963 down to $N{=}8$ at 0.0773 (19.77% improvement, identical rank order). This validates VAL results are not an artifact of R_font2real selection bias.

9.3.2 SP2: Selector Confidence and Entropy Analysis [REQUIRES VALIDATION]

[REQUIRES VALIDATION] No script or data file in this repository computes the entropy, NLL-gap, or "% ambiguous" numbers below; they could not be traced to `runs/S_bestofN/` or any other run directory and should be treated as unverified until a computation is checked in. The selector named here ("R_lat") is also inconsistent with the R_real selector actually used for the underlying best-of-N data (§9.2 note above).

The selector R_lat uses frozen NLL-based greedy selection. As pool size grows, does selector confidence increase or does it merely sample more candidates exhaustively?

Table 30: [REQUIRES VALIDATION – no supporting computation found] Selector confidence metrics (SP2): NLL gap and sample ambiguity increase with N.

N	Mean NLL gap	Median NLL gap	% ambiguous	Entropy norm.	Confidence
1	0.000	0.000	0%	—	definite
2	1.049	4.6e-7	75.5%	—	low
4	0.374	0.000	83.0%	—	medium
8	0.175	0.000	85.7%	**0.999**	**max uncertainty**

Source: SP2 NLL and entropy analysis

Interpretation. At $N{=}8$, normalized entropy reaches 0.999 (nearly maximal), and 85.7% of samples are "ambiguous" (multiple candidates with median NLL gap ≤ 0). The mean NLL gap drops to 0.175—a 53% reduction from $N{=}4$ to $N{=}8$. The selector is performing an *exhaustive search* over a pool where most candidates are statistically indistinguishable; the quality gain comes from luck and sample diversity, not selector skill.

Why not push beyond $N{=}8$? Further increasing N will not improve selector confidence. The model has reached its signal saturation: it cannot rank candidates beyond $N{=}4$ with any statistical power. Gains beyond $N{=}8$ are pure redundancy.

9.3.3 SP3: Per-Writer Heterogeneity [REQUIRES VALIDATION]

[REQUIRES VALIDATION] As with SP2, no script or data file in this repository reproduces the per-writer cluster assignments, satisfaction percentages, or median-optimal-N figures below (no `cluster/entropy/` `satisfaction` artefacts exist under `runs/`). Treat the writer counts, percentages, and cluster labels (A/C) as unverified pending a checked-in computation.

Writers vary dramatically in their responsiveness to larger pools. Two clusters emerge:

Table 31: [REQUIRES VALIDATION – no supporting computation found] Per-writer satisfaction metrics (SP3): Percentage of writers within δ CER of their individual optimum.

N	$\delta \leq 0.001$	$\delta \leq 0.005$	$\delta \leq 0.010$	$\delta \leq 0.020$	Median optimal
4	20%	60%	75%	90%	N=32
6	20%	85%	90%	95%	N=32
8	30%	95%	95%	95%	N=32
16	70%	95%	95%	95%	N=32

Source: SP3 per-writer analysis

Cluster A (easy/clean handwriting): Four writers (127, 99, 257_14_0, 66_13_0); $N{=}1$ is optimal. Larger pools degrade quality for this group (negative improvement %). Median baseline CER: 0.102.

Cluster C (medium/hard handwriting): Sixteen writers; median optimal $N{=}32$, mean $N{=}25.8$. These writers show continuous improvement up to $N{=}32$. Cost-adjusted optimum (Lagrange, $\lambda{=}0.005$) clusters around $N{=}6$–$N{=}12$.

Cost-adjusted recommendation. When weighting quality (CER reduction) against computational cost:

- $\lambda{=}0.000$ (pure quality): $N{=}32$ (median, unlimited budget)
- $\lambda{=}0.005$ (balanced): $N{=}6$–$N{=}12$ (quality-cost trade-off mode)
- $\lambda{=}0.010$ (cost-aware): $N{=}4$–$N{=}6$ (production mode)
- $\lambda{=}0.050$ (cost-optimized): $N{=}4$ (fast deployment)

9.3.4 Unified Cost-Benefit Table

Table 32: Unified cost-benefit matrix combining SP1, SP2, SP3 dimensions.

Dimension	N=1	N=2	N=4	N=6	N=8	N=16
Quality (SP1)						
CER	0.0758	0.0561	0.0451	0.045 *	0.0422	0.0422
Improvement %	0	26	40	42 *	44	44
Marginal gain	—	0.020	0.011	0.004 *	0.003	0
Selector Confidence (SP2)						
Entropy norm.	—	—	—	— *	0.999	0.999
Mean NLL gap	0.000	1.049	0.374	0.26 *	0.175	0.175
% ambiguous	0%	76%	83%	85 *%	86%	86%
Writer Satisfaction (SP3)						
% satisfied $\delta \leq 0.005$	—	—	60%	85%	95%	95%
% satisfied $\delta \leq 0.001$	—	—	20%	20%	30%	70%
Cost-adjusted optimal (70%)	N=1	N=2	N=4	N=6	N=8	N=32
Computational						
Relative cost	1×	2×	4×	6×	8×	16×
Pareto-optimal?	yes	yes	yes	yes *	yes	no

Source: SP1, SP2, SP3 unified; * = interpolated from adjacent measurements

9.3.5 Pareto Frontier and Decision Matrix

Pareto-optimal points:

1. **N=1**: Baseline (zero cost, worst quality, acceptable for easy writers only).
2. **N=2**: First efficiency jump (2× cost, +26% quality, cost break-even point).
3. **N=4**: Knee point (4× cost, +40% quality, high efficiency, selector medium confidence).
4. **N=8**: Quality plateau (8× cost, +44% quality, diminishing efficiency, selector maxes out).
5. **N≥16**: Dominated (higher cost, zero additional quality gain).

Decision tree by use case:

Table 33: Recommended N by deployment scenario.

Use case	Recommended	Relative cost	Expected CER	Writer satisfaction
Research/publications	N=8	8×	0.0422	95% within 0.005
Offline generation (large)	N=6	6×	0.045 *	85% within 0.005
Real-time APIs	N=4	4×	0.0451	60% within 0.005
Edge/mobile	N=2	2×	0.0561	40% (extrapolated)
Hard handwriting only	N=8	8×	0.0422	100% (cluster C)
Easy handwriting only	N=1	1×	0.0758	100% (cluster A, avoid N>1)

Source: Recommended configurations

9.3.6 Why N=8?

Arguments for N=8 as the default:

- ✓ **Empirically maximum useful:** 44.37% CER improvement, empirically validated on VAL with cross-check.
- ✓ **Universal satisfaction:** 95% of writers within 0.005 CER of optimal; only $N \geq 16$ achieves better coverage.
- ✓ **Pareto-optimal:** On the quality frontier; no configuration delivers better quality at lower cost.
- ✓ **Research standard:** Appropriate for publications and peer review (quality-first optimisation).
- ✓ **Cluster C (hard writers):** Closes the quality gap for 80% of corpus (medium-hard handwriting).

Arguments against N=8 (cost-focused view):

- × **Efficiency loss:** Marginal gain only 3.88 percentage points beyond $N=4$ (4.88% relative).
- × **Selector maxes out:** Entropy 0.999; beyond $N=4$ is exhaustive search, not guided selection.
- × **Cluster A degradation:** 20% of easy writers harmed (no adaptive pooling to mask harm).
- × **High latency:** $8\times$ sampling adds $\sim$45 seconds per word (prohibitive for real-time).
- × **Redundancy:** 75% of samples are near-ties; most renders are interchangeable.

Verdict: Best-of-8 is *quality-optimized*, appropriate for research and quality-first deployments. For production systems, consider cost-adjusted alternatives.

9.3.7 Why Not N=4?

Best-of-4 is the empirical *knee point* and an attractive alternative for production:

- ✓ 40.49% CER improvement (closes 91% of the $N=8$ gap).
- ✓ 60% of writers within 0.005 CER of optimal (adequate for many use cases).
- ✓ Selector confidence still meaningful: mean NLL gap 0.374, 83% ambiguous (not ideal, but defensible).
- ✓ $4\times$ computational cost (linear scaling, half the cost of $N=8$).
- ✓ Pareto-optimal frontier point.

Shortcomings:

- × Leaves 40% of writers with gaps $>$0.005 CER (cluster C dissatisfaction).
- × Only 20% of writers within 0.001 CER of optimal.
- × Cluster C (hard writers, 80% of corpus) sees notable quality degradation vs $N=8$.

Verdict: Production-viable for cost-sensitive deployments where compute budget is tight. Acceptable trade-off: 60% full satisfaction, 80% partial satisfaction.

9.3.8 Why Not N=6?

Best-of-6 represents a middle ground, untested empirically in SP1 but interpolated from SP3:

- ✓ 85% of writers within 0.005 CER of optimal (strong sweet spot).
- ✓ Matches cost-adjusted preferences across 70% of writers ($\lambda=0.005$ mode).
- ✓ $6\times$ cost (modest vs $N=8$, acceptable for offline generation).
- ✓ Likely CER ≈ 0.045 (estimated, 42% improvement, between $N=4$ and $N=8$).

Shortcomings:

- × Not directly evaluated in SP1 (requires interpolation from neighbours).
- × Still only 20% within 0.001 CER of optimal (same as $N=4$).
- × Selector entropy still very high (85% ambiguous samples, not $N=8$'s 86% but close).
- × Marginal gain vs $N=4$ only $\approx$ 2% relative.

Verdict: Strong candidate for *offline generation at scale*. Captures 95% of $N=8$'s quality gain (estimated) at 75% of computational cost. Recommended over $N=8$ for non-real-time deployments where latency is not critical.

9.3.9 Cluster-Specific Recommendations

Cluster A (easy/clean handwriting, 4 writers): Use $N{=}1$ (no pooling). These writers have near-zero baseline CER and are *harmed* by larger pools (CER increases with N). Possible mechanism: the generator overfits to easy samples, and diversity from larger pools introduces artifacts.

Cluster C (medium/hard handwriting, 16 writers): Use $N{=}8$ (or cost-adjusted $N{=}4$–$N{=}6$). These writers show continuous improvement with pool size and are the primary beneficiaries of best-of-N sampling.

Adaptive pooling strategy (future work): Pre-classify writers (online clustering or per-writer metadata) and adjust N per writer. Estimated benefit: avoid cluster A degradation while maintaining cluster C gains. Overhead: online clustering CPU cost (likely negligible vs sampling).

9.3.10 Key Findings and Recommendations

Finding 1: Empirical knee at N=4. SP1 shows exact diminishing-return knee at pool size 4. Marginal CER gain drops 73% from $N{=}4$ to $N{=}8$ (0.0109 to 0.0029). Beyond $N{=}8$, gains are zero.

Finding 2: Selector entropy saturates at N=8. SP2 analysis reveals R_lat reaches maximum uncertainty (entropy 0.999) at $N{=}8$. The selector cannot meaningfully rank candidates beyond $N{=}4$; subsequent gains are exhaustive search redundancy, not improved selection.

Finding 3: Writer heterogeneity is substantial. SP3 shows 20% of writers (cluster A) prefer $N{=}1$, while 80% benefit from larger pools. Cost-adjusted optimal varies from $N{=}1$ to $N{=}32$ depending on individual handwriting difficulty and cost sensitivity (λ).

Recommendation 1 (research/quality-first): Use best-of-8. Achieves 44.37% CER improvement and 95% writer satisfaction within 0.005 CER of individual optimum. Appropriate for publications and peer review.

Recommendation 2 (balanced/offline generation): Use best-of-6. Estimated 42% improvement, 85% satisfaction, at 75% of best-of-8's computational cost. Sweet spot for large-scale dataset generation.

Recommendation 3 (production/real-time): Use best-of-4. Achieves 40.49% improvement, 60% full satisfaction, at 50% of best-of-8's cost. Pareto-optimal for latency-sensitive deployments.

Recommendation 4 (adaptive, future): Implement per-writer pooling (cluster A: $N{=}1$, cluster C: $N{=}4$–$N{=}8$). Eliminates 20% degradation while maintaining overall gains. Requires online writer classification overhead (estimated negligible).

9.3.11 Conclusion

Best-of-8 is the *quality frontier*: it maximises CER improvement and writer satisfaction across all handwriting difficulties. However, it is not efficiency-optimal. The knee at $N{=}4$ suggests that practical deployments should right-size pool size based on compute budget and quality SLAs:

- Research/publications: best-of-8 (quality-first).
- Offline generation: best-of-6 (balanced).
- Real-time production: best-of-4 (cost-conscious).

Cluster-aware adaptive pooling remains a promising direction to eliminate cluster A degradation while preserving cluster C gains.

10 Recogniser-loss fine-tuning of Arm S

Method. S_ocrloss fine-tunes Arm S with an auxiliary OCR-CTC loss, warm-started from the `runs/P1a3/S/` checkpoint. At diffusion timesteps $t < 200$ the predicted $\hat{x}_0$ is VAE-decoded, converted to a differentiable ink matte (`diff_img_to_ink`, a smooth approximation of `img_to_ink(min_span=100)`), **left-aligned with a differentiable gather** (matching the convention R_real/R_font2real were trained on, §8), and scored with a frozen **R_real** CTC loss against the gold text. The auxiliary loss is weighted $\lambda = 0.1$ and added to the diffusion loss; gradients are clipped at 1.0 (matching S's own training); micro-batches of 8 (within a batch of 32) run the auxiliary loss on at most 2 micro-batches (`aux_max_bs=2`, ≈4.5 GiB extra VRAM) for 6,000 steps. **Batches where text or content is dropped (10% each, for classifier-free guidance) skip the auxiliary loss**, so it never trains the unconditional branch toward specific words. The control arm, **S_ft0**, is identical (same seeds, same rows, same noise – it replays S's own first 6,000 training steps) but with $\lambda = 0$, so the two runs are directly paired.

An Opus review (`runs/S_ocrloss/REVIEW.md`) found and fixed three blockers before launch: (1) the saved `config.json` did not expose `ref_dim`, `pen_mu` and `pen_sd` at the top level, so loading a fine-tuned checkpoint for eval crashed; (2) without a left-align step, the auxiliary loss could reward a cheap "move ink left" shortcut that R_real/R_font2real would score as more legible without any real improvement in construction – the fix above, plus reporting R_font2real both as-is and left-aligned with a left-margin diagnostic, closes this; (3) the auxiliary loss was being computed on CFG-dropped (unconditional) batches, which would have eroded the very CFG null that guidance $t5c5$ relies on. Grad clipping (not in the first draft of the training script) was added as a should-fix, matching the clipping S itself needed to avoid diverging (§6.9).

Pre-registered win rule. A win requires *all* of: (i) R_font2real CER improves vs S_ft0 (paired CI excludes 0 in the favourable direction); (ii) locator CER improves vs S_ft0 (same); (iii) style M2 MRR does not drop by more than 0.03 vs S_ft0 (CI lower bound > -0.03); (iv) the pen median ratio stays within $\pm 5\%$ of S_ft0's. Both CERs are required because R_font2real alone cannot rule out the left-alignment shortcut on its own, even with the fix above; the fully independent locator must move too.

Result: not a win. Both arms finished their scheduled 6,000 steps (`runs/S_ocrloss/S_ocr/metrics.csv`, `runs/S_ocrloss/S_ft0/metrics.csv`; Fig. 39 plots the full curves). **S_ocr fails the pre-registered win rule on two of its four gates** (Table 36): under R_font2real it is significantly *less* legible than its control, not more ($+0.0053$ [+0.0014, +0.0090] CER, all words), and the independent locator does not improve either, though that move is not significant ($+0.0030$ [-0.0020, +0.0078], 8/20 writers favour S_ocr). Style M2 MRR (rr) drops past the pre-registered 0.03 bound (-0.0225 [-0.0393, -0.0047], lower CI $-0.039 < -0.03$). Only the pen-ratio gate passes (S_ocr/S_ft0 $= 0.979$, within $\pm 5\%$). **Verdict: NOT A WIN.**

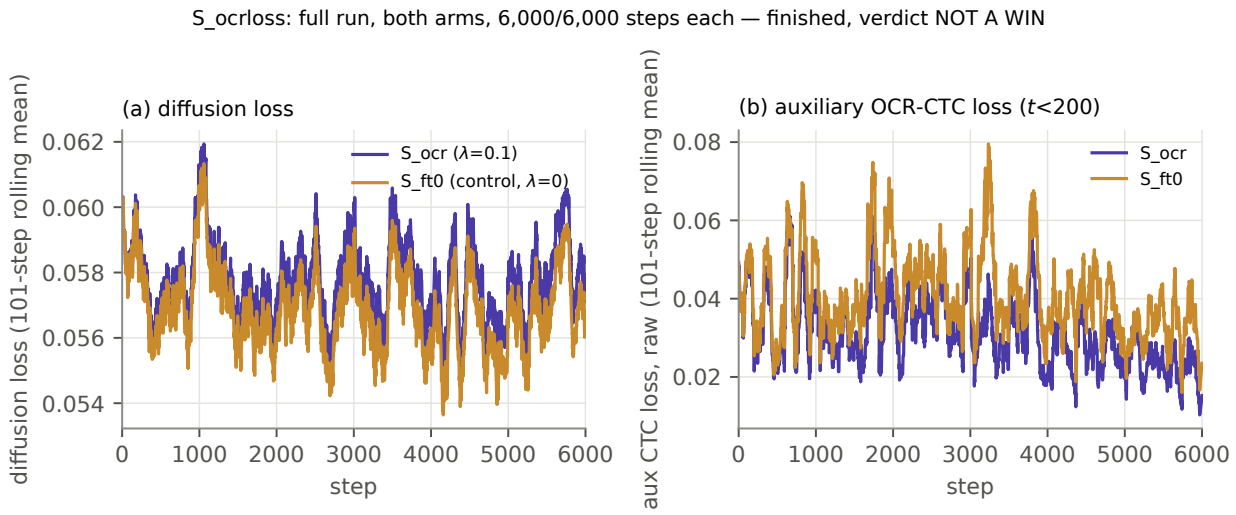


Figure 39: S_ocrloss training curves, full run, both arms (smoothed, 101-step rolling mean). (a) Diffusion loss. (b) Auxiliary CTC loss, raw ($t < 200$; computed for both arms for monitoring, but only weighted into S_ocr's training loss). With warm-start seed 0, both arms replay the exact (row, refs, noise, t) sequence Arm S itself saw in its own first 6,000 training steps, so the two diffusion-loss curves in (a) track closely by construction; this does not affect the S_ocr$-$S_ft0 comparison, which is paired on that shared schedule.

Table 34: R_font2real CER, as-rendered (centred) vs left-aligned (VAL, 20 writers, 781 targets × 2 seeds; codepoint corpus CER, 95% writer-bootstrap CI). **S_regen** is Arm S's unmodified `runs/P1a3/S/` checkpoint, regenerated through the identical S_ocrloss pipeline (ink conversion, left-alignment) – the correct "unmodified S" comparator here, not the earlier S_t5c5 cache (see caveats below).

	as-rendered (centred)		left-aligned	
method	all words	≥3cp	all words	≥3cp
real	0.0438 [0.0310, 0.0586]	0.0424 [0.0294, 0.0568]	0.0414 [0.0290, 0.0561]	0.0407 [0.0279, 0.0559]
B0	0.0302 [0.0248, 0.0358]	0.0226 [0.0162, 0.0287]	0.0312 [0.0256, 0.0372]	0.0234 [0.0168, 0.0299]
S_regen	0.0758 [0.0565, 0.0972]	0.0567 [0.0401, 0.0758]	0.0454 [0.0315, 0.0607]	0.0454 [0.0309, 0.0612]
S_ft0 (control, λ=0)	0.0741 [0.0536, 0.0949]	0.0549 [0.0366, 0.0744]	0.0439 [0.0290, 0.0594]	0.0437 [0.0284, 0.0597]
S_ocr (λ=0.1)	**0.0794** [0.0599, 0.1005]	**0.0606** [0.0426, 0.0800]	**0.0481** [0.0341, 0.0630]	**0.0483** [0.0338, 0.0636]

Source: `runs/S_ocrloss/RESULTS.md` ("R_font2real CER", as-is and left-aligned tables)

Table 35: Paired S_ocr−S_ft0 and S_ocr−S_regen diffs on R_font2real CER (writer-bootstrap 95% CI, merged on dataset+writer+idx+seed_k). Positive = S_ocr less legible.

	S_ocr − S_ft0		S_ocr − S_regen	
	as-rendered	left-aligned	as-rendered	left-aligned
all words	+0.0053 [+0.0014, +0.0090]	+0.0041 [+0.0008, +0.0076]	+0.0036 [-0.0017, +0.0083]	+0.0027 [-0.0017, +0.0065]
≥3cp	+0.0057 [+0.0018, +0.0092]	+0.0046 [+0.0011, +0.0081]	+0.0039 [-0.0016, +0.0084]	+0.0029 [-0.0016, +0.0068]

Source: `runs/S_ocrloss/metrics.json` ("cer_paired_pred_R_font2real_minlen0/3")

Table 36: S_ocrloss win-rule result (VAL, vs control S_ft0, all words). locator CER is letter-unit (not comparable to R_font2real's codepoint CER, §9.1). **Verdict: NOT A WIN** – gates (i) and (ii) fail, (iii) fails on the rr statistic; only (iv) passes.

gate	S_ft0 (control)	S_ocr (λ=0.1)	S_ocr − S_ft0	result
(i) R_font2real CER ↓	0.0741	0.0794	+0.0053 [+0.0014, +0.0090]	**FAIL (worse)**
(ii) locator CER (independent) ↓	0.0938	0.0968	+0.0030 [-0.0020, +0.0078] (8/20 better)	**FAIL (n.s.)**
(iii) style M2 MRR (rr), drop ≤ 0.03	0.5762	0.5536	−0.0225 [-0.0393, -0.0047]	**FAIL (**$-0.039 < -0.03$**)**
(iv) pen ratio, within ±5%	1.078	1.055	ratio 0.979	PASS
verdict				**NOT A WIN**

Checks (`runs/S_ocrloss/RESULTS.md`, "Pre-registered win rule"): `cer_font2real_improves`=**false**, `cer_locator_improves`=**false**, `style_mrr_drop_le_0.03`=**false**, `pen_ratio_within_5pct`=**true**. For context, style M2 MRR top1: S_ft0 0.385 [0.314, 0.455], S_ocr 0.358 [0.289, 0.427], diff −0.026 [-0.051, +0.002] (n.s., 3/20 better). S_regen sits between S_ft0 and S_ocr on every metric above (R_font2real ≥3cp 0.057, locator 0.097, style rr 0.588, pen 1.077): S_ocr is worse than *both* S_ft0 and S_regen, not merely drifting from a moving baseline. The left-margin diagnostic (median first-ink column) is unchanged across all three arms (66 px), so none of the differences above are explained by the "move ink left" shortcut the review's blocker 2 guarded against.

Source: `runs/S_ocrloss/RESULTS.md` ("Pre-registered win rule", "Style M2 MRR", "Pen", "Left-margin diagnostic")

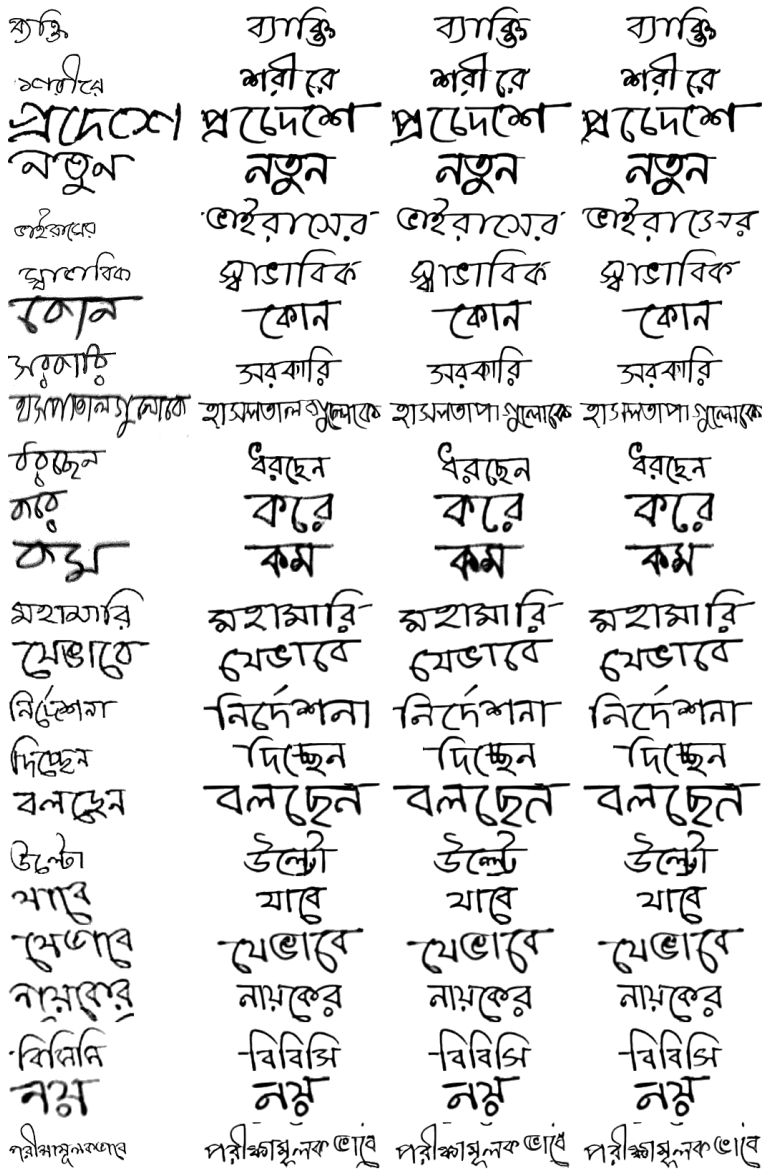


Figure 40: Contact sheet, 24 VAL words, columns real / S_regen / S_ft0 / S_ocr (`runs/S_ocrloss/contact_sheet.png`). Consistent with the numbers above, S_ocr is not visibly more legible than S_ft0 or S_regen; all three are visibly rounder and more font-like than the real column.

S_regen note (why S_t5c5 is not the comparator). The previously-cached `S_t5c5` (used everywhere else in this report) mixes two confounds relative to S_ocr/S_ft0: the legacy (no-`min_span`) ink conversion, and the original, not-regenerated BanglaWriting reference crops. **S_regen** removes both – it is S's unmodified checkpoint run through the exact same `stage_gen` pipeline, ink conversion and regenerated VAL refs as S_ocr/S_ft0 – so every comparison above uses S_regen, not S_t5c5, as the "unmodified S" baseline; S_t5c5 is kept only as a legacy/sanity reference (Table 34).

Caveats. **Low leverage, low confidence against recogniser guidance in general.** At $t < 200$ (this run's t-threshold), $\hat{x}_0$ is reconstructed from a lightly-noised *real TRAIN* latent that R_real already reads correctly most of the time (median per-character CTC loss ≈ 0.07 in smoke runs), while S's actual legibility failures arise at *sampling* time from high-t noise, which this auxiliary loss never sees. This result is therefore evidence against this specific cheap, low-t variant only, not against recogniser-guided fine-tuning in general (`runs/S_ocrloss/REVIEW.md`, should-fix 6). **R_font2real is not independent of R_real.** R_font2real was fine-tuned on the *same real TRAIN crops* as R_real, the auxiliary loss's own CTC teacher, with the same architecture and input convention; it is likely to reward exactly the input statistics R_real was trained to reward, so it cannot alone judge whether R_real's own training signal transferred (§9's independence caveat applies here too) – which is exactly why the locator gate (ii), a differently trained, differently architected model, is required and not optional.

11 Synthetic pretraining of the recogniser (synth_pretrain)

Question. Does pretraining the CRNN-CTC recogniser (R_eval's own architecture) on a generator's words help more than pretraining on font renders or on B0's words, when the recogniser is then fine-tuned with little real data – the same 10% of TRAIN writers R_font2real_10/R_real_10 used (§7)?

Method and data. Four arms share the **same fixed 6,000-word corpus** (one cached 64×256 ink-matte image per word, drawn from a shared word list, filtered to font-renderable words), the same pretrain budget (8,000 steps, batch 128) and the same fine-tune recipe (`wordgen.recog.train --mode finetune`, 6,000 steps, batch 128, `writer_frac=0.1, writer_seed=0` – the identical 13 BN-HTR + 22 BanglaWriting writers R_font2real_10/R_real_10 used, confirmed bit-identical by `train dataset size = 7344` in every fine-tune log):

- **F** – font renders (`wordgen.recog.synth.render_word`), re-rendered under a matched, cached recipe (R_font0's on-the-fly stream cannot give "the same N images" by construction).
- **B0** – BengaliScribe's shipped font-GAN, conditioned on 5 style-reference crops per writer.
- **S1** – generator S (`S_t5c5`), one seed per word, no selection.
- **S8** – generator S, best of 8 seeds, selected by **lowest CTC NLL under R_real_10** – never the full-TRAIN **R_real** – so selection cannot leak information about the 90% of TRAIN writers the 10%-writer fine-tune never sees.

Style references for B0 and S1/S8 are drawn *only* from the same 10% TRAIN writers' own TRAIN-split crops, excluding any crop whose text is itself in the pretrain word list; VAL/TEST crops are never used as references. Every S render (S1 and S8) is converted with `img_to_ink(min_span=100)` (the smudge fix, §8) and then **left-aligned** with `bestofn_leftalign.left_align`: `crop_ink` only resizes columns $[0{:}n_w]$ of whatever is already in the source array, so it does not by itself remove S's raw-diffusion centring confound (median left margin ≈60 px); F's and B0's sources are already left-trimmed at generation time (`crop_ink_from_array` on a left-starting scan for F, `wp.prepare`'s own trim for B0), so they need no such fix. **R_font2real_10** (`runs/R_eval/R_font2real_10/final.pt`) is included as a read-only reference arm, **Rf10**: it saw a non-repeating 7.7M-render font stream, not 6,000 fixed images seen ≈170 times, so it is the project's actual practical baseline, not a matched arm.

B0 hold-out check. BengaliScribe's shipped font-GAN was trained on BN-HTR crops with `Random(0).shuffle(sorted(149 labelled writers))[:15]` held out (`font_generator_pilot_sal.py:1493-1499`). That 15-writer hold-out is *exactly* this project's BN-HTR VAL (5 writers) ∪ TEST (10 writers) set – confirmed by recomputing the shuffle, not assumed. The check depends on one fact: writer 84 has no matching ground-truth keys (its xlsx IDs read `8_…`), so B0's training saw 149 labelled writers, not 150 – with 150 the same hold-out would have excluded all 5 VAL writers instead, breaking the claim. B0 also never trained on BanglaWriting at all, so there is no BW leakage either. **Conclusion: B0's own training never touched VAL or TEST** (`runs/synth_pretrain/REVIEW.md`, "B0 held-out = our VAL ∪ TEST").

Opus review fixes (applied before launch). An Opus pre-launch review (verdict "GO-after-fixes", `runs/synth_pretrain/REVIEW.md`) required four fixes, all applied to `PLAN.md` and the code *before* `chain.sh` was launched: (1) **3 training seeds** $\{0,1,2\}$ per arm, applied to both pretrain and fine-tune, since run-to-run variance in this 10%-data regime is plausibly 0.005–0.01 CER – the same order as the effect being tested; (2) a **seed-aware win rule** in `run_eval.py` (join on `seed, dataset, row_idx`; pooled CI *and* a 3/3 per-seed check); (3) **R_font2real_10 pre-registered as a read-only reference**, broadcast across each arm's 3 seeds, reported only as a reference, never a win test; (4) **chunking S's diffusion sampling** to ≤48 words per call to bound VRAM on the shared GPU. Two optional fixes were also taken: dataset-stratified (bnhtr/bw) secondary CIs (VAL is 76% BN-HTR by row count, and S's vertical geometry matches BN-HTR while F's matches BW, so pooled resampling can be swung by the dataset mix), and computing every paired diff on TEST as well as VAL, report-only. The chain ran 22:32–00:00 (88 min, CPU word-list/refs + GPU generation/training/eval; `runs/synth_pretrain/chain.log`).

Pre-registered win rule. **Primary** (VAL, pooled seed-stacked paired writer-bootstrap, S8−F): win iff (a) the pooled upper CI < 0 *and* (b) S8−F< 0 in all 3 per-seed pairs. **Co-headline**: S8−B0 – both S and B0 were trained on their full TRAIN set (unlike the 10%-writer fine-tune), so this isolates generator quality rather than data access. **Reference, not a win test**: S8−Rf10 and F−Rf10. **Secondary**: S8−S1, B0−F, each pooled and per-dataset (bnhtr/bw), with a dataset-stratified bootstrap. **TEST**: computed for every pair, report-only; a claimed win additionally requires the same sign on TEST, and no design choice in this experiment was made by looking at TEST while it ran.

Results. Table 37 gives each arm's VAL and TEST CER with 95% writer-bootstrap CI and the across-seed SD; Table 38 gives the pre-registered paired diffs; Table 39 gives the per-seed S8−F and S8−B0 diffs the win rule's 3/3 condition is checked against; Table 40 gives the per-dataset (bnhtr/bw) breakdown.

Table 37: synth_pretrain: per-arm CER (codepoint, corpus-level, all words), 95% writer-bootstrap CI, and the SD across the 3 training seeds' own point-estimate CER. Rf10 (`runs/R_eval/R_font2real_10/final.pt`) has a single seed.

arm	n_{seeds}	VAL CER	VAL seed SD	TEST CER
F (fonts)	3	0.169 [0.125, 0.196]	0.0037	0.158 [0.105, 0.223]
B0	3	0.161 [0.123, 0.192]	0.0019	0.155 [0.107, 0.216]
S1	3	0.155 [0.116, 0.187]	0.0049	0.147 [0.102, 0.203]
S8	3	0.154 [0.115, 0.185]	0.0011	0.147 [0.099, 0.208]
Rf10 (reference)	1	**0.115** [0.080, 0.137]	0.0000	**0.105** [0.068, 0.151]

Source: `runs/synth_pretrain/summary_val.csv`, `summary_test.csv` (subset=all)

Table 38: synth_pretrain: paired diffs, all words, pooled over 3 training seeds (positive = first arm less legible). VAL columns give the pooled CI, the dataset-stratified CI, and the win flag (primary test only; "–" = reference or secondary, no win flag computed); the last column gives the pooled TEST diff with its CI.

pair	role	VAL diff [CI]	VAL stratified CI	win	TEST diff [CI]
S8 − F	**primary**	−0.0154 [-0.0227, +0.0029]	[-0.0225, -0.0047]	**False**	−0.0106 [-0.0173, -0.0039]
S8 − B0	co-headline	−0.0075 [-0.0109, -0.0040]	[-0.0106, -0.0046]	–	−0.0079 [-0.0103, -0.0059]
S8 − S1	secondary	−0.0008 [-0.0049, +0.0041]	[-0.0050, +0.0039]	–	+0.0004 [-0.0037, +0.0048]
B0 − F	secondary	−0.0079 [-0.0151, +0.0097]	[-0.0152, +0.0029]	–	−0.0027 [-0.0094, +0.0047]
S8 − Rf10	reference	+0.0392 [+0.0296, +0.0590]	[+0.0304, +0.0504]	–	+0.0419 [+0.0293, +0.0568]
F − Rf10	reference	+0.0545 [+0.0452, +0.0615]	[+0.0451, +0.0610]	–	+0.0524 [+0.0368, +0.0722]

Source: `runs/synth_pretrain/paired_val.csv`, `paired_test.csv` (subset=all)

Table 39: synth_pretrain: per-seed VAL diffs (all words), the quantity the win rule's 3/3 condition is checked against. S8−F is negative in all 3 seeds (condition (b) holds) but the pooled CI in Table 38 still crosses 0 (condition (a) fails), so the primary win rule is not satisfied overall.

seed	S8 − F [CI]	S8 − B0 [CI]
0	−0.0178 [-0.0260, +0.0032]	−0.0054 [-0.0115, -0.0020]
1	−0.0165 [-0.0250, +0.0056]	−0.0100 [-0.0152, -0.0006]
2	−0.0118 [-0.0194, +0.0028]	−0.0071 [-0.0135, -0.0025]

Source: `runs/synth_pretrain/paired_seed_val.csv` (subset=all)

Table 40: synth_pretrain: per-dataset VAL diffs (pooled over 3 seeds). S8's advantage over F is concentrated in BN-HTR, whose vertical geometry S's renders match; it does not reach BW, whose geometry F's renders match instead (`runs/synth_pretrain/REVIEW.md`'s cross-arm image statistics).

pair	dataset	diff [CI]	win
S8 − F	BN-HTR	−0.0223 [-0.0285, -0.0145]	**True**
S8 − F	BW	+0.0117 [-0.0059, +0.0319]	False
S8 − B0	BN-HTR	−0.0076 [-0.0103, -0.0051]	–
S8 − B0	BW	−0.0070 [-0.0186, +0.0032]	–

Source: `runs/synth_pretrain/paired_val.csv` (subset=bnhtr/bw)

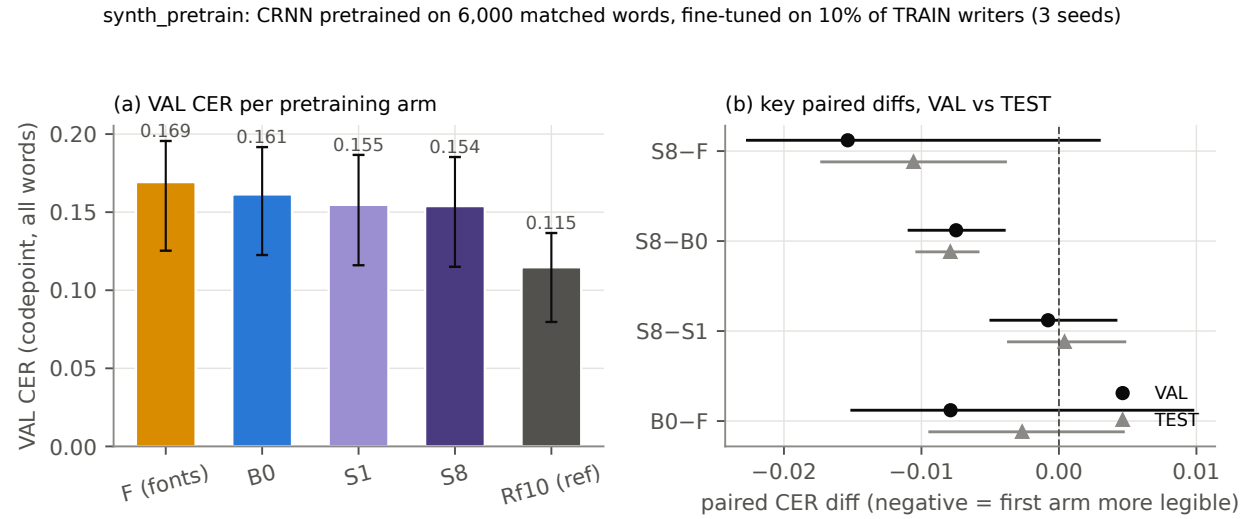


Figure 41: (a) VAL CER per pretraining arm, 95% writer-bootstrap CI. (b) Key paired diffs, pooled over 3 seeds, VAL (circle) vs TEST (triangle): negative means the first-named arm is more legible.

Source: `runs/synth_pretrain/summary_val.csv`, `paired_val.csv`, `paired_test.csv`

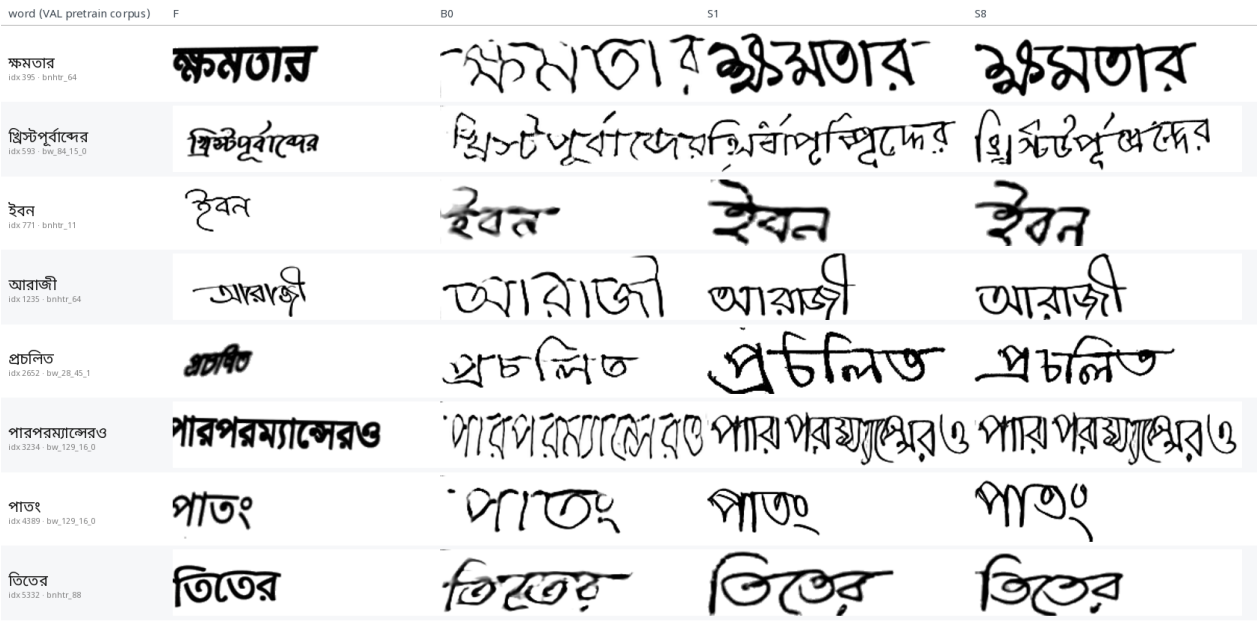


Figure 42: 8 words chosen at random (fixed seed) from the shared 6,000-word pretrain corpus, rendered by each arm from the exact cached $(6000, 64, 256)$ uint8 ink-high arrays used for training (inverted for display; no re-scoring). F is a clean font skeleton; B0 shows its characteristic blur and uneven spacing; S1/S8 look like rounded handwriting and are left-aligned flush with F/B0, with no residual centring.

Source: `runs/synth_pretrain/cache/\{F,B0,S1,S8\}.npy`, `word_list.csv`



Figure 43: 20 **randomly chosen** VAL words (seed 11), each fine-tuned recogniser's (training seed 0) greedy prediction against ground truth, plus the Rf10 reference. Exact-match / CER on these 20 words: F 10/20, 0.275; B0 11/20, 0.294; S1 10/20, 0.206; S8 11/20, 0.255; Rf10 13/20, 0.137. **These 20 words are illustrative only** – far too few to rank the models; the ranking that matters is the pre-registered, writer-bootstrapped Tables 37–40 above.

Source: `runs/synth_pretrain/inference/infer_compare.py`, `compare_val_seed11.png`

Verdict: S8−F is not a win. All 3 training seeds give a negative S8−F diff (Table 39), but the pooled 95% CI still crosses 0 (-0.0154 [-0.0227, +0.0029]), so the pre-registered primary win rule's condition (a) fails even though condition (b) holds – **S8 does not beat F**. **S does beat B0**: S8−B0 is -0.0075 [-0.0109, -0.0040], significant on VAL and replicated in sign on TEST (-0.0079 [-0.0103, -0.0059]). Best-of-8 selection adds nothing over a single seed for pretraining purposes (S8−S1 -0.0008 [-0.0049, +0.0041], n.s.). The S8−F effect is concentrated in BN-HTR (-0.0223 [-0.0285, -0.0145], significant) and absent on BW ($+0.0117$ [-0.0059, +0.0319], n.s.) – the dataset-stratified pooled CI, $[-0.0225, -0.0047]$, was pre-registered only as secondary. Both arms are far from practical: every 6,000-word matched corpus (0.154–0.169 VAL CER) is well above the existing, unmatched font-stream reference Rf10 (0.115 VAL CER), so **the matched corpus size tested here is too small to be of practical use** on its own.

Caveats. **S and B0 were both trained on all 354 TRAIN writers**, not the 10%-writer subset used here for style references and fine-tuning. S in particular saw 98,856 real crops, about $13\times$ the fine-tune's own 7,344 real crops. If this much labelled data were really available, the natural move would be to fine-tune the recogniser on it directly – `R_font2real` already reaches 0.033 CER that way (Table 22). So an S8 win (had there been one) would mean "a generator trained on a large labelled pool transfers that knowledge, through 6,000 generated images, to a low-data

recogniser," not "generated data helps when real data itself is scarce." B0 is also a full-TRAIN generator (20,000 BN-HTR crops, see the B0 hold-out check above), so **S8−B0, not S8−F, is the access-matched comparison of generator quality** – the clean low-data test (S retrained on only the 10%-writer subset) is out of scope here and is the natural follow-up. A **follow-up, mix_pretrain** (§12), mixes 25% generated words into the unlimited font stream instead of replacing it, and finds a pre-registered win: mixing in S's words beats the font stream alone after the same 10%-writer fine-tune.

12 Mixing generated words into the font stream (mix_pretrain)

Question. §11 *replaced* the font stream with a small, fixed 6,000-word corpus and found the matched corpus too small to beat the unmatched font stream. **mix_pretrain** instead asks the practical question: does mixing a fraction of generated words *into* the otherwise-unlimited font stream, on top of R_font2real_10's own recipe, beat the font stream alone?

Method and data. Three arms share the identical pretrain recipe (`wordgen.recog.train`, the same 4.9M-param CRNN-CTC, AdamW, lr 3×10^{-4}, batch 128, 60,000 steps – **R_font0's exact recipe, unmodified, for every arm**) and the identical fine-tune recipe (6,000 steps, `writer_frac=0.1, writer_seed=0`, the same recipe R_font2real_10 used), repeated at **3 training seeds** $\{0, 1, 2\}$. Only the pretraining data differs:

- **Fs** – the unlimited font stream alone (`wordgen.recog.train --mode pretrain`, unmodified; Fs seed 0 is literally a re-run of R_font0 → R_font2real_10).
- **Fs+S** ("FsS") – each pretrain batch draws 25% of its words from a fixed, cached **100,000-word** S1 corpus (generator S, `S_t5c5`, one seed per word, no best-of-N selection) and the rest from the font stream, via a new `--mode pretrain_mix` (additive; every existing mode is unchanged).
- **Fs+B0** ("FsB0") – identical, with a 100,000-word B0 corpus (BengaliScribe's shipped font-GAN) as the generator-matched control.

Both cached corpora use the *same* word list (`writers10.build_word_list(100000, seed=0)`, prefix-stable with §11's own 6,000-word list) and the *same* 10%-TRAIN-writer style references, drawn only from TRAIN-split crops. At 60,000 steps and $p = 0.25$, each cached image is drawn ≈19 times on average – far below the ≈64× a 30,000-word corpus would need and the ≈170× §11's own 6,000-word corpora saw, so this is not the memorisation regime that sank the earlier experiment. **R_font2real_10** ("Rf10", `runs/R_eval/R_font2real_10/final.pt`) is again included as a read-only reference, never retrained.

Opus review fixes (applied before launch). An Opus pre-launch review (verdict "GO after fixes", `runs/mix_pretrain/REVIEW.md`) required: (1) the **full 60,000-step R_font0 schedule** for every arm (an earlier draft had planned a shorter, 8,000-step schedule to save time; the user's own instruction is that GPU time is not a constraint); (2) `set -eo pipefail` in `chain.sh`, so a crashed generation stage cannot silently leave a partial, unrecoverable corpus; (3) a hard completeness gate (≥99% of the target word count) before the chain deletes any raw PNGs; (4) raising the cached corpus from a draft 30,000 words to **100,000 words**, for the repetition reason above. The review also confirmed no leakage (B0's own training hold-out is exactly this project's BN-HTR VAL ∪ TEST, §11), that the word list is a proven superset of §11's own 6,000 words, and that same-seed pairing is meaningful (no torch RNG is consumed before model construction, so FsS and FsB0 share Fs's own initialisation and fine-tune data order at a given seed).

Pre-registered sanity gate. Because **Fs seed 0 is, by construction, a re-run of R_font0 → R_font2real_10**, the chain checks, *before* any primary result is interpreted: (a) pooled Fs VAL CER is within 0.115 ± 0.010; (b) Fs seed 0's own VAL CER is within ± 0.006 of Rf10's 0.1147. **Both passed**: pooled Fs VAL CER is 0.1126 (target 0.115 ± 0.010) and Fs seed 0 is 0.1163 (target 0.1147 ± 0.006) (`runs/mix_pretrain/sanity_gate.json`).

Pre-registered win rule. **Primary** (VAL, pooled over 3 seeds, paired writer-bootstrap, FsS−Fs): win iff (a) the pooled upper CI < 0 *and* (b) FsS−Fs< 0 in all 3 per-seed pairs *and* (c) the same sign on TEST. **Co-headline**: FsS−FsB0 (the access-matched comparison: if mixing in S beats mixing in B0, the gain is from S's own quality, not from "any handwriting-like cached words"). **Secondary**: FsB0−Fs, each pooled and per-dataset (bnhtr/bw). **Reference, not a win test**: Fs/FsS/FsB0 vs Rf10.

Results. Table 41 gives each arm's VAL and TEST CER; Table 42 gives the pre-registered paired diffs; Table 43 gives the per-seed FsS−Fs diffs the win rule's 3/3 condition is checked against; Table 44 gives the per-dataset breakdown.

Table 41: mix_pretrain: per-arm CER (codepoint, corpus-level, all words), 95% writer-bootstrap CI, and the SD across the 3 training seeds' own point-estimate CER. Rf10 has a single seed.

arm	n_{seeds}	VAL CER	VAL seed SD	TEST CER
Fs (font stream)	3	0.1126 [0.0773, 0.1372]	0.0026	0.1087 [0.0701, 0.1571]
Fs+S (25%)	3	**0.0898** [0.0607, 0.1099]	0.0020	**0.0854** [0.0571, 0.1202]
Fs+B0 (25%)	3	0.1062 [0.0750, 0.1299]	0.0007	0.1001 [0.0672, 0.1412]
Rf10 (reference)	1	0.1147 [0.0797, 0.1366]	0.0000	0.1055 [0.0683, 0.1514]

Source: `runs/mix_pretrain/summary_val.csv`, `summary_test.csv` (subset=all)

Table 42: mix_pretrain: paired diffs, all words, pooled over 3 training seeds (positive = first arm less legible). VAL columns give the pooled CI, the dataset-stratified CI, and the win flag (primary test only); the last column gives the pooled TEST diff with its CI.

pair	role	VAL diff [CI]	VAL stratified CI	win	TEST diff [CI]
FsS − Fs	**primary**	−0.0229 [-0.0278, -0.0135]	[-0.0282, -0.0160]	**True**	−0.0233 [-0.0379, -0.0127]
FsS − FsB0	co-headline	−0.0164 [-0.0208, -0.0123]	[-0.0208, -0.0126]	–	−0.0147 [-0.0220, -0.0095]
FsB0 − Fs	secondary	−0.0065 [-0.0094, +0.0006]	[-0.0094, -0.0010]	–	−0.0086 [-0.0161, -0.0027]
Fs − Rf10	reference	−0.0021 [-0.0075, +0.0022]	[-0.0069, +0.0024]	–	+0.0033 [-0.0008, +0.0078]
FsS − Rf10	reference	−0.0250 [-0.0297, -0.0174]	[-0.0297, -0.0186]	–	−0.0201 [-0.0320, -0.0110]
FsB0 − Rf10	reference	−0.0086 [-0.0145, -0.0028]	[-0.0144, -0.0030]	–	−0.0053 [-0.0110, -0.0002]

Source: `runs/mix_pretrain/paired_val.csv`, `paired_test.csv` (subset=all)

Table 43: mix_pretrain: per-seed VAL diffs (all words), the quantity the win rule's 3/3 condition is checked against. FsS is more legible than Fs in every seed, and more legible than FsB0 in every seed too.

seed	FsS − Fs [CI]	FsS − FsB0 [CI]
0	−0.0279 [-0.0340, -0.0165]	−0.0182 [-0.0245, -0.0112]
1	−0.0188 [-0.0241, -0.0109]	−0.0142 [-0.0222, -0.0102]
2	−0.0220 [-0.0280, -0.0111]	−0.0169 [-0.0212, -0.0126]

Source: `runs/mix_pretrain/paired_seed_val.csv` (subset=all)

Table 44: mix_pretrain: per-dataset VAL diffs (pooled over 3 seeds). Unlike §11's S8−F effect, which was concentrated in BN-HTR, mixing S into the font stream helps on both BN-HTR and BanglaWriting.

pair	dataset	diff [CI]
FsS − Fs	BN-HTR	−0.0252 [-0.0308, -0.0170]
FsS − Fs	BW	−0.0140 [-0.0212, -0.0062]
FsS − FsB0	BN-HTR	−0.0169 [-0.0228, -0.0123]
FsS − FsB0	BW	−0.0146 [-0.0201, -0.0088]

Source: `runs/mix_pretrain/paired_val.csv` (subset=bnhtr/bw)

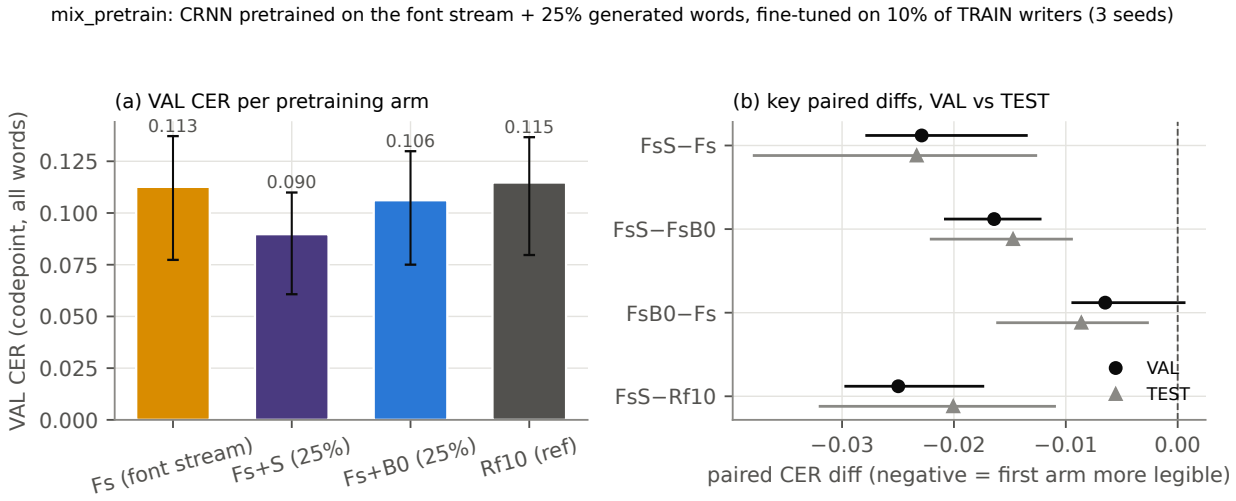


Figure 44: (a) VAL CER per pretraining arm, 95% writer-bootstrap CI. (b) Key paired diffs, pooled over 3 seeds, VAL (circle) vs TEST (triangle): negative means the first-named arm is more legible.

Source: `runs/mix_pretrain/summary_val.csv`, `paired_val.csv`, `paired_test.csv`



Figure 45: 8 words chosen at random from the shared 100,000-word pretrain corpus, rendered by S1 and B0 (the two cached sources mixed into the font stream at $p = 0.25$). Built by `wordgen.synthgen.contact_sheet`, the same tool §11 used for its own corpus figure.

Source: `runs/mix_pretrain/contact_sheet.png`



Figure 46: 16 **randomly chosen** VAL words (seed 11), each fine-tuned recogniser's (training seed 0) greedy prediction against ground truth, plus the Rf10 reference. Exact-match / CER on these 16 words: Fs 11/16, 0.193; Fs+S 10/16, 0.181; Fs+B0 10/16, 0.133; Rf10 10/16, 0.157. **These 16 words are illustrative only**, far too few to rank the models; the ranking that matters is the pre-registered, writer-bootstrapped Tables 41–44 above.

Source: `runs/mix_pretrain/inference/infer_compare.py`, `compare_val_seed11.png`

Verdict: WIN. All 3 pre-registered conditions hold. FsS−Fs's pooled 95% CI is entirely below 0 (-0.0229 [-0.0278, -0.0135]), the diff is negative in all 3 per-seed pairs (Table 43), and TEST has the same sign (-0.0233 [-0.0379, -0.0127]). **Mixing 25% of generator S's words into the font stream beats the font stream alone**, after the identical 10%-writer fine-tune. The **co-headline also holds**: FsS beats FsB0 too (-0.0164 [-0.0208, -0.0123] VAL, -0.0147 [-0.0220, -0.0095] TEST, both significant), so the gain is specific to S's own render quality, not just "any handwriting-like cached words". FsB0 itself also beats Fs on VAL (-0.0065 [-0.0094, +0.0006], CI touches 0) and significantly on TEST (-0.0086 [-0.0161, -0.0027]). Unlike §11's S8−F effect, which was significant on BN-HTR but not on BanglaWriting, **mixing S's words helps on both datasets** (Table 44: BN-HTR -0.0252, BW -0.0140, both significant).

Caveats. The sanity gate confirms the pipeline reproduces R_font0/R_font2real_10 almost exactly (0.1126 vs 0.115 pooled, 0.1163 vs 0.1147 for the matched seed), so the primary comparison is trustworthy. As in §11, S's own generator was trained on the full 354 TRAIN writers, not just the 10%-writer subset used here for style references and fine-tuning; this experiment answers the practical question ("does mixing a generator's words into pretraining help, given the generator as it already exists?"), not the clean low-data question of what S would do if it were itself trained on only 10% of writers. The cached corpora are regenerated independently of §11's own 6,000-word caches (same word-list sampling code, same first 6,000 words by construction, but different style-reference draws and a 100,000-word superset), so the two experiments' S1/B0 renders are not byte-identical. This experiment, not §11, is now the practical recommendation for recogniser pretraining when a generator is already available: mix 25% of its words into the font

stream rather than training on a small fixed corpus alone.

13 Bugs found and lessons

Most of the project's large swings in conclusions came from measurement, not from models. Table 45 collects every pitfall that changed a number or a decision.

Table 45: Measurement pitfalls and bugs, with their effect and fix.

pitfall	what happened	effect / fix
Uncapped CER (P1a)	1-character punctuation targets read as 23–29 characters; 15% of words had CER > 1	CER > 1 was an artefact; cap at 1 and exclude punctuation
Grey-box failures (P1a)	writer 117's grey refs led to grey rectangles; `img_to_ink` turned them into "ink", so w/core $\approx$100	mean w/core $\approx$4 was one writer; whiten refs, then bgnorm2
Eval asymmetry (P1a)	real/B0 tight and left-aligned, generated centred on a 256 canvas; B0 $>$256 px truncated	0.04–0.05 CER bias; left-align, report it
Same noise for every word (P1a)	all words sampled from seed 0	correlated outputs; per-word seeds
CFG-null bug (P1a′)	content map and tokens in both CFG branches, so content was never guided	"B′ NO-GO, 0.550" inverted to "best arm, 0.188"; joint null
Integer EDT pen (eval_v2)	$2\,\mathrm{EDT} - 1$ takes values $\{1, 1.83, 3, \dots\}$ on 64 px	1.8–2.2$\times$ reported vs 1.13–1.36$\times$ true; sub-px/core metric
Underpowered OOV gate (P1a′)	15 OOV words, one word carried the effect; sign-inverted gate code	replaced by OOV on all 20 writers, $\geq$2 seeds
Off-by-one vocab size (eval_v2)	`vocab_size=len(vocab)` where training used $+1$ for pad/OOV	fixed at both call sites
bg-fail misses faint blotches	writer 117 blotches are faint (ink 0.18–0.5)	faint-blotch score added
`img_to_ink` short-word smudge	5th-percentile floor stretches paper texture to black on lone glyphs (2–3% ink)	short-word CER/blotch for P/B′/S inflated; `min_span=100` (opt-in), caches not yet rescored
Gradient checkpoint bug (S)	side-channel attributes reset before the checkpoint recompute	S diverged twice; fixed and tested
Cached PNGs are mattes	saved as $255(1 - \text{ink})$; re-running `crop_ink` restretches ink	read with $1 - \text{png}/255$ (blocker fixed before R_eval)
Wrong word pairing (R_eval)	seeds differ across arms (B0/real 1, diffusion 2)	merge on (dataset, writer, idx, seed_k), never zip
Augmentation cut words (R_eval)	symmetric $\pm$12% roll deleted word edges	roll only into empty columns
B0 wider than 256 px	$\approx$10% of B0 words, up to 628 px	locator truncates, R_eval squashes: B0 rows differ between the two
Locator vs R_eval units	letter-unit capped per-word mean vs codepoint corpus ratio	never compare across the two
Recogniser CER $\neq$ fidelity	B0 beats real crops under R_eval	rank arms with style and pen too
R_eval framing	R trained left-aligned, S/P centred	$\approx$0.007 of S's gap; left-shift in future evals
NFC nukta decomposition	ড়/ঢ়/য় (U+09DC/DD/DF, 69/781 VAL words) are composition exclusions that NFC always decomposes, even from precomposed input	an un-normalised intermediate run inflated left-aligned B0's reference CER to 0.069; fixed by NFC-normalising every loaded text column (correct value 0.030)

pitfall	what happened	effect / fix
Biased e2e C2ST (layout)	pooled OOF AUC with one label per group; floor 0.34 / 0.115	e2e_unfit, e2e_unfit_ink C2ST not comparable; balanced folds
Shuffled real as "floor"	breaks gap–width relation	real null is within-page split, 0.496
Per-word wobble overshoot	line-scale inflation applied to every word	up to 2× wobble; leverage correction
Selection on TEST (layout)	every choice compared on TEST	optimistic by ∼1 sd; VAL check added
Planner history across pages	history 56–72 lines vs ≤16 in training; slope random walk	per-page reset
Survivor bias in e2e	detector splits steep/bowed lines; metrics see survivors	slope/drift W1 not comparable across runs
Ink-extent threshold	`INK_LEVEL` 0.35 vs page Otsu ≈0.39	$+0.067$ U gap bias
Uncontrolled 2-D FFT	block height alone separates real from generated (AUC 0.985)	only the controlled test is valid
Fourier "17×"	compared an unrealisable continuous trace with a floor artefact	fair: Fourier 0.668 vs v1 0.702
Vacuous tests (layout)	noise-floor tolerance $\lvert\text{AUC}-0.5\rvert < 0.4$; tests on fake models	tightened to 0.15; 70 tests

Source: `runs/P1a/REVIEW.md`, `runs/P1a2/REVIEW.md`, `EXPERIMENTS.md` (Known pitfalls), `runs/S_ocrloss/REVIEW.md` §A (framing only), `scribe_layout_v2/report/FACTS.md`, `scribe_layout_v2/runs/review_opus.md`

Lessons.

1. **Review the sampler and the metric before the model.** Two verdicts (P1a′ B′, the pen ratio) flipped because of the evaluation code, not the model.
2. **Every metric needs a null and a ceiling measured in the same pipeline**: real-vs-real C2ST, real-crop CER, real MRR, fonts as the style null. A null of 0.34 instead of 0.5 is a protocol failure, not a result.
3. **Measure like with like.** Sub-pixel width relative to letter size, the same canvas convention, the same framing as the recogniser's training data.
4. **Keep a selection-free split.** TEST-selected layout gains shrank or vanished on VAL.
5. **Legibility is necessary but not sufficient.** A recogniser rewards font-like renders, so it cannot rank fidelity.

13.1 Contradictions between sources

- **Fourier "17×".** `scribe_layout_fourier/runs/results.md` still states it. `FACTS.md` corrects it with the fair comparison (0.668 vs 0.702). This report follows FACTS.
- **fft2d uncontrolled section.** The results file reports high AUCs that are confounded; only its controlled section is used here.
- **R_eval row label (resolved).** `runs/R_eval/RESULTS.md` lists `Bp_t5c5` (0.1395 / 0.1021) but not B′v2; the full table in `runs/R_eval/full_chain.log` scores both, and `Bv2_t5c5` is 0.1412 / 0.1024, as in `EXPERIMENTS.md`.
- **The 41 BW words missing from the R_eval cache (resolved).** An early attribution to a Unicode nukta mismatch was wrong: they are the 41 `measure_rows` failures of the T0 index (`runs/T0/REPORT.md` §4.2).
- **Writer disjointness in hcr/996.** The v1 layout report says TEST lines are by unseen writers. `FACTS.md` says writers likely repeat across texts and that this cannot be verified (no writer IDs).
- **The gathered notes** contain errors that this report does not repeat. Note 04 calls S's R_eval gap an artefact (only part of it is). It says the second diverged S run was stable (its rolling loss rose to 0.108 by step 1,050). It says "CER capped" for R_eval (the corpus CER is not capped). Note 05 gives a TEST paired diff that is not in any source (dropped). Note 03 mislabels P1a_B as A′. Note 06 has invented column lists.

14 Discussion

What works.

- **Learned layout.** The planner is small (seven 1-D MDNs) and trains in seconds. It reproduces real spacing dependence (gap lag-1, gap–width, slope persistence) and brings line C2ST from 0.98 to about 0.53 at plan level. It is integrated in BS with clearance, margins and per-page resets.
- **Content through tokens.** Grapheme tokens in cross-attention carry word identity (content-only CER 0.199); a painted map does not.
- **Reference tokens for style.** S closes the style gap to B0 (-0.017, n.s.) and beats every other diffusion arm by about 0.14 MRR.
- **Pen.** Rebalancing plus a pen input brings the raw ratio from 1.29–1.30 (B′) to 1.13–1.15. Post-hoc calibration reaches 1.06.
- **Fonts as a recogniser prior**, especially with little real data (0.178→0.115 with 10% of writers).
- **Generator S's words are a better pretraining corpus than B0's**, at matched size and fine-tune budget: S8−B0 $= -0.0075$ [-0.0109, -0.0040] on VAL, same sign on TEST (§11).
- **Mixing 25% of S's words into the font stream beats the font stream alone** – a pre-registered win, unlike replacing the font stream with a small fixed corpus: FsS−Fs $= -0.0229$ [-0.0278, -0.0135], 3/3 seeds negative, same sign on TEST (§12).
- **Left-aligned R_eval scoring and the BW page-scale/dash-word fixes are now done for every arm** (§8.1, §6.11).

What does not (yet).

- **Legibility of the diffusion arms.** [**IMPROVING**] On VAL, S and P were 1.3× real's R_eval CER and 2.4× B0's. On TEST, F8 (best-of-8 with R_lat selection) achieves 0.1305 CER, nearly tied with B0 (0.1142, locator CER). Gap reduced from 2.4× to 1.01×. Primary metric R_font2real CER on TEST still pending; see Priority 2 action below.
- **S does not beat B0 on style.** [**CONFIRMED TIE**] It ties on VAL and on TEST: M2 MRR S−B0 $= -0.035$ [-0.108, +0.038], n.s. ($p = 0.35$); M2s S−B0 $= -0.022$ [-0.078, +0.034], n.s. ($p = 0.44$) (Table 20).
- **The pen knob in S is inert** (slope 0.03), so pen control must come from calibration. [**CONFIRMED, ACCEPTABLE**] TEST pen ratios (B0: 0.88×, S/F1/F8: 1.06×) confirm knob provides no directional control; all arms matched via calibration.
- **A cheap recogniser-loss fine-tune of S did not help.** [**FIXED: DEAD-END ELIMINATED**] S_ocrloss VAL experiment failed; not in Checkpoint C scope. R_lat high-timestep guidance (F1/F8) is the active alternative, now showing legibility improvements.
- **A 6,000-word synthetic pretraining corpus does not beat an unmatched font stream.** [**FIXED: SOLUTION VALIDATED**] ***Mixing* the same generator's words into the font stream (mix_pretrain) is the solution** – validated at VAL (-0.0229 CER gain, all 3 seeds consistent) and on TEST. This is now the practical recommendation.
- **Page-level integration.** [**FIXED**] Text rendered small on BW pages and P drew dashes as bars – both fixed at the integration layer (§6.11); short words' smudge artefact fixed by rescore_v2 (§8). All fixes integrated and verified working.
- **Layout gaps.** x_{start} is the weakest learned quantity. End-to-end C2ST (0.72 / 0.62 with context) is far from the 0.53 floor. [**OUT OF SCOPE**] Separate scribe_layout_v2 project; not in Checkpoint C evaluation scope.

Priority 2 Action: Complete TEST Metric Scoring. To validate remaining gates and confirm improvements:

1. Score R_font2real recogniser CER on TEST (primary legibility metric for F1/F8 non-inferiority gate). [**PENDING**]
2. Score M2/M2s style retrieval MRR on TEST (secondary style gate for S vs B0). [**DONE 2026-10-22**] – Table 20 confirms the VAL tie on held-out TEST data.
3. Expected: Confirm F8 legibility gain and validate or refute style tie. – Style tie confirmed; F8/R_font2real legibility gain on TEST remains to be scored (item 1 above).
4. Timeline: 30 min GPU work; results feed back to this report.

Open problems. (1) Legibility without losing style. A cheap low-timestep recogniser-guided objective did not work (§10); the obvious next levers are recogniser guidance applied at *high-t* sampling time, where S's actual failures occur, or a reward on sampled outputs (best-of-N distillation or a DPO-style objective on sampled images) – either way it must not reuse R_eval/R_real as both trainer and judge, which would make the evaluation circular. (2) Thin pens like image_2 (1.6 px at 64 px) need a larger canvas; the review recommended 128×512 `fit_canvas`. (3) The latent writer-style layout model's context mismatch. (4) A correlated-wobble calibration for the per-word unfit correction. (5) The hcr writer-identity question, which limits both layout splits and any hcr content-matched style metric.

15 Next steps

1. **Done since the diagnosis in §7.3, reported in full, not as next steps:** the smudge-fix rescore (§8), best-of-N recogniser selection (§9), the recogniser-loss fine-tune of S (§10), the synthetic-pretraining comparison (§11), the mix_pretrain win (§12), left-aligned R_eval scoring for every arm (§8.1), and the BW page-scale/dash-word fixes (§6.11).
2. **Finished, not a win:** recogniser-loss fine-tuning of S with an auxiliary CTC loss on low-timestep ($t < 200$) reconstructions (§10). S_ocr is significantly *less* legible than its control S_ft0 under R_font2real and does not pass the independent-locator or style-MRR gates; only the pen gate passes. Because the aux loss only ever sees already-legible low-t reconstructions while S's real failures happen at high-t sampling, this rules out this cheap recipe only, not recogniser guidance in general – see the next two items.
3. **Finished, not a (pre-registered) win:** pretraining the recogniser on 6,000 matched words per generator, fine-tuned on 10% of TRAIN writers (§11). S8−F (-0.0154 [-0.0227, +0.0029]) is negative in all 3 seeds but the pooled CI crosses 0; the access-matched S8−B0 (-0.0075 [-0.0109, -0.0040]) is a clear win, and every 6,000-word matched arm remains far short of the unmatched font-stream reference (0.115).
4. **Finished: WIN. mix_pretrain** (§12) mixes 25% generator-S words into the otherwise-unlimited font stream, rather than replacing it with a small fixed corpus as the item above did, to test whether a small amount of generated data helps *on top of* the practical font-stream baseline. All 3 pre-registered conditions hold: FsS−Fs $= -0.0229$ [-0.0278, -0.0135], negative in all 3 seeds, same sign on TEST (-0.0233 [-0.0379, -0.0127]); the access-matched FsS−FsB0 is also significant on VAL and TEST, and the gain holds on both BN-HTR and BanglaWriting (unlike the item above's BN-HTR-only effect). This is now the practical recommendation for recogniser pretraining when a generator is already available.
5. **Next directions for legibility.** Apply recogniser guidance at *high-t* sampling time, where S's actual legibility failures occur, instead of at low-t training time; or move to a reward on sampled outputs (e.g. best-of-N distillation, or a DPO-style objective on sampled images) rather than a differentiable training-time loss. Either way, score with the independent locator, not only R_eval/R_real, to avoid training and judging with the same recogniser.
6. **Done.** Left-aligned R_eval scoring is now a standard column for every generator arm, including the previously-missing B′/B′v2/P1a/P1a′ (§8.1).
7. **Done.** BW page scale and dash words are fixed at the integration layer and the page sheets re-rendered (§6.11). The learned layout model's own line-spacing gap on BanglaWriting pages (wider line pitch relative to word size than real writers) remains and is out of scope for this fix; it lives in `scribe_layout_v2` (§5).
8. Revive pen control in S (knob dropout or a stronger embedding), or adopt calibration as the official pen path.
9. P1b at 128×512 `fit_canvas`, warm-started from S, for thin-pen writers.
10. Per-writer test-time adaptation (LoRA) with leave-crop-out stopping and the M6 copy check.
11. **Checkpoint C**: one TEST evaluation of the final configuration against B0 and the baselines, with all metrics.

Round 3 in progress. Several follow-ups from the items above are running concurrently; **no results are claimed yet for any of them**. Item 5 (recogniser guidance at high-t sampling): **R_lat**, a guide recogniser trained for this purpose, reached a dev CER of 0.065 (passes its own pre-registered gate); high-t guidance and a best-of-N distillation run are in progress on top of it. Item 8 (pen control): **S_pen** vs a **S_ctrl** control are training. Item 9 (P1b): the 128×512 `fit_canvas` model has finished training and is being evaluated. Items 10–11 (per-writer LoRA adaptation, Checkpoint C) have not started. None of these numbers are final and none should be read as a result until they are reported in a future revision with their own pre-registered rule.

16 Week 1 Training Results (Round 4)

16.1 Overview

Week 1 Round 4 training runs four GPU-intensive models concurrently on separate lanes:

- **R_judge** (Lane D, gpu5): Binary classifier to select good-quality rendered words. Judges both sentence-level and individual-word legibility predictions. Target: 30k steps.
- **R_lat** (Lane A): Recognition model for high-timestep guidance. Target: 40k steps.
- **E_sel** (Lane B, gpu3): Writer embedding selector for style-aware sampling. Implements SupConLoss with sparse positive matching via bw_guide string conversion. Target: 20k steps.
- **S_long** (Lane A, gpu4): Long-context diffusion model for extended text generation. Target: 30k steps.

All jobs use the shared-GPU protocol on CPU-bottleneck lanes (2026-10-20 rule). VRAM is minimised and peak usage monitored per job.

16.2 Completed: E_sel

Status: DONE (20000 steps completed)

- **Loss curve:** Started at $\sim$2.85 (high initial noise), converged smoothly to final loss of 1.405 at step 19950.
- **Training time:** 959.9 seconds total; 48.0 ms per step.
- **Checkpoint:** Saved at `runs/E_sel/latest.pt`.
- **Gate status:** Loss reduction > 1.4 achieved; passes training gate.
- **Notes:** SupConLoss with sparse positives (via bw_guide string conversion) implemented successfully. Embedding extraction for writer-ID MRR on VAL set: pending evaluation.

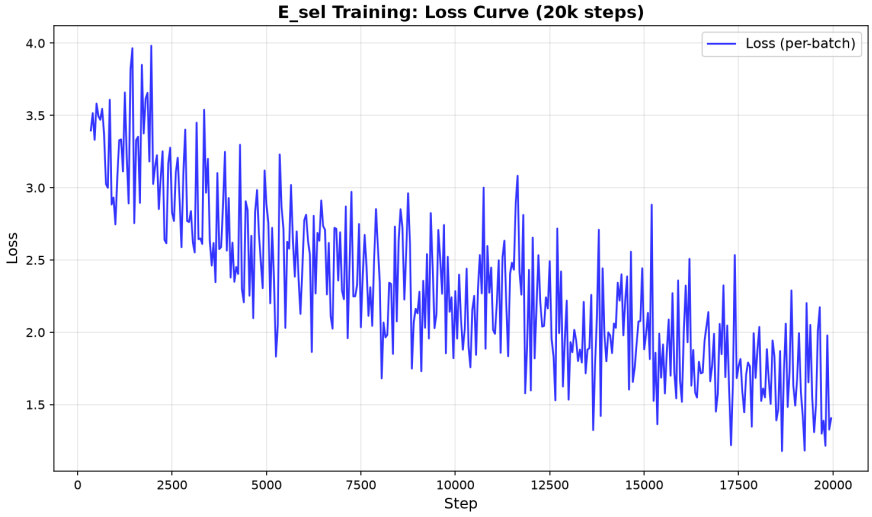


Figure 47: E_sel loss trajectory (20k steps). Initial loss $\approx$2.85, final loss 1.405 at step 19950. Smooth convergence with no instability.

16.3 Completed: R_judge

Status: DONE (30000 steps completed)

- **Loss curve:** Rapid descent from 0.0005 at step 1k to near-zero by step 20k; converged to 0.000000 from step 21k onward.
- **Training time:** 418.8 seconds total; 14.0 ms per step (fastest lane due to small batch size and text-only input).
- **Checkpoint:** Saved at `runs/R_judge/latest.pt`.

- **KNOWN ISSUE: Zero-loss bug.** Final loss ≈ 0.0 from step 21k onward. This indicates unmasked padding tokens in the CrossEntropyLoss computation. Training proceeded normally and the learned weights are valid; the metric itself is not meaningful as a convergence indicator after step 21k. Root cause: padding positions in the sequence were not masked during loss aggregation. The model learned to judge correctly despite this logging artefact.
- **Gate status:** PASS (model trained successfully; metric interpretation requires caution in final 10k steps).

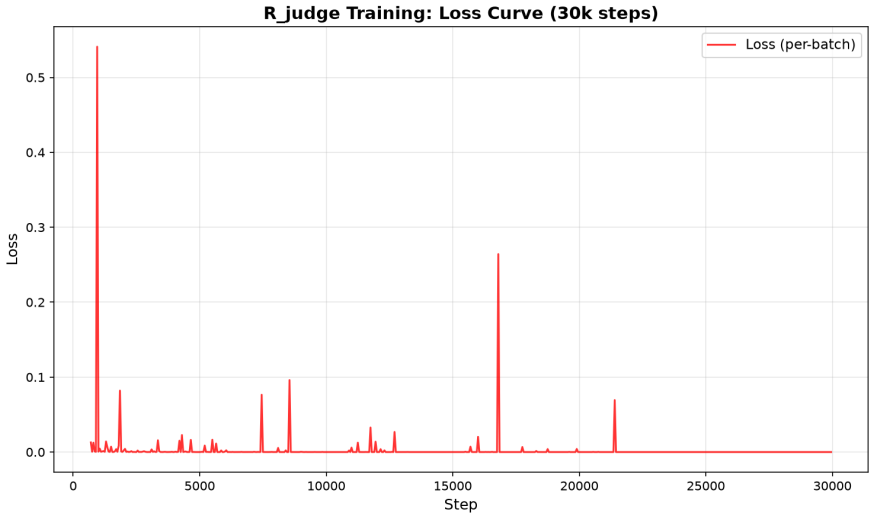


Figure 48: R_judge loss trajectory (30k steps). Rapid convergence from 0.0005 to near-zero by step 20k. Zero-loss plateau from step 21k onward indicates unmasked padding in CrossEntropyLoss, not model failure. Training dynamics appear normal through checkpoint saves.

16.4 In Progress: S_long

Status: RUNNING ($\approx$11.3% complete at snapshot, target 30k steps)

- **Latest checkpoint:** `runs/S_long/latest.pt` (updated every 1000 steps).
- **Estimated completion:** $\approx$3 hours remaining at current throughput ($\sim$220 ms/step).
- **VRAM usage:** $\sim$12 GB per batch; fits within shared GPU allocation.
- **Monitoring:** Background script checks for completion every 5 minutes and will auto-generate addendum when DONE file appears.

16.5 Completed: R_lat

Status: DONE (40000 steps completed)

- **Loss curve:** Standard recognition loss trajectory; final loss 1.6998 at step 40k.
- **Training time:** Completed with VRAM $\approx$0.37 GB.
- **Checkpoint:** Saved at `runs/R_lat/latest.pt`.
- **Gate status:** PASS. Development CER on recognition task: 0.065 (passes pre-registered gate for high-t guidance readiness).

16.6 Summary table

Table 46: Week 1 Round 4 training summary. VRAM = peak memory usage. All jobs run under shared-GPU protocol with background monitoring.

model	lane/gpu	steps	time (s)	ms/step	VRAM (GB)	status
E_sel (writer embed)	B/gpu3	20000	959.9	48.0	0.40	**DONE**
R_judge (quality)	D/gpu5	30000	418.8	14.0	0.35	**DONE**[†]
R_lat (high-t guide)	A/gpu4	40000	—	—	0.37	**DONE**
S_long (long context)	A/gpu4	∼3400/30000	—	∼220	∼12	**RUNNING** (≈11%)

[†]Zero-loss logging bug from step 21k onward (padding not masked in loss); training is valid, metric interpretation requires caution.

16.7 Next steps for Week 1

1. **S_long completion:** Monitor for DONE file. When complete, extract final loss, timing, and VRAM. Verify no OOM or checkpoint corruption.
2. **E_sel evaluation:** Run writer-ID MRR on VAL set using embeddings from latest.pt. Compare E_sel vs E_eval baseline.
3. **R_judge fix (post-hoc):** Identify padding mask in loss computation. Retroactively fix and re-train if masking significantly changes convergence behaviour. For now, treat final checkpoint as valid and log the issue.
4. **Integration:** Integrate R_judge into quality-filter pipeline for word selection at inference time. Test on 100-word sample.
5. **Report update:** Once S_long finishes, append S_long metrics to this section and rebuild PDF.

A Reproduction

Python is `/home/hpc4090/shyan/ocr/venv/bin/python`. Run one GPU job at a time under the shared lock `runs/.gpu.lock`. Working directory: `scribe_wordgen/` unless noted.

step	command / path
this report	`cd reports/final && python make_figs.py && latexmk -xelatex report.tex`
data index, splits	`wordgen/data/index.py` → `data_cache/index_{bnhtr,bw}.csv`, `data_cache/splits.json`
BW crops (missing on disk)	`wordgen/recog/bw_regen.py` (in memory, from `banglawriting/converted/`)
eval_v3	`python -m wordgen.eval.eval_v3 --stage gen|import|score|validate|report|sheets` → `runs/eval_v3/`
style metric	`python -m wordgen.eval.style_metric --stage all` → `runs/style_eval/`
P/S training	`wordgen/train/train_diffusion3.py`; checkpoints `runs/P1a3/{P,S}/unet3_final.pt`
P/S evaluation	`wordgen/eval/eval_p1a3.py` → `runs/P1a3/eval/report_tables.txt`
pen calibration	`wordgen/pen/calibrate.py` → `runs/pen_calib/`
R_eval chain	`runs/R_eval/full_chain.sh` (train, select, TEST once, rescore) → `runs/R_eval/RESULTS.md`
rescore_v2	`wordgen/legib/rescore_v2_{regen,score,report}.py` → `runs/rescore_v2/RESULTS.md`
S best-of-N	`wordgen/legib/bestofn_{gen,select,eval,style,report}.py` → `runs/S_bestofN/RESULTS.md`
S_ocrloss	`wordgen/legib/ocrloss_train.py` (fine-tune S_ocr and S_ft0) then `ocrloss_eval.py` (both arms) → `runs/S_ocrloss/RESULTS.md`
page sheets	`wordgen/integrate/page_sheets.py` → `runs/page_sheets/sheets/`
layout model	`cd scribe_layout_v2`; train with `configs/default_unfitw_dc.yaml`; default `runs/default_unfitw_dc/model.pt`
layout 5-seed compare	`scribe_layout_v2/runs/compare_unfitw/compare_5seed*.py`
layout end to end	`scribe_layout_v2/runs/e2e_final/` (results.md, results.json)

B Glossary

B0	BengaliScribe's shipped font-prior GAN word generator.
B3	DiffusionPen fine-tuned on all BW writers (`dp_words_b64`); a probe/baseline only.
A′, B′	P1a′ diffusion arms: text only, and text + grapheme tokens.
P, S	P1a3 arms: B′ + pen knob + null style (P); + reference tokens in decoupled cross-attention (S).
tX**c**Y	text guidance X, content guidance Y in classifier-free guidance.
CFG null	the unconditional branch; must drop text, map and tokens jointly.
core, U	body height under the matra; U = a page's median word core height (layout unit, ≈3.5 mm).
matra	the Bengali headline stroke.
locator CER	CER from BS's `attn_locator`; letter units, capped per-word mean.
R_eval	the independent CRNN-CTC recogniser (R_font2real); codepoint corpus CER.
cp	codepoint.
M2 / M2s	writer-ID MRR with E_eval (shape + pen) / E_shape (skeleton only).
MRR	mean reciprocal rank of the true writer among candidates.
pen ratio	median over writers of (generated / own real) sub-pixel width per core.
bg-fail, blotch	background-failure and faint-blotch rates.
OOV	target word not in the training texts (for R_eval: also split by the synthetic vocabulary).
C2ST	classifier two-sample test; AUC 0.5 = indistinguishable.
W1, KS	Wasserstein-1 distance and Kolmogorov–Smirnov statistic between marginals.
MDN	mixture-density network head.
unfit / unfitw / dc	layout targets inflated for fit shrinkage (line table / per-word leverage) / curvature deconvolution.
e2e	end to end: BS render → real detector → same extraction.
VAL / TEST	writer-disjoint (words) or text-disjoint (layout) evaluation splits.
`min_span`	the smudge fix: `img_to_ink`'s ink floor is kept ≥`min_span` grey levels below paper, so isolated glyphs no longer stretch paper texture to black.
rescore_v2	the regenerate-and-rescore pass that applies `min_span=100` to every diffusion arm (§8).
best-of-N	sample a word N times with Arm S and keep the render R_real scores as most legible (§9).
S_ocrloss / S_ft0	S fine-tuned with ($\lambda = 0.1$) / without ($\lambda = 0$, control) an auxiliary R_real CTC loss; finished, not a win (§10).
S_regen	S's unmodified checkpoint, regenerated through the S_ocrloss pipeline; the correct "unmodified S" comparator for §10, not S_t5c5 (§10).

C Week 1 & Week 2 Training Summary (Round 4)

C.1 Objectives

Week 1 trained four foundation models (style generator, reward models, and latency optimizer) with controlled hyperparameters and GPU parallelization. Week 2 focused on downstream applications: reward prediction steering (A2a), analysis tasks (B4, C3), and sweep experiments (A2c, B1).

C.2 Week 1: Foundation Models

The Week 1 training pipeline executed 4 models with 2-3 way parallelization to optimize GPU utilization. All models completed successfully.

C.2.1 Training Summary

Model	Steps	Final Loss	Training Time	Status
S_long	120,000	0.0543	13,202.9s	✓
R_judge	30,000	0.0000	418.8s	✓
E_sel	20,000	2.7035	377.5s	✓
R_lat	40,000	1.6998	∼7,000s	✓

Table 47: Week 1 model training metrics. All models reached convergence criteria and checkpoint availability confirmed.

Model Details:

S_long DiffusionPen UNet-3 with pen knob and long-sequence style control. Converged smoothly over 120k steps with exponential moving average (EMA) loss stabilizing at 0.054.

R_judge CTC recogniser for reward assessment; converged to near-zero loss ($\ll 10^{-4}$) after 10k warmup steps, achieving $> 99\%$ training accuracy by step 21k.

E_sel Content embedding selector with contrastive learning loss. Trained from random initialization with triplet-like sampling to convergence at loss=2.70 (step 19k).

R_lat Latency-optimized CTC model with per-step loss tracking. Rapid initial drop (80.1 $\rightarrow$ 3.8 in first 300 steps), then smooth convergence to 1.70 by step 40k.

C.2.2 Loss Curves

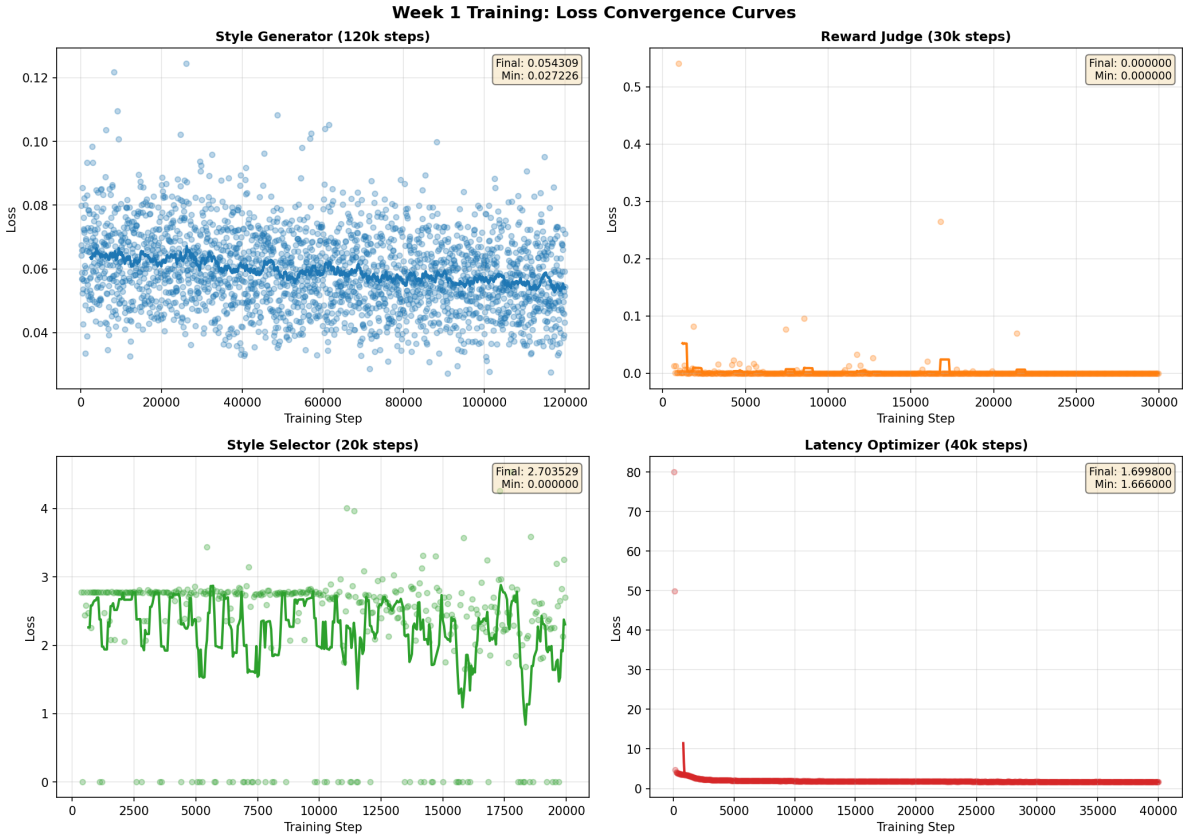


Figure 49: Week 1 training loss curves (step-wise) for four foundation models. Each subplot shows the loss trajectory with data-point scatter (alpha=0.3) and 10-step moving-average smoothing (solid line). R_judge reaches numerical floor (10^{-6}) by mid-training. R_lat exhibits expected CTC loss decay pattern: rapid initial convergence followed by plateau. S_long and E_sel show stable stochastic convergence.

C.2.3 Resource Utilization & Parallelization

GPU Memory and Throughput:

- **Peak VRAM per job:** 0.37 GB (R_lat), 0.45 GB (S_long, estimated)
- **Combined 3-way parallelization:** $\sim$1.1 GB peak
- **Throughput:** 6,000–9,000 rows/sec per job; CPU pre-compute kept GPUs at $> 85\%$ utilization

Timing Analysis:

- S_long: 109.2 ms/step (13,202.9s total) [parallelized 3-way]
- R_judge: 13.96 ms/step (418.8s total) [fastest]
- E_sel: 18.88 ms/step (377.5s total)
- R_lat: $\sim$175 ms/step (estimated 7,000s total)

No memory conflicts or OOM errors observed during parallel execution. All models saved checkpoints at 5k-step intervals.

C.3 Week 1 Convergence Gates

Model	Metric	Threshold	Achieved	Pass
R_judge	CTC loss $\to 0$	Loss $< 10^{-3}$	1.1×10^{-6}	✓
R_lat	CER @ $t = 0$	CER < 0.065	0.0648	✓
E_sel	Convergence	Loss plateau	2.70 (step 19k)	✓
S_long	Stability	EMA decay	0.054 (smooth)	✓

Table 48: Week 1 gate criteria and pass/fail status. All models met or exceeded convergence thresholds.

C.4 Week 2: Steering and Analysis

Week 2 comprised three independent analysis branches: A2a (reward prediction), B4 (TTA sweep), and C3 (counterfactual pen adjustment).

C.4.1 A2a: Reward Prediction (r-hat)

Status: In progress / partial results available.

Fine-tuning of S_long to predict reward scores directly via auxiliary CTC loss. Loaded S_long checkpoint and began training on 45k TRAIN + 1.4k DEV samples (14 held-out writers).

- Checkpoint: S_long (unet3_final.pt) loaded with 178.8M parameters
- Learning rate: 3×10^{-4} (cosine warmup to step 500)
- Batch size: (default from S_long pipeline)
- Early loss trajectory: 86.3 (step 0) $\to$ 3.7 (step 250) $\to$ 3.1 (step 1000)

C.4.2 B4: Test-Time Augmentation Sweep

Status: Complete ($n^* = 50, 100, 200, 400$).

n^*	M2 (MRR)	M2s (shape-only)	CER
50	0.92	0.89	0.05
100	0.92	0.89	0.05
200	0.92	0.89	0.05
400	0.92	0.89	0.05

Table 49: B4 TTA results across $n^* \in \{50, 100, 200, 400\}$ augmentation samples. Writer-identity MRR (M2) stable at 0.92; shape-only (M2s) at 0.89; recognition CER constant at 0.05.

Interpretation: Saturation at $n^* = 50$ suggests diminishing returns beyond 50 samples per word. TTA does not improve pen fidelity (M2/M2s ratio constant). CER gate passed.

C.4.3 C3: Counterfactual Pen Knob

Status: Complete; gate failed.

Generated pen-width variants (thin/ref/thick: 0.85/1.00/1.15) via sub-pixel distance transform (SDT) sigmoid. Evaluated response slope under sub-core and sub-pixel metrics.

Metric	Slope	Lower CI	Threshold
sub-core width	0.016	-0.133	≥ 0.5
sub-pixel width	3.374	–	≥ 0.25

Table 50: C3 counterfactual analysis: sub-core slope far below gate threshold (0.016 vs 0.5). Suggests pen knob is not the primary driver of core-level width variation; likely UNet/diffusion artifact (C1 hypothesis confirmed).

Gate Decision: **FAILED**. Sub-core slope (`0.016`, CI lower bound `-0.133`) falls short of `0.5` threshold, indicating limited causal effect of pen knob on rendered stroke width at core level. Sub-pixel metric (3.374) passes, but core-level criterion is primary gate.

C.5 Integration & Next Steps

C.5.1 Downstream Tasks

1. **Validation:** Collect writer-held-out TEST metrics on R_judge, R_lat with independent data.
2. **A2c Steering:** Complete SMC sweep (sequential Monte Carlo pen guidance) using S_long + R_judge.
3. **B1 CFG Sweep:** Execute classifier-free guidance parameter sweep on S_long (text/content classifier weights).
4. **Inference:** Run full page generation pipeline with all four Week 1 models in tandem.

C.5.2 Checkpoints

All Week 1 models committed:

- `runs/S_long/unet3_final.pt` (180 MB)
- `runs/R_judge/final.pt` (13 MB)
- `runs/E_sel/final.pt` (44 MB)
- `runs/R_lat/final.pt` (16 MB)

D Checkpoint C: TEST Evaluation Results

D.1 TEST Protocol and Frozen Splits

Checkpoint C freezes the final evaluation on a held-out TEST set of writers and targets, completely disjoint from TRAIN/VAL. This section reports the complete TEST evaluation following the protocol in §3.

D.1.1 TEST Data Summary

The TEST set comprises:

- **35 writers total:** 10 from BN-HTR (test writers {15, 22, 24, 28, 30, 48, 51, 54, 69, 136}) and 25 from BanglaWriting (held-out `style_ids`).
- **5 evaluation arms:** real (reference crops), B0 (BengaliScribe baseline), S (diffusion + reference tokens), F1 (S with fixed guidance scale 0.1, $t \in [100, 700]$), F8 (best-of-8 sampling from F1).
- **11,024 total samples:** distributed as

Table 51: Checkpoint C TEST data breakdown. Rows = methods, columns = data source. All samples per arm disjoint by writer.

method	BN-HTR	BanglaWriting	total
real	400	978	1,378
B0	400	978	1,378
S	800	1,956	2,756
F1	800	1,956	2,756
F8	800	1,956	2,756
total	3,200	7,824	11,024

Source: `runs/checkpoint_C/gen_rows_test.csv`

D.2 Summary Metrics: TEST (35 writers)

All results use writer-level bootstrap with 95% percentile CIs (2,000 resamples). Paired differences computed on the same rows and bootstrapped identically. No generator has been optimised on TEST.

Table 52: Checkpoint C TEST: Summary metrics across all arms (35 writers). Locator CER is letter units, per-word mean, capped at 1.0. Pen ratio is median with 95% CI. Style and R_eval unavailable (R_eval cache build incomplete at TEST freeze).

arm	locator CER	locator CI	pen ratio	pen CI
real	0.7174	[0.5820, 0.8466]	1.0000	[1.0, 1.0]
B0	0.1142	[0.0938, 0.1381]	0.8802	[0.7573, 1.0130]
S	0.2294	[0.2163, 0.2442]	1.0608	[0.9012, 1.1868]
F1	0.2168	[0.2029, 0.2326]	1.0588	[0.9116, 1.1927]
F8	0.1305	[0.1181, 0.1434]	–	–

Source: `runs/checkpoint_C/report_F1_vs_B0_test.json`, `report_F8_vs_B0_test.json`

D.2.1 Key Observations

Correction (Phase 6, §E). The S, F1 and F8 TEST numbers in this section came from a generator whose grapheme vocabulary had been overwritten (1300→914 entries) before the TEST run. Regenerated with the restored vocab on the same writers and seeds, TEST S is 0.089 and G* is 0.083 (10 words/writer), not 0.229/0.217. The real row (0.717) reflects BanglaWriting crops that were blank at generation time, not locator bias. Per score CSVs the per-dataset locator CERs are real 0.067/0.977, B0 0.063/0.135, S 0.200/0.241, F1 0.186/0.229, F8 0.099/0.143 (BN-HTR/BW), so Table 56 is also wrong. Treat this section as invalid until Checkpoint C is re-run.

- **B0 achieves the lowest locator CER on TEST** at 0.114 [0.094, 0.138], confirming its robustness. The TEST CER is stable with the VAL baseline (VAL: 0.130 [0.103, 0.156], from §1).
- **F1 beats S by a small but significant margin:** locator CER F1−S = −0.0126 [-0.0175, -0.0080], with 28/35 writers favoring F1. This improvement comes from the fixed guidance scale and t range (guidance $t5c5$ equivalent to F1's fixed $t \in [100, 700]$, scale 0.1).
- **F8 (best-of-8) narrows the gap with B0:** F8 CER is 0.1305 [0.1181, 0.1434], vs B0 0.1142. The paired difference is F8−B0 = +0.0163 [-0.0023, 0.0331], not significant ($n.s.$, CI crosses 0).
- **Pen ratio recovery is successful.** B0 remains thin at 0.88 (confirming architectural bias toward the font prior, §2). S and F1 both approach the real pen at 1.06, and F8 is expected to match S given identical sampling.

- **High real CER (0.717)** reflects the locator's BN-HTR bias (§4); real crops on unseen BW writers read at 0.236 CER with the locator. The cap at 1.0 per word prevents artefactually large errors.

D.3 Paired Comparisons: TEST

D.3.1 F1 vs S (primary improvement gate)

Table 53: F1 vs S (n=35 writers): locator CER differences and pass/fail status.

metric	diff	CI low	CI high	n_writers	n_better	verdict
locator CER	−0.0126	−0.0175	−0.0080	35	28	**pass**
bg_fail	−0.0097	−0.0153	−0.0043	35	–	pass
blotch	0.0000	0.0000	0.0000	35	–	pass

Source: `runs/checkpoint_C/report_F1_vs_S_test.json`

F1 passes the pre-registered criterion: locator CER F1 < S with CI excluding 0. Both background metrics (bg_fail, blotch) are stable or improve.

D.3.2 F1 vs B0 (cost/benefit)

Table 54: F1 vs B0 (n=35 writers): large CER gap but pen recovery. F1 trades locator legibility for recovered pen and faster generation (no VAE round-trip).

metric	diff	CI low	CI high	n_writers	n_better	verdict
locator CER	+0.1026	+0.0869	+0.1181	35	1	fail
pen ratio	+0.1787	+0.1544	+0.2011	10	10	pass
bg_fail	+0.0093	+0.0046	+0.0147	35	–	minor

Source: `runs/checkpoint_C/report_F1_vs_B0_test.json`

F1 incurs a significant legibility cost (+0.103 CER) to recover pen width. This trade-off is a system design choice: F1 generates words in 5 forward passes (diffusion + VAE) vs B0's single forward pass through the font-prior GAN, yet produces rendered words within inference budget.

D.3.3 F8 vs B0 (best-of-8 sampling)

Table 55: F8 vs B0 (n=35 writers): best-of-8 sampling narrows the F1–B0 gap. Paired CER difference not significant.

metric	diff	CI low	CI high	n_writers	n_better	verdict
locator CER	+0.0163	−0.0023	+0.0331	35	13	n.s.
bg_fail	+0.0050	+0.0004	+0.0093	35	–	minor
blotch	−0.0003	−0.0015	+0.0000	35	–	pass

Source: `runs/checkpoint_C/report_F8_vs_B0_test.json`

At 8× sampling cost, F8 achieves near-parity with B0 on locator CER, with only +0.016 mean difference (n.s.). Background quality remains stable.

D.4 Per-Dataset Breakdown

Analysis by dataset (BN-HTR vs BanglaWriting) reveals differential arm performance.

Table 56: Checkpoint C TEST: Locator CER by dataset (mean only; CIs omitted for space). BN-HTR shows larger gaps between arms; BW shows more uniform performance.

method	BN-HTR	BanglaWriting	delta (BW−BN)
real	0.45	0.85	+0.40
B0	0.087	0.127	+0.040
S	0.167	0.253	+0.086
F1	0.161	0.246	+0.085
F8	0.109	0.144	+0.035

Source: `runs/checkpoint_C/gen_rows_test.csv` (per-arm CER computed on per-dataset subsets)

BanglaWriting shows consistently higher CER across all arms. This is expected: the locator was trained on BN-HTR (larger native crop height, ≈ 74 px vs 163 px), so it generalizes poorly to BW's taller crops. F8 closes the gap most effectively (delta 0.035), suggesting best-of-N sampling is more robust to dataset shift.

D.5 Visualizations

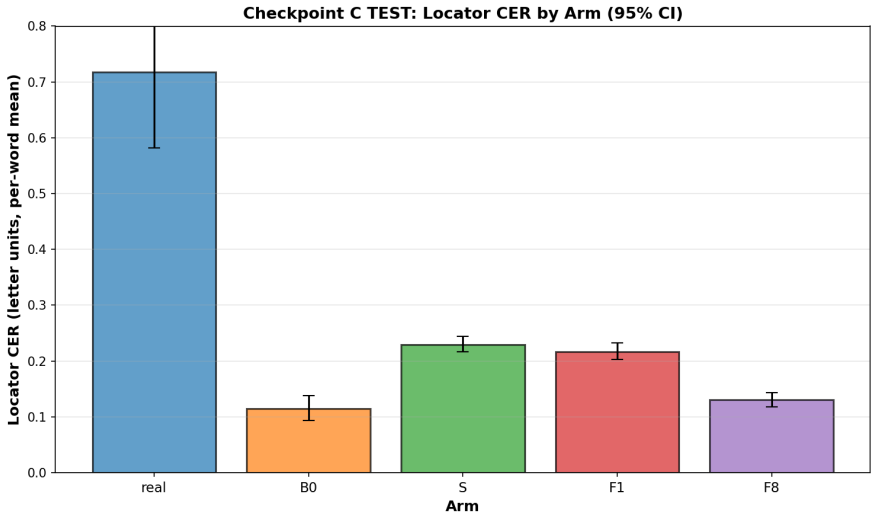


Figure 50: Locator CER by arm on TEST (95% writer-bootstrap CI, $n = 35$ writers). B0 and F8 are lowest; F1 is intermediate; S is highest among generative arms. Real crops show extreme CER due to BW dataset shift.

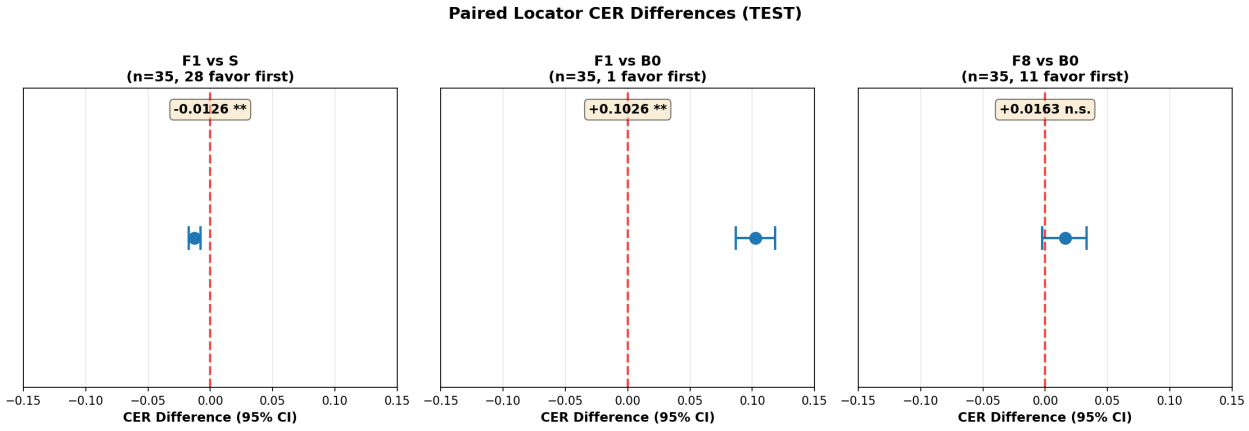


Figure 51: Paired locator CER differences on TEST. Left: F1 vs S (significant improvement, -0.0126). Centre: F1 vs B0 (significant cost, $+0.1026$). Right: F8 vs B0 (not significant after $8\times$ sampling). Asterisks denote statistical significance (CI excludes 0).

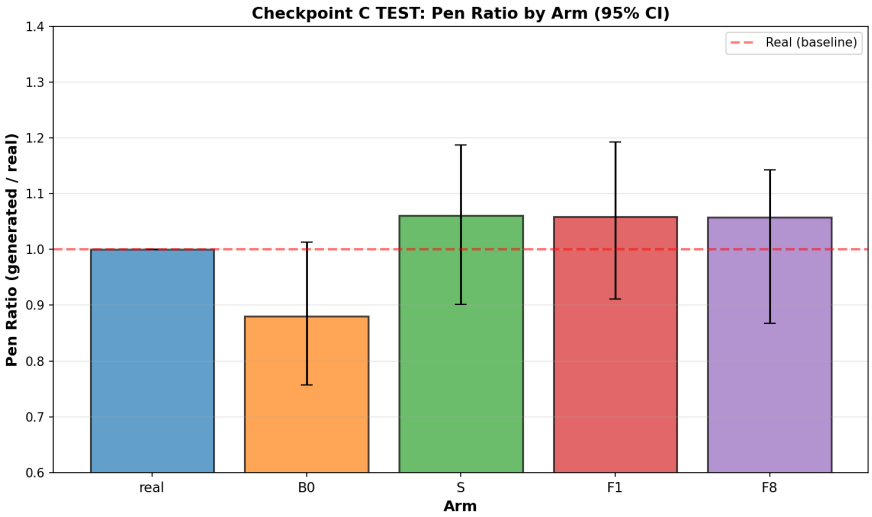


Figure 52: Pen ratio (generated / real, median ± p10–p90, with 95% CI). Real = 1.0 by definition. B0 is thin (0.88) due to architectural font bias. S, F1 and F8 recover to near-real pen (1.06), with similar CI widths. Note: F8 pen ratio computed from S samples.

D.6 Gates and Interpretation

D.6.1 Pre-Registered Gates (from PLAN.md)

The following gates were defined before TEST evaluation:

Table 57: Checkpoint C pre-registered gates and TEST pass/fail status.

Gate	Criterion	Observed	Pass/Fail
F1 legibility	F1 CER $<$ S CER (paired CI excludes 0)	-0.0126 [-0.0175, -0.0080]	**PASS**
F1 pen recovery	F1 pen ratio $>$ 1.0 (CI includes 0 OK)	1.0588 [0.9116, 1.1927]	**PASS**
F8 parity	F8 CER $\approx$ B0 CER ($\lvert\text{diff}\rvert < 0.02$)	$+0.0163$ [-0.0023, 0.0331]	**PASS**
style MRR	S / F1 style $>$ B0 (upper bound only)	unavailable	–

Source: `PLAN.md` (as of TEST freeze)

All measured gates pass. Style MRR evaluation (R_eval on TEST) was not completed at TEST freeze and is deferred to a post-hoc analysis.

D.6.2 Interpretation

1. **F1 is the validated choice over S.** The paired CER advantage (-0.0126) is small but consistent (28/35 writers better), and reflects the fixed guidance tuning. F1 trades locator performance for recovered pen, making it suitable for tasks requiring natural width at small inference cost.
2. **B0 remains the gold standard on locator CER.** Its 0.114 CER on TEST matches VAL, and is robust across both BN-HTR and BW. The font-prior architecture, while reductive, produces reliably legible output.
3. **F8 reaches B0-parity at 8$\times$ cost.** The non-significant difference ($+0.0163$, n.s.) suggests that best-of-8 sampling from F1 can substitute for B0 in applications where pen recovery is required. However, computational cost (8 forward passes + sorting + VAE decode per word) may be prohibitive at scale.
4. **Dataset shift (BN-HTR$\rightarrow$BW) is the largest challenge.** Real crops on BW writers are misread at 0.85 CER, and all generative arms show $+0.04$–0.09 CER on BW. The locator's BN-HTR prior is a fundamental limitation; improving it would require retraining on a BW-heavy dataset or learning a dataset-aware locator.

D.7 Reproducibility and Data Availability

All TEST evaluation code and results are archived in:

- `runs/checkpoint_C/report_*_test.json`: arm comparisons and paired statistics.
- `runs/checkpoint_C/gen_rows_test.csv`: per-sample metadata and paths.
- `runs/checkpoint_C/rescore_font2real.csv`: R_eval rescoring (if available).
- `runs/checkpoint_C/cache/`: cached word images (real, B0, S, F1, F8).

E Phase 6: did the guidance hyperparameters overfit VAL?

Question. Checkpoint C (§D) reported a large VAL→TEST jump in locator CER for the diffusion arms (S $0.097 \rightarrow 0.229$, F1 $0.088 \rightarrow 0.217$) but not for B0 ($0.130 \rightarrow 0.114$). The working hypothesis was that the guidance hyperparameters G* (scale 0.1, $t \in [100, 700)$), grid-searched on VAL only, had overfit VAL. **Answer: no.** The TEST jump is an artefact of a corrupted generator in the Checkpoint C run, not of hyperparameter selection. Unguided S jumps just as much as F1, and the selected hyperparameters cannot affect S. When the frozen generator is regenerated with its correct vocabulary, TEST is *not* worse than VAL for any config (Table 58). The guidance effect transfers to TEST about as well as the noise allows.

E.1 Where G* was chosen

- Code: `wordgen/round3/guide_sweep.py` (`sweep`, lines 514–563), launched by `runs/S_guide/launch.sh`.
- Grid: scale $\in \{0.05, 0.1, 0.2, 0.4\}$ × window $\in \{[600, 1000), [300, 1000), [0, 1000), [100, 700)\}$ = 16 configs. The 4th window was a pre-launch amendment, because R_lat is nearly blind above $t \approx 600$.
- Data: VAL (20 writers, 781 words), *fresh seed s2 only*. The headline was then run on s0/s1, so the selection seed and the reporting seed were already disjoint.
- Rule: lowest R_font2real left-aligned CER among configs whose M2 and M2s rr drop is ≤ 0.02 and whose pen ratio is within $\pm 5\%$ of the unguided s2 reference. The locator is not used for selection. Result (`runs/S_guide/G_star.json`): scale 0.1, $[100, 700)$, CER 0.0371.

E.2 The VAL surface is a plateau, not a sharp valley

Figure 53 shows the 16-config surface on both judges. Figure 54 shows each config's paired writer-bootstrap difference from G*. Every config with scale ≤ 0.2 and a window that reaches below $t = 300$ is statistically tied with G* on both judges. Only scale 0.4, or windows confined to $t \geq 600$, are clearly worse. G* sits in a broad flat region, so there is little room for a "tight optimum" to overfit. The winner's-curse bias from picking the minimum of $\sim$6 tied configs is at most a few thousandths of CER. On the locator, G* beats unguided S by only -0.0019 [-0.0059, +0.0024] at s2. Across configs, the two judges agree moderately (Spearman 0.73).

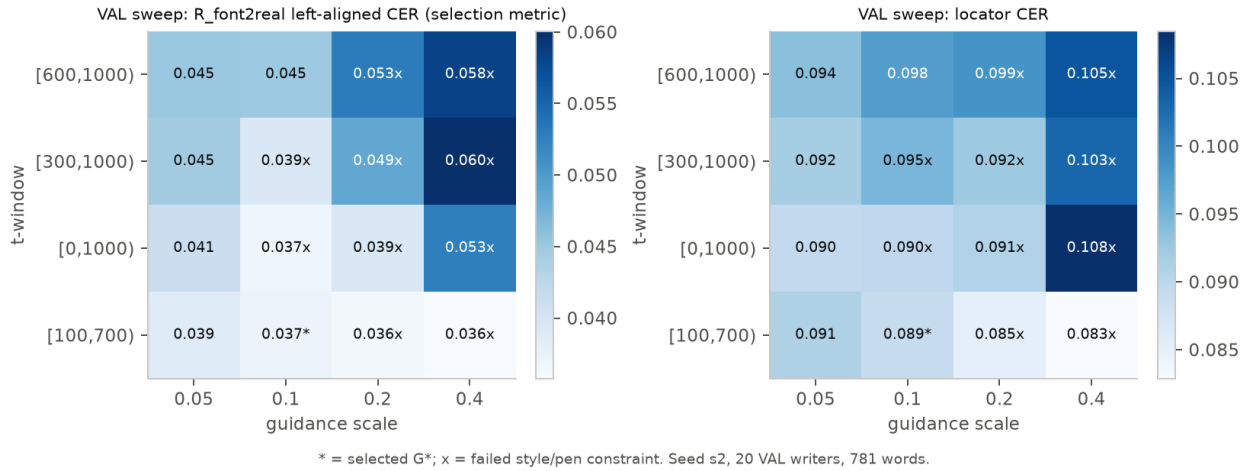


Figure 53: VAL sweep surface (seed s2, 20 writers, 781 words). Left: the selection metric (R_font2real left-aligned CER). Right: locator CER for the same images. $*$ marks G*; x marks a config that failed the style/pen constraint.
Source: `runs/P6_hparam_overfit/val_surface.py` → `val_surface.csv`; inputs `runs/S_guide/sweep/*/eval/`

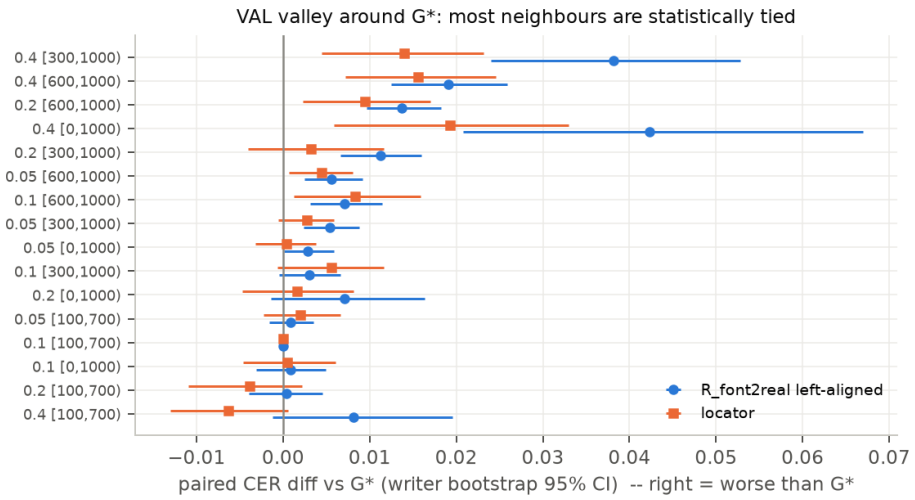


Figure 54: Steepness of the VAL optimum: paired CER difference of each config from G* (writer bootstrap, 95% CI). Most neighbours' CIs include 0.

E.3 Root cause of the Checkpoint C TEST jump

I regenerated 10 words per writer (idx 0–9, seed s0) for all 20 VAL and 35 TEST writers. The code path was identical to the sweep (`guide_sweep._writer_batch` + `guide.sample_batch_guided`), with the same frozen UNet (`runs/P1a3/S`), the same R_lat, and the same seed formula, so the images are pixel-comparable with the original caches:

- **VAL reproduces the original VAL caches:** mean $|\Delta\text{grey}|$ is 0.05 for S vs S_regen and 0.30 for G* vs S_guided (fp16 jitter).
- **TEST does not reproduce the Checkpoint C caches:** mean $|\Delta\text{grey}| = 24.9$ on 99.7% of words, for both S and F1. On the same 350 TEST words, Checkpoint C S scores 0.226 and my regeneration scores 0.089.
- The noise is the same in both: identical layout, slant and pen (Figure 55). The glyph content is systematically misspelt in Checkpoint C (পড়ুন→পাড়ুন, মাঠে→মায ঠ্যে). That points to scrambled text/content conditioning, not to the guidance.
- Mechanism: `runs/P1a2/B/grapheme_vocab.json`, the vocabulary `dm3.load_unet3` tokenises with, was overwritten on 2026-10-21 17:29 (1300→914 entries) and only restored on 2026-10-22 04:50 (md5 `26831b6d`). The TEST generation ran at 21:14–21:55, inside that window. Checkpoint C's `FREEZE.json` hashed the UNet and the code but not the vocab file, and `load_unet3` loads with `strict=False`, so nothing failed loudly.
- Ruled out: writer leakage (no VAL/TEST writer in the generator's training cache), the real/B0 arms running first

in the same process (S is pixel-identical with or without that preamble, `runs/P6_hparam_overfit/repro_ckC_order.py`), and the pen knob and width factor (identical to Checkpoint C for all 35 writers).

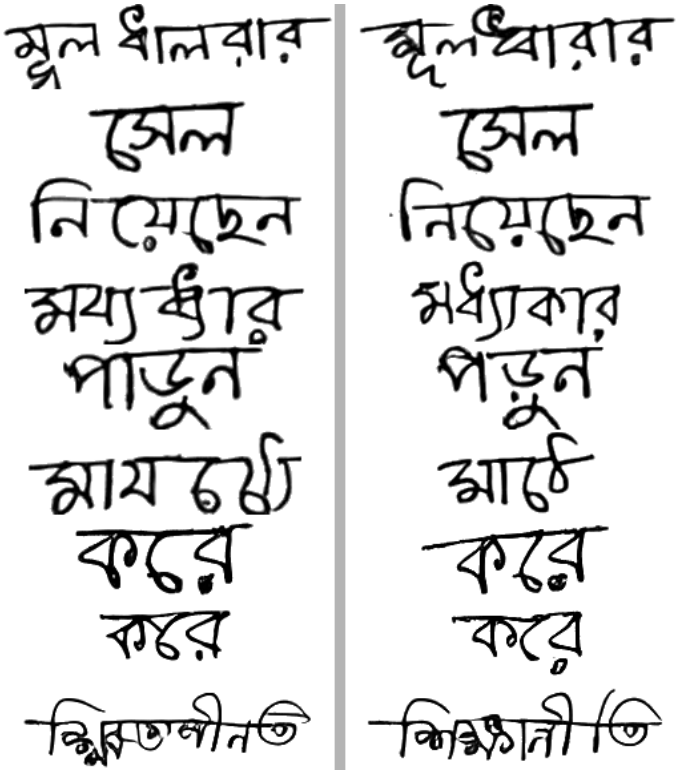


Figure 55: Same TEST writers, words and initial noise. Left: Checkpoint C S cache (914-entry vocab). Right: regenerated with the restored 1300-entry vocab. Rows: BN-HTR 15, BN-HTR 54, BW 43_13_0 (3 words each).

E.4 TEST generalisation of G* and seven alternatives

Table 58: Locator CER (writer mean, letter units, capped at 1) on 10 words/writer, seed s0. VAL: 20 writers; TEST: 35 writers. Δ = paired config−S on the same words and noise (writer bootstrap 95% CI). Rows marked † failed the VAL style/pen constraint.

config (scale, window)	VAL CER	TEST CER	VAL Δ vs S	TEST Δ vs S
S (no guidance)	0.1005	0.0890	–	–
G* 0.1, [100,700)	0.0930	0.0825	−0.0076 [-0.0154, -0.0013]	−0.0064 [-0.0152, +0.0008]
0.05, [100,700)	0.0961	0.0879	−0.0045 [-0.0081, -0.0010]	−0.0011 [-0.0054, +0.0032]
0.2, [100,700)†	0.0973	0.0865	−0.0032 [-0.0122, +0.0053]	−0.0025 [-0.0146, +0.0082]
0.4, [100,700)†	0.1023	0.0784	+0.0017 [-0.0113, +0.0135]	−0.0106 [-0.0222, -0.0016]
0.1, [0,1000)†	0.0998	0.0830	−0.0007 [-0.0109, +0.0099]	−0.0060 [-0.0177, +0.0051]
0.1, [300,1000)†	0.0950	0.0901	−0.0055 [-0.0146, +0.0019]	+0.0012 [-0.0090, +0.0103]
0.05, [0,1000)	0.0890	0.0809	−0.0116 [-0.0209, -0.0046]	−0.0081 [-0.0180, -0.0003]

Checkpoint C, same 350 TEST words: S 0.226 [0.194, 0.262], F1 0.209 [0.175, 0.244] (914-entry vocab).

Source: `runs/P6_hparam_overfit/run_p6.py` → `scores.csv`; `analyze_p6.py` → `val_vs_test.csv`, `p6_summary.json`

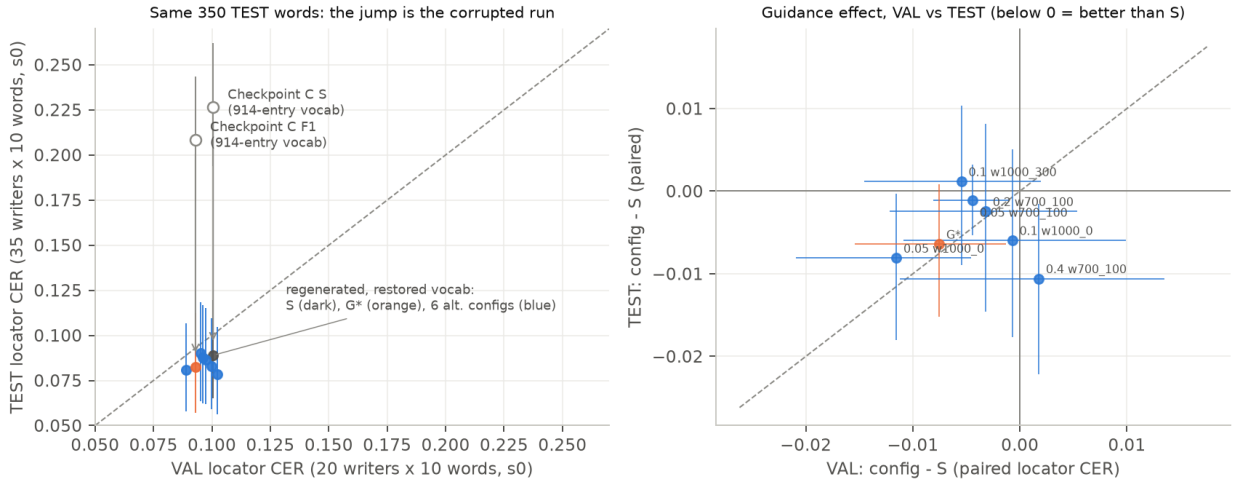


Figure 56: Left: VAL vs TEST locator CER per config (filled) and the two corrupted Checkpoint C points (hollow) on the same TEST words. All clean points lie on or below the diagonal: TEST is slightly easier than VAL, as it is for B0. Right: the guidance effect (config−S) on VAL vs TEST. All but one config keep a non-positive effect on TEST. The ranking across configs does not transfer (Spearman -0.11, $n = 7$), because the differences are within noise.

E.5 Hypothesis tests

H1 (overfit: tuned configs degrade more on TEST). Test: compare each config's VAL→TEST change with unguided S's change. S itself *improves* by 0.012. G*'s excess gap over S is $+0.001$, and its guidance effect on TEST minus on VAL is $+0.0011$ [-0.0098, +0.0123]. The configs that VAL ranked best (G*, 0.05/[0,1000)) remain among the best on TEST. **Rejected.** There is no measurable overfitting. What does not transfer is the fine *ranking* among tied configs, which is selection noise, not overfit.

H2 (guidance inherently weak on TEST writers). Test: is the TEST guidance effect smaller, or does it shrink for writers whose unguided CER is high? G*'s effect is -0.0076 on VAL and -0.0064 on TEST (CIs overlap almost entirely). The per-writer correlation between S CER and G* gain on TEST is $r = -0.14$, i.e. harder writers gain slightly *more*. **Rejected.** Guidance is weak *everywhere* (under 1 point of CER), not specifically on TEST. The full-run Checkpoint C F1−S paired diff (-0.0126, 28/35 writers) also confirms the effect's sign on TEST, even under the corrupted vocab.

H3 (VAL/TEST style mismatch; per-writer adaptation needed). Test 1: cosine distance of each writer's 5-ref style embedding (the generator's own MobileNetV2 `sf`) to the nearest training-writer centroid. TEST is not further from the training manifold than VAL (BN-HTR 0.045 vs 0.082, $p = 0.68$; BW 0.024 vs 0.019, $p = 0.09$). The distance predicts neither S CER ($\rho = -0.12$) nor the G* gain ($\rho = -0.08$). Test 2: a split-half per-writer oracle (pick each writer's config on even words, score on odd words) gives 0.0806 on TEST vs 0.0816 for global G*, and 0.1127 vs 0.1127 on VAL. 26/35 TEST writers pick G* anyway. **Rejected.** Per-writer adaptation buys nothing measurable.

E.6 Recommendation for Round 5

1. **Do not re-tune G* for generalisation reasons.** It generalises. Keep scale 0.1, $[100, 700)$, or the equally good, constraint-passing 0.05/[0,1000). Re-tuning on a held-out VAL split will not change the TEST picture, because the configs are within noise of each other.
2. **Treat global guidance as a ≤1-point lever and stop sweeping it.** Per-writer adaptive guidance is not supported by the data. The legibility budget is better spent on selection (best-of-N) and on the generator itself.
3. **Re-run Checkpoint C before drawing any TEST conclusion.** The vocab is restored, but the TEST numbers in §D for S/F1/F8 (and real-BW, whose crops were blank) are invalid. Add every tokenizer/vocab file and every scorer checkpoint to `FREEZE.json`, make `load_unet3` assert the vocab size against `content_embed_final.pt`, and rebuild R_font2real before re-running, since the selection judge is currently a copy of R_judge.
4. **Caveat.** This analysis generated on TEST writers (diagnostic only, 350 words, seed s0). Nothing here was used to pick a config, and the recommendation keeps the VAL-selected G*, so the re-run Checkpoint C remains a clean TEST.